

3

Rock properties

This chapter describes a number of properties of the rocks which form the reservoir. This includes the manner in which the pore space varies with pressure, how fluids interact with rocks, and how fluids interact with each other within the rock.

- The [Saturation functions](#) chapter explains the key concepts of relative permeability and capillary pressure modeling. These data specify how fluids block each other in the pore volume and how the rock interacts with the fluids.
- The [Saturation table scaling](#) chapter explains the details of the end point scaling facilities. This tool allows saturation tables to be adjusted from cell to cell in a straightforward manner without defining many saturation tables.
- The [Miscibility and surface tension effects](#) chapter gives information on how to model the miscibility of hydrocarbon phases with one another in a compositional simulation.
- The [Hysteresis](#) chapter explains the facility to add hysteretic saturation tables. This allows the saturation function to be specified differently when water saturation is rising and when water saturation is falling.
- The [Hysteresis in WAG floods](#) chapter explains the facility to alter saturation functions in water alternating gas injection scenarios to model effects such as the trapping of gas.
- The [Rock compaction](#) chapter explains a number of facilities which model the response of the rock matrix to changes in pressure, from simple tables of pore volume to variations in transmissibility, hysteretic effects and coal matrix shrinkage.

Saturation functions

x	Eclipse 100
x	Eclipse 300

Saturation functions are used to describe properties that depend on the phase saturations, namely the relative permeabilities and capillary pressures. The phase saturations are linked by the relationship:

$$S_o + S_g + S_w = 1$$

Eq. 3.1

In two-phase systems there is only one independent saturation and the properties are one dimensional functions. For example, in an oil-water system $S_g = 0$ and properties are a function of either S_w or $S_o = 1 - S_w$. Data for these one dimensional functions are entered as tables (keywords [SWOF](#) and [SGOF](#) / [SLGOF](#) or the alternative family [SWFN](#), [SGFN](#) and [SOF2](#)).

In three-phase systems there are two independent saturations and properties can be either one or two dimensional functions. A number of ways of modeling three phase properties are available:

- Capillary pressures and the water and gas relative permeabilities are treated as one dimensional functions, but the oil relative permeability is treated as a two dimensional function by interpolating the oil-gas and oil-water relative permeabilities. Data is entered as one dimensional tables using either of the families of the two phase keywords, except that [SOF3](#) is needed instead of [SOF2](#) to describe the two oil relative permeabilities. Interpolation of the oil relative permeability is performed using either the default method or one of Stone's methods.
- Two dimensional tables [SOF32D](#), [SGF32D](#) and [SWF32D](#) can be entered for any (or all) of the phase relative permeabilities. The capillary pressures are described either by using two dimensional tables [PCG32D](#), [PCW32D](#) (Eclipse 100) or by using the one dimensional keywords (Eclipse 300).
- The Baker methods can be used to describe any (or all) of the phase relative permeabilities as two dimensional functions. Data is entered as one dimensional tables using the keywords [SOF3](#), [SGFN](#) (or [SGF3](#)) and [SWFN](#) (or [SWF3](#)).

The methods described above can be applied to individual phases, and different methods can be used for each phase. Two further methods are available that apply to all phases:

- The IKU three-phase method. Data is entered as one dimensional tables using the keywords [SOF3](#), [SGF3](#) and [SWF3](#)
- The ODD3P three phase method. Data is entered as one dimensional tables using the keywords [SOF3](#), [SGF3](#) and [SWF3](#).

Saturation table keywords

All the saturation functions may be entered in multi-table form; tables are associated with cells using the [SATNUM](#) keyword and its hysteresis counterparts.

One dimensional tables (for two or three phase systems)

There are two families of keywords that define one-dimensional saturation-dependent properties of the reservoir fluids. The difference between them is that the first family allows you to enter the oil relative permeabilities in the same tables as the water and gas relative permeabilities, while in the second family, the oil relative permeabilities must be entered in a separate table versus oil saturation.

The keywords for the first family of tables are:

Keyword	Description
SWOF	Sets water relative permeability $k_{rw}(S_w)$, oil relative permeability in water $k_{row}(S_w)$, and water-oil capillary pressure $P_{cow}(S_w)$ as a function of water saturation. This can be used for two or three-phase systems with both water and oil.
SGOF	Sets gas relative permeability $k_{rg}(S_g)$, oil relative permeability in gas at connate water $k_{rog}(S_g)$, and oil-gas capillary pressure $P_{cog}(S_g)$ as a function of gas saturation. Either SGOF or SLGOF can be used for two or three-phase systems with both gas and oil.
SLGOF	Sets gas relative permeability $k_{rg}(S_l)$, oil relative permeability in gas at connate water $k_{rog}(S_l)$, and oil-gas capillary pressure $P_{cog}(S_l)$ as a function of liquid saturation $S_l = S_o + S_w = 1 - S_g$. Either SGOF or SLGOF can be used for two or three-phase systems with both gas and oil.

Table 3.1: Family (i) saturation table keywords

The keywords for the second family of tables are:

Restriction	Keyword	Description
	SWFN	Sets water relative permeability $k_{rw}(S_w)$ and capillary pressure $P_{cow}(S_w)$ as a function of water saturation. This can be used for two or three-phase systems with water.
	SGFN	Sets gas relative permeability $k_{rg}(S_g)$ and capillary pressure $P_{cog}(S_g)$ as a function of gas saturation. This can be used for two or three-phase systems with gas.
	SOF2	Sets oil relative permeability $k_{ro}(S_o)$ as a function of oil saturation. This can be used for two-phase systems with oil.
	SOF3	Sets relative permeability of oil in water $k_{row}(S_o)$, and oil in gas $k_{rog}(S_o)$ at the connate water saturation, as a function of oil saturation. This can be used for three-phase systems with oil.
Eclipse 100	SGWFN	Sets the gas $k_{rg}(S_g)$ and water $k_{rw}(S_g)$ relative permeabilities and the gas-water capillary pressure $P_{cgw}(S_g)$ for two-phase systems with only gas and water present as a function of gas saturation $S_g = 1 - S_w$. (This keyword is optional, as the gas and water saturation functions could alternatively be supplied through the SWFN and SGFN keywords, where the water capillary pressure supplied in SWFN represents the true gas-water capillary pressure and the capillary pressure is set to zero in the SGFN table.)

Table 3.2: Family (ii) saturation table keywords

Two dimensional tables (for three phase systems)

For three phase systems, two dimensional tables can be used to describe saturation properties. The keywords are:

Restriction	Keyword	Description
	SOF32D	Sets the oil relative permeability $k_{ro}(S_w, S_g)$ in a three-phase system as a function of both water saturation and gas saturation.
	SWF32D	Sets the water relative permeability $k_{rw}(S_o, S_g)$ in a three-phase system as a function of both oil saturation and gas saturation.
	SGF32D	Sets the gas relative permeability $k_{rg}(S_o, S_w)$ in a three-phase system as a function of both oil saturation and water saturation.
Eclipse 100	PCW32D	Sets the water/oil capillary pressure $P_{cow}(S_o, S_g)$ in a three-phase system as a function of both oil saturation and gas saturation.
Eclipse 100	PCG32D	Sets the gas/oil capillary pressure $P_{cog}(S_o, S_w)$ in a three-phase system as a function of both oil saturation and water saturation.

Table 3.3: Two dimensional saturation tables

One dimensional tables for three phase models (Baker, IKU and ODD3P)

These are required for three-phase systems which use the Baker (see [BAKER1](#) and [BAKER2](#) keywords), IKU (see [IKU3P](#) keyword) or the ODD3P (see the [ODD3P](#) keyword) relative permeability models.

The keywords are:

Restriction	Keyword	Description
Eclipse 300	SWF3	Sets water relative permeability to oil $k_{rwo}(S_w)$ and gas $k_{rwg}(S_w)$ and capillary pressure $P_{cow}(S_w)$ as a function of water saturation.
Eclipse 300	SGF3	Sets gas relative permeability to oil $k_{rgo}(S_g)$ and water $k_{rgw}(S_g)$ and capillary pressure $P_{cog}(S_g)$ as a function of gas saturation.
Eclipse 300	SOF3	Sets relative permeability of oil in water $k_{row}(S_o)$, and oil in gas $k_{rog}(S_o)$ at the connate water saturation, as a function of oil saturation.

Table 3.4: One dimensional tables for three phase models

Saturation table compatibility

Sufficient data must be provided to allow the simulators to calculate relative permeabilities and capillary pressures for all phases present. In two-phase systems, the one dimensional tables must be used. In general, either family can be used, however family (ii) must be used if running

Eclipse 100

- Miscible Flood

Eclipse 100

- Solvent options

Eclipse 100

- Two-phase gas-water systems

Keywords from the different families must not be mixed in the same run.

In three-phase systems, both one and two dimensional tables can be used. If one dimensional tables (of either family) are supplied, then two dimensional interpolation methods, such as those of Stone can be used.

Eclipse 300

When saturation functions are defined using keywords from the first keyword family, these are converted internally into the equivalent keywords in the second keyword family before any consistency checks are performed. Hence any consistency warnings or errors will be reported in terms of the second family of keywords. For example, if the keywords **SWOF** and **SGOF** are specified, any consistency warnings or errors will be reported in terms of the **SWFN**, **SGFN** and **SOF3** keywords.

Eclipse 300

Other models for three phase systems, Baker, IKU and ODD3P require specific tables: **SOF3**, **SGF3** and **SWF3**.

If a two-dimensional table **SOF32D** is used to describe the oil relative permeability $k_{ro}(S_w, S_g)$, either one or two dimensional tables can be used to describe the gas and water relative permeabilities. If the family (i) one dimensional tables (**SWOF** / **SGOF**) are used, then the one dimension oil relative permeabilities in these tables ($k_{row}(S_w)/k_{rog}(S_g)$) will be ignored.

If two-dimensional tables are used to describe the water or gas relative permeability, then capillary pressure data can be supplied as follows:

- Eclipse 100

Two-dimensional tables (**PCG32D** / **PCW32D**)

Eclipse 300

- One dimensional table, in which case the one dimensional relative permeability in the table will be ignored.

Saturation table lookup

Saturation values need not be specified at equal intervals, and are used in the specified form, rather than being interpolated onto regular intervals.

Default values represented by 1* may be inserted as required in one or more rows for columns 2 onwards for the one-dimensional table keywords **SWFN**, **SWF3**, **SWOF**, **SGFN**, **SGWFN**, **SGF3**, **SGOF**, **SLGOF**, **SOF2** and **SOF3**. Where default values have been entered, the actual values will be calculated by linear interpolation using the nearest non-default neighbors and substituted into the table.

However, where only one non-default value is entered in an entire column or where default values are entered at the start or end of the tables, the interpolation behavior for Eclipse 100 and Eclipse 300 can differ. For example, for the **SGOF** keyword, if the capillary pressure column is defaulted throughout, the values are replaced with 0.0 throughout by both simulators. However, if only one value is not defaulted and this is not the first value in the table, in Eclipse 100, the first value is set to 0.0, linear interpolation is performed to replace any default values up to the defined value, and all subsequent values are assigned to the defined value. In contrast, in Eclipse 300, all defaulted values in the column are set equal to the single non-defaulted value. It is recommended, therefore, that when defaults are entered into a table, the table is reported using the **RPTPROPS** keyword and the resultant values inspected to verify the intended interpolation behavior.

Tolerance for critical saturations

Critical saturations are assigned from the saturation tables by establishing the highest value of saturation for which the associated relative permeability is zero. In practice, the exceeding of a threshold (tolerance) value is used to test whether the relative permeability value is zero or non-zero. For Eclipse 100 and 300 the default thresholds are 1×10^{-6} and 1×10^{-20} respectively. These default values may be modified using the **TOLCRIT** keyword.

Water saturation properties

One dimensional tables

An example data table for the [SWFN](#) keyword is given below. The data for the [SWOF](#) keyword is similar but with an additional column containing oil relative permeabilities.

```
-- SWAT KRW PCOW
--      (PSIA)
SWFN
.22 .0 7.0
.3 .07 4.0
.4 .15 3.0
.5 .24 2.5
.6 .33 2.0
.8 .65 1.0
.9 .83 .5
1.0 1.0 .0 /
```

The water saturations must be entered in ascending order.

The third column is the oil-water capillary pressure ($P_{cow} = P_o - P_w$). The values can become negative with increasing water saturation, if required. They should all be set to zero if there is no water-oil capillary pressure. In saturation tables intended for well completions only (that is, not referenced for any grid block in keyword [SATNUM](#)) the capillary pressure data is not used, and the values entered are disregarded. In gas-water problems, the water capillary pressure is interpreted as the gas-water capillary pressure ($P_{cgw} = P_g - P_w$).

Three saturation values in the table are of special interest:

S_{wcr} The **critical** water saturation. This is the highest saturation value for which the water relative permeability is zero. At saturations above this value, water is mobile. Note that a critical water saturation **must** be defined (the [SWFN](#) or [SWOF](#) table must contain a water saturation for which $k_{rw} = 0$). In the example above, $S_{wcr} = 0.22$.

S_{wco} The minimum water saturation value in the table. The equilibration calculation (see keyword [EQUIL](#)) sets the water saturation to this value in grid blocks that lie above the water contact (or water transition zone). In this respect S_{wco} is the **connate** water saturation. In the example shown above, $S_{wco} = S_{wcr}$. If for any reason the connate water saturation is less than the critical value, the table should begin with the connate value. For example for $S_{wco} = 0.2$ and $S_{wcr} = 0.22$, the table should begin with the lines

```
.20 .0 7.0
.22 .0 7.0
.3 .07 4.0
...
...
```

(Note that Eclipse can handle zero gradients in the capillary pressure versus saturation, as shown in the above example between the S_w values 0.2 & 0.22.)

S_{wmax} The maximum water saturation value in the table. The equilibration calculation sets the water saturation to this value in grid blocks that lie below the water transition zone. In the example shown above, $S_{wmax} = 1.0$, which results in the water zone being fully saturated with water.

In an **oil-wet system**, in order to use correctly the Killough wetting phase model on the oil phase (option 7, item 2 of **EHYSTR** keyword), $S_{w\max}$ should equal $1.0 - S_{oco}$, S_{oco} being the connate oil saturation.

Two dimensional tables

A full 2-dimensional table of water relative permeability versus oil and gas saturations can be entered in three-phase runs using the **SWF32D** keyword.

When using **SWF32D** the connate water is defined as $S_{wco} = 1.0 - S_{o,max}$.

Keyword **PCW32D** can be used in conjunction with **SWF32D** to enter the water/oil capillary pressure if required.

Gas saturation properties

One dimensional tables

An example data table for the **SGFN** keyword is given below. The data for the **SGOF** keyword is similar but with an additional column containing oil relative permeabilities. The data for the **SLGOF** keyword is similar to the **SGOF** keyword, except that the saturation column contains liquid saturation values in ascending order.

--	SGAS	KRG	PCOG
--			(PSIA)
SGFN			
	.0	.0	0.0
	.04	.0	.2
	.1	.022	.5
	.2	.1	1.0
	.3	.24	1.5
	.4	.34	2.0
	.5	.42	2.5
	.6	.5	3.0
	.7	.81	3.5
	.78	1.0	3.9 /

The gas saturations must be entered in ascending order.

The third column is the oil-gas capillary pressure ($P_{cog} = P_g - P_o$). The values can start negative if required. They should all be set to zero if there is no gas-oil capillary pressure. In saturation tables intended for well completions only (that is not referenced for any grid block in keyword **SATNUM**) the capillary pressure data is not used, and the values entered are disregarded. In gas-water problems, the gas-water capillary pressure is defined in the water saturation table, so the capillary pressure values in the gas saturation table should be set to zero.

Three saturation values in the table are of special interest:

S_{gcr} The critical gas saturation.	This is the highest saturation value for which the gas relative permeability is zero. At saturations above this value, gas is mobile. Note that a critical gas saturation must be defined (the SGFN or SGOF {or SLGOF } table must contain a gas {or liquid} saturation for which $k_{rg} = 0$). In the example above, $S_{gcr} = 0.04$.
---	--

S_{gco} The minimum gas saturation value in the table. The equilibration calculation (see keyword [EQUIL](#)) sets the gas saturation to this value in grid blocks that lie below the gas contact (or gas transition zone). In this respect S_{gco} is the **connate** gas saturation.

Eclipse 100

Normally in Eclipse 100 $S_{gco} = 0.0$, as in the example above.

Eclipse 300

In Eclipse 300 $S_{gco} = 0.0$.

$S_{g \max}$ The maximum gas saturation value in the table. The equilibration calculation sets the gas saturation to this value in grid blocks that lie above the gas transition zone. Normally $S_{g \max} = 1.0 - S_{wco}$, as in the example above, with $S_{wco} = 0.22$ as in the first [SWFN](#) table example in this chapter.

In an **oil-wet system**, in order to use correctly the Killough wetting phase model on the oil phase (option 7, Item 2 of [EHYSTR](#) keyword), $S_{g \max}$ should equal $1.0 - S_{oco}$, S_{oco} being the connate oil saturation.

Two dimensional tables

A full 2-dimensional table of gas relative permeability versus oil and water saturations can be entered using the [SGF32D](#) keyword, in three-phase runs.

Eclipse 100

Keyword [PCG32D](#) can be used in conjunction with [SWF32D](#) to enter the gas/oil capillary pressure if required.

Three-phase gas relative permeability models

In Eclipse 300, a number of formulae are available for calculating the two-dimensional three-phase gas relative permeability at particular oil and water saturations, from the input one-dimensional relative permeabilities:

- Default method
- Baker 1 (can also be used for oil and water)
- Baker 2 (can also be used for oil and water)
- IKU (used for all phases)
- ODD3P (used for all phases)

The formulae are not needed if the [SGF32D](#) keyword has been used to input the three-phase oil relative permeability as a two-dimensional table versus oil and gas saturations (see above).

Default formula

Eclipse 300

The gas relative permeability is given by

$$k_{rg} = \frac{S_o k_{rgo} + (S_w - S_{wco}) k_{rgw}}{S_o + S_w - S_{wco}} \quad \text{Eq. 3.2}$$

The formula is essentially the same as the default formula for oil in "[Three phase oil relative permeability models](#)".

Here

k_{rgo} is the gas relative permeability for a system with oil, gas and connate water

k_{rgw} is the gas relative permeability for a system with gas and water only

k_{rgo} is calculated by lookup of the one-dimensional gas relative permeability table. The value is modified if the [MISCIBLE](#) option is used. The value is further modified if the default HCSCAL option is used. For more information, see [FORMOPTS](#).

The k_{rgw} calculation depends on whether the [NOMIX](#) keyword has been specified. If it has, the value is calculated by lookup of the one-dimensional gas relative permeability table. If [NOMIX](#) has not been specified, the oil and gas relative permeability tables are interpolated. See "[Near critical oil and gas relative permeabilities](#)" for details.

Oil saturation properties

One dimensional tables

The relative permeabilities of oil in water, and oil in gas and connate water, are entered either in keywords [SWOF](#) and [SGOF](#) (or [SLGOF](#)), or with keyword [SOF3](#) (or [SOF2](#) in a two-phase run).

The relative permeability tables are not used in the calculation of the initial saturations. The Eclipse equilibration calculation sets oil saturations using the formula $S_o = 1 - S_w - S_g$.

An example data table for keyword [SOF3](#) is given below. If family (i) were being used, the k_{row} data would be included in keyword [SWOF](#) and the k_{rog} data would be included in either keyword [SGOF](#) or [SLGOF](#).

-- SOIL	KROW	KROG	
SOF3			
.0000	.0000	.0000	
.2000	.0000	.0000	
.3800	1*	.0000	
.4000	.0050	1*	
.4800	1*	.0200	
.5000	.0649	1*	
.5800	1*	.1000	
.6000	.1250	1*	
.6800	1*	.3300	
.7000	.4000	1*	
.7400	1*	.6000	
.7800	1.0000	1.0000	/

The oil saturations must be entered in ascending order in column 1. Columns 2 and 3 contain the corresponding oil relative permeabilities for oil-water systems, and for oil-gas-connate water systems respectively. Columns 2 and 3 may contain default specifications (represented by 1*). When the table is read in, defaults are replaced by values obtained by linear interpolation (in the above example, the first default in column 2 is replaced by the value 0.0045).

Five saturation values in the table are of special interest:

S_{oco} The minimum oil saturation value in the table.

This is the **connate** oil saturation. Normally, $S_{oco}=0.0$, as in the example above. This value is particularly used in **oil-wet system** modeling, applying the Killough wetting model to the oil phase (option 7, item 2 of [EHYSTR](#) keyword), to calculate both $S_{g\ max}$ and $S_{w\ max}$. In these cases, normally S_{oco} is a non-zero value.

S_{ocr} The critical oil saturation.	This is the highest saturation value for which both the oil-water and oil-gas relative permeabilities are zero. At saturations above this value, oil is mobile in the three-phase region. Note that a critical oil saturation must be defined (the SOF3 table must contain an oil saturation for which both k_{row} and k_{rog} are 0). In the example above, $S_{ocr}=0.20$.
S_{orw} The Residual Oil Saturation in the oil-water system.	This is the saturation at which the oil relative permeability in the oil-water system becomes zero (S_o at which $k_{row}=0$). In the example above, $S_{orw}=0.20$.
S_{org} The Residual Oil Saturation in the gas-oil system.	This is the saturation at which the oil relative permeability in the gas-oil system becomes zero (S_o at which $k_{rog}=0$). In the example above, $S_{org}=0.38$.
$S_{o\ max}$ The maximum oil saturation value in the table.	This should be equal to $1-S_{wco}-S_{wco}$ as determined from the SWFN table. The two oil relative permeabilities should be the same at $S_{o\ max}$ (both represent a case with $S_o = S_{o\ max}$, $S_w = S_{wco}$ and $S_g = 0$). Eclipse reports an error if this condition is not satisfied. In an oil-wet system , normally $S_{o\ max} = 1.0$.

Two-dimensional tables

A full two-dimensional table of oil relative permeabilities versus water and gas saturations can be input using the SOF32D keyword, in three-phase runs. This keyword replaces SOF3, or overrides the oil relative permeability data supplied in the SWOF and SGOF or SLGOF keywords. An example of the SOF32D keyword is given below:

```

SOF32D
--
-----SWAT-----
0.22 0.27 0.32 0.37 0.42 0.47 0.52 0.57 0.72 0.77 0.78 /
--SGAS
0.00 1.000 0.625 0.345 0.207 0.113 0.083 0.053 0.023 0.002 0.001 0.000 /
0.05 0.555 0.337 0.210 0.110 0.078 0.047 0.021 0.004 0.001 0.000 /
0.10 0.330 0.212 0.106 0.074 0.042 0.019 0.003 0.002 0.000 /
0.15 0.215 0.103 0.069 0.036 0.017 0.003 0.002 0.001 0.000 /
0.20 0.100 0.065 0.031 0.015 0.002 0.002 0.001 0.000 /
0.25 0.060 0.025 0.014 0.002 0.001 0.001 0.000 /
0.30 0.020 0.012 0.001 0.001 0.001 0.001 0.000 /
0.35 0.010 0.001 0.001 0.001 0.000 /
0.40 0.000 0.000 0.000 0.000 /
/

```

The first record contains the water saturation values, and the first column of the subsequent records contains the gas saturation values. At present the SOF32D keyword cannot be used with the end-point scaling, hysteresis and miscible options.

Three-phase oil relative permeability models

A number of formulae are available for calculating the two dimensional three-phase oil relative permeability at particular water and gas saturations, from the input one dimensional relative permeabilities of oil in water and oil in gas and connate water:

- Default method
- Stone 1 (modified)

- Stone 2 (modified)
- Eclipse 300: Baker 1 (can also be used for gas and water)
- Eclipse 300: Baker 2 (can also be used for gas and water)
- Eclipse 300: IKU (used for all phases)
- Eclipse 300: ODD3P (used for all phases)

The formulae are not needed if the [SOF32D](#) keyword has been used to input the three-phase oil relative permeability as a two-dimensional table versus water and gas saturations (see above).

Default formula

The default model for the three-phase oil relative permeability is based on an assumption of complete segregation of the water and gas within each grid cell. The model provides a simple but effective formula which avoids the problems associated with other methods (for example poor conditioning and negative values).

The default model assumed by Eclipse is shown in figure 3.1. The oil saturation is assumed to be constant and equal to the block average value, S_o , throughout the cell. The gas and water are assumed to be completely segregated, except that the water saturation in the gas zone is equal to the connate saturation, S_{wco} . The full breakdown, assuming the block average saturations are S_o , S_w and S_g (with $S_o + S_w + S_g = 1$) is as follows:

In a fraction $S_g / (S_g + S_w - S_{wco})$ of the cell (the gas zone),

the oil saturation is S_o

the water saturation is S_{wco}

the gas saturation is $S_g + S_w - S_{wco}$

In a fraction $(S_w - S_{wco}) / (S_g + S_w - S_{wco})$ of the cell (the water zone),

the oil saturation is S_o

the water saturation is $S_g + S_w$

the gas saturation is 0

The oil relative permeability is then given by

$$k_{ro} = \frac{S_g k_{rog} + (S_w - S_{wco}) k_{row}}{S_g + S_w - S_{wco}} \quad \text{Eq. 3.3}$$

where

k_{rog} is the oil relative permeability for a system with oil, gas and connate water tabulated as a function of S_o ,

k_{row} is the oil relative permeability for a system with oil and water only, also tabulated as a function of S_o .

For the default model, the values of $k_{rog}(S_o)$ and $k_{row}(S_o)$ are looked up as functions of the oil saturation $S_o = 1 - S_w - S_g$. Both two-phase oil relative permeability functions are tabulated as functions of oil

saturation in the input data either directly if the Family (ii) set of keywords is used or indirectly via internal tabular conversion if the Family (i) set of keywords is used.

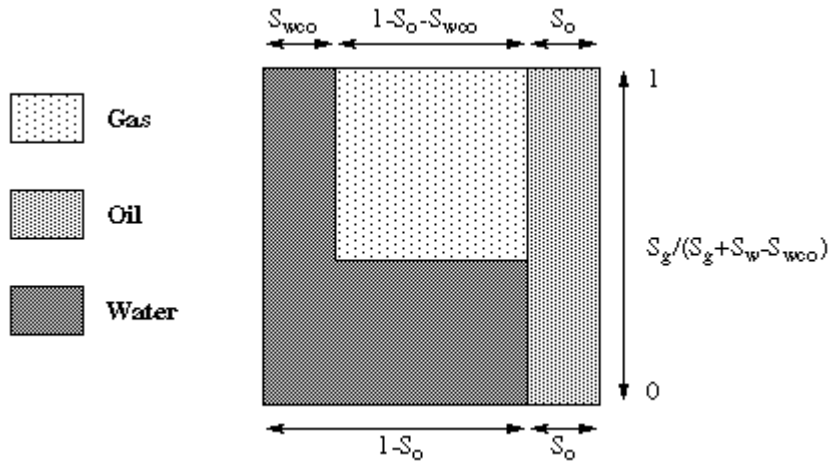


Figure 3.1. The default three-phase oil relative permeability model assumed by Eclipse

Stone's first model (modified)

A modified version of Stone's first three-phase oil relative permeability model ([Stone, 1973](#)), summarized in [Aziz and Settari \(1979\)](#), is available with keyword **STONE1** in the PROPS section. The formula is

$$k_{ro} = k_{rocw} SS_o F_w F_g \quad \text{Eq. 3.4}$$

where

k_{rocw} is the value of the oil relative permeability in the presence of connate water only

$$SS_o = (S_o - S_{om}) / (1 - S_{wco} - S_{om}) \text{ when } S_o > S_{om}$$

$$F_w = k_{row} / (k_{rocw} \cdot (1 - SS_w))$$

$$F_g = k_{rog} / (k_{rocw} \cdot (1 - SS_g))$$

where

$$SS_w = (S_w - S_{wco}) / (1 - S_{wco} - S_{om}) \text{ when } S_w > S_{wco}$$

$$SS_g = S_g / (1 - S_{wco} - S_{om})$$

where

S_o , S_w and S_g represent the block averaged values for the oil, water and gas saturations in a grid cell

k_{rog} denotes the oil relative permeability for a system with oil, gas and connate water

k_{row} denotes the oil relative permeability for a system with oil and water only

k_{rocw} denotes the oil relative permeability in the presence of connate water only

S_{om} is the minimum residual oil saturation

For Stone's first model, the values of $k_{rog}(S_{og})$ and $k_{row}(S_{ow})$ are looked up as two-phase saturation functions where $S_{og} = 1 - S_{wco} - S_g$ and $S_{ow} = 1 - S_w$. Both two phase oil relative permeability functions are tabulated as functions of oil saturation in the input data either directly if the family (ii) set of keywords is used or indirectly via internal tabular conversion if the family (i) set of keywords is used.

By default S_{om} is taken to be the minimum of the critical oil-to-water saturation and the critical oil-to-gas saturation, $\min(S_{owcr}, S_{ogcr})$. Alternatively, S_{om} can be input either as a function of water saturation (**SOMWAT** keyword) or as a function of gas saturation (**SOMGAS** keyword).

If the SOMWAT keyword is used, S_{om} is interpolated from the water saturation as shown in the following figure:

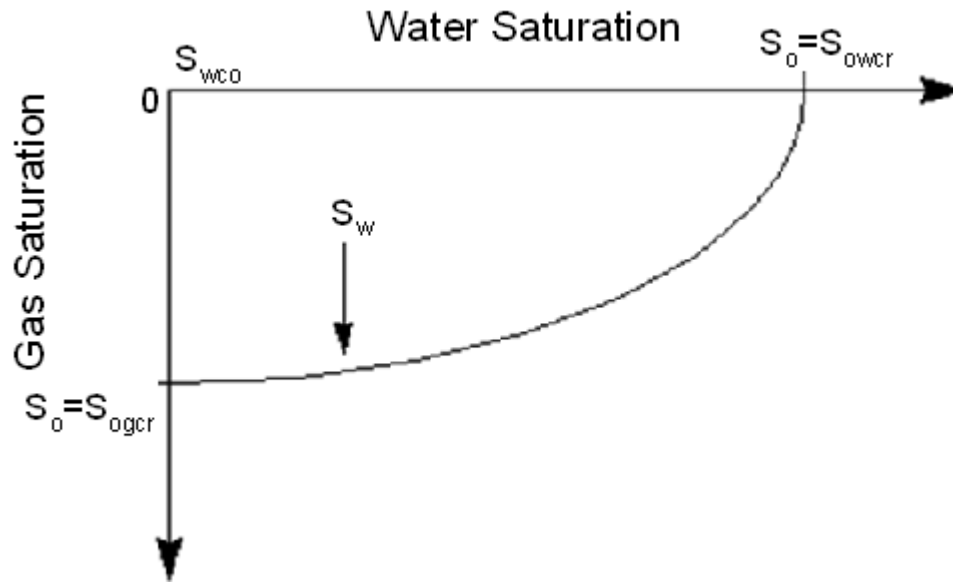


Figure 3.2. Calculating the minimum residual oil saturation from the water saturation

If the SOMGAS keyword is used, S_{om} is interpolated from the gas saturation as shown in the following figure:

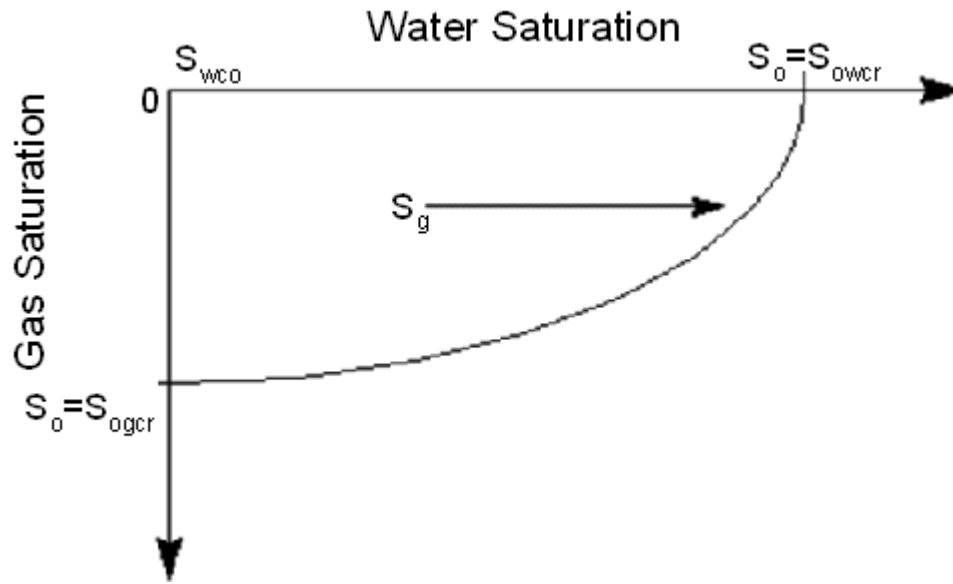


Figure 3.3. Calculating the minimum residual oil saturation from the gas saturation

If the hysteresis option is active, the drainage and imbibition S_{om} values are taken from the [SATNUM](#) and [IMBNUM](#) tables. For example in the SOMWAT case:

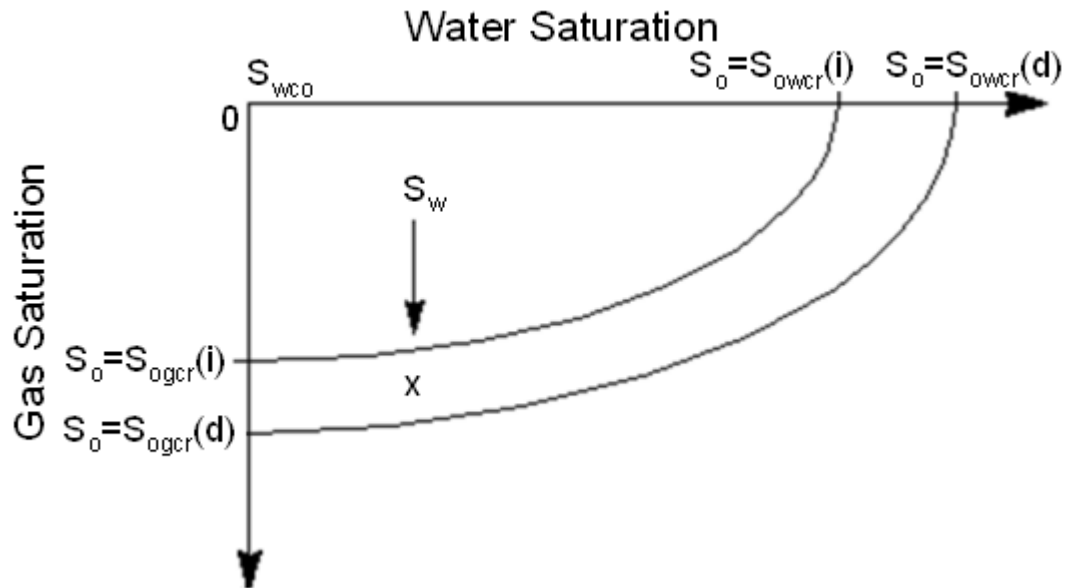


Figure 3.4. Calculating the minimum residual oil saturation for drainage and imbibition with hysteresis

The S_{om} value (point x in the picture above) is calculated as follows:

$$S_{om} = S_{om}^{(i)} + F \cdot (S_{om}^{(d)} - S_{om}^{(i)}) \quad \text{Eq. 3.5}$$

$$F = F_{sw}F_w + (1 - F_{sw})F_g \quad \text{Eq. 3.6}$$

$$F_{sw} = \frac{S_w - S_{wco}}{1 - S_{wco} - S_{owcr}^{(h)}} \quad \text{Eq. 3.7}$$

$$F_w = \frac{S_{owcr}^{(i)} - S_{owcr}^{(h)}}{S_{owcr}^{(i)} - S_{owcr}^{(d)}} \quad \text{Eq. 3.8}$$

$$F_g = \frac{S_{ogcr}^{(i)} - S_{ogcr}^{(h)}}{S_{ogcr}^{(i)} - S_{ogcr}^{(d)}} \quad \text{Eq. 3.9}$$

$S_{owcr}^{(i)}$ is the critical oil-to-water imbibition

$S_{owcr}^{(d)}$ is the critical oil-to-water drainage

$S_{owcr}^{(h)}$ is the critical oil-to-water actual

$S_{ogcr}^{(i)}$ is the critical oil-to-gas imbibition

$S_{ogcr}^{(d)}$ is the critical oil-to-gas drainage

$S_{ogcr}^{(h)}$ is the critical oil-to-gas actual

The calculation of three phase oil relative permeability using Stone's method 1 is enabled by means of the [STONE1](#) keyword in the [PROPS](#) section of the input data. Tabulated output of the three phase oil relative permeability values is controlled by the mnemonic [SOF](#) in the [RPTPROPS](#) keyword.

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If [STONEPAR](#) is used, this is obtained as a function of the A and B parameters:

$$S_{om} = S_{ogcr} X_g^A + S_{owcr} (1 - X_g)^A + B S_{gm}^2 X_g (1 - X_g) \quad \text{Eq. 3.10}$$

where:

$$S_{gm} = 1 - S_{wco} - S_{ogcr} \quad \text{Eq. 3.11}$$

Stone's first model exponent

The formula for Stone's first model may be rearranged such that an exponent may be applied to the combination of saturation terms:

$$k_{ro} = \beta (k_{row} k_{rog}) / k_{rocw} \quad \text{Eq. 3.12}$$

where

$$\beta = \left(\frac{SS_o}{(1 - SS_w) \cdot (1 - SS_g)} \right)^\eta \quad \text{Eq. 3.13}$$

This formulation is due to [Hustad and Holt \(1992\)](#). The β term may be interpreted as a variable that varies between zero and one for low and high oil saturations respectively. If the exponent η is equal to 1.0, the formula corresponds to Stone's first model. Values above 1.0 cause the oil isoperms to spread within the saturation space, whereas values below 1.0 have the opposite effect. The value of the exponent may be used, therefore, to match the predicted oil recovery to the observed data and may be specified using the [STONE1EX](#) keyword.

Stone's second model (modified)

A modified version of Stone's second three phase oil relative permeability model ([Stone, 1973](#)), summarized in [Aziz and Settari \(1979\)](#) can be used. The formula is

$$k_{ro} = k_{rocw} \left[\left(\frac{k_{row}}{k_{rocw}} + k_{rw} \right) \left(\frac{k_{rog}}{k_{rocw}} + k_{rg} \right) - k_{rw} - k_{rg} \right] \quad \text{Eq. 3.14}$$

where

k_{rog} denotes the oil relative permeability for a system with oil, gas and connate water

k_{row} denotes the oil relative permeability for a system with oil and water only

k_{rocw} is the oil relative permeability in the presence of connate water only

Both sets of two-phase oil relative permeability functions are tabulated as functions of oil saturation in the input data either directly if the Family (ii) set of keywords is used or indirectly via internal tabular conversion if the Family (i) set of keywords is used. For Stone's second model, the values of $k_{rog}(S_{og})$ and $k_{row}(S_{ow})$ are looked up as two-phase saturation functions where $S_{og} = 1 - S_{wco} - S_g$ and $S_{ow} = 1 - S_w$.

Note the values of k_{ro} produced by this formula can be negative. Eclipse automatically changes any negative k_{ro} values produced to zero.

The calculation of three phase oil relative permeability using Stone's method 2 is enabled by means of [STONE2](#) keyword in the PROPS section of the input data. Tabulated output of the three phase oil relative permeability values is controlled by the mnemonic SOF in the [RPTPROPS](#) keyword.

Three phase relative permeability models

Baker method

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Baker's two interpolation methods ([Baker, 1988](#)) can be used to determine three-phase relative permeability. The methods are selected by using the [BAKER1](#) or [BAKER2](#) keywords in the PROPS section. The Baker methods can be applied to any (or all) phases.

The keyword [BAKER1](#) is used to perform linear interpolation based on the relations:

$$k_{ro} = \frac{(S_g - S_{gc})K_{rog} + (S_w - S_{wc})K_{row}}{(S_g - S_{gc}) + (S_w - S_{wc})} \quad \text{Eq. 3.15}$$

$$k_{rg} = \frac{(S_o - S_{oc})K_{rgo} + (S_w - S_{wc})K_{rgw}}{(S_o - S_{oc}) + (S_w - S_{wc})} \quad \text{Eq. 3.16}$$

$$k_{rw} = \frac{(S_o - S_{oc})K_{rwo} + (S_g - S_{gc})K_{rwg}}{(S_o - S_{oc}) + (S_g - S_{gc})} \quad \text{Eq. 3.17}$$

The keyword [BAKER2](#) is used to perform linear interpolation based on the relations:

$$k_{ro} = \frac{S_g K_{rog} + S_w K_{row}}{S_g + S_w} \quad \text{Eq. 3.18}$$

$$k_{rg} = \frac{S_o K_{rgo} + S_w K_{rgw}}{S_o + S_w} \quad \text{Eq. 3.19}$$

$$k_{rw} = \frac{S_o K_{rwo} + S_g K_{rwg}}{S_o + S_g} \quad \text{Eq. 3.20}$$

$K_{rog}(S_o)$ and $K_{row}(S_o)$ are looked up from the [SOF3](#) table.

$K_{rgo}(S_g)$ and $K_{rgw}(S_g)$ are looked up from the [SGF3](#) table.

$K_{rwo}(S_w)$ and $K_{rwg}(S_w)$ are looked up from the [SWF3](#) table.

This linear interpolation may be shown geometrically on a ternary diagram similar to figure 3.5.

IKU method

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The IKU model, which is due to [Hustad and Hansen \(1996\)](#) and [Hustad et al \(2002\)](#), is an alternative three-phase relative permeability model to those of Stone and is selected by specifying the [IKU3P](#) keyword in the PROPS section. This model applies to all three phases symmetrically and data are entered for water and gas with respect to the other two phases using the [SWF3](#) and [SGF3](#) keywords. Normalized saturations S_{in} are used to look up, in normalized tables, the pairs of two-phase relative permeabilities $k_{rij}(S_{in})$ and $k_{rik}(S_{in})$ where $i, j, k = g, o, w$ subject to the constraint that $i \neq j \neq k$. These are then combined to calculate the three-phase relative permeability k_{ri} . The normalized saturation S_{in} is given by:

$$S_{in} = \frac{S_i - S_{imn}}{S_{imx} - S_{imn}} \quad \text{Eq. 3.21}$$

where

S_i is the grid block saturation for phase i

S_{imn} and S_{imx} is given by:

$$S_{imn} = \frac{S_j S_{irj} + S_k S_{irk} + S_{irj} S_{irk} (S_i - 1)}{S_j (1 - S_{irk}) + S_k (1 - S_{irj})} \quad \text{Eq. 3.22}$$

$$S_{imx} = \frac{S_j S_{kri} + S_k S_{jri} + S_{jri} S_{kri} (S_i - 1)}{S_j S_{kri} + S_k S_{jri}} \quad \text{Eq. 3.23}$$

where

S_j and S_k are the grid block saturations for phases j and k respectively and

S_{irj} , S_{irk} , S_{jri} , S_{kri} , S_{krj} and S_{jrk} are the residual saturations of phases i to j , i to k , j to i , k to i , k to j and j to k respectively

The normalized table look-up for $k_{rij}(S_{in})$ is accomplished by normalizing the tabulated data for $k_{rij}(S_i)$ between the lower limit S_{irj} and the upper limit $1 - S_{jri}$ such that $S_i = S_{irj}$ and $S_i = 1 - S_{jri}$ correspond to 0 and 1 respectively. The value of $k_{rij}(S_{in})$ is then determined for the supplied value of S_{in} . A similar process is employed to calculate $k_{rik}(S_{in})$, $k_{rji}(S_{jn})$, $k_{rjk}(S_{jn})$, $k_{rki}(S_{kn})$ and $k_{rkj}(S_{kn})$.

Each pair of two-phase relative permeabilities is then combined to calculate the three-phase relative permeability:

$$k_{ri} = \frac{S_j}{S_j + S_k} \cdot k_{rij} + \frac{S_k}{S_k + S_j} \cdot k_{rik} \quad \text{Eq. 3.24}$$

and similarly for k_{rj} and k_{rk} . This formulation is symmetrical and is applied to all three phases.

This calculation process may be illustrated for the case of relative permeability to oil k_{ro} with the aid of a ternary diagram (as illustrated in the following diagram):

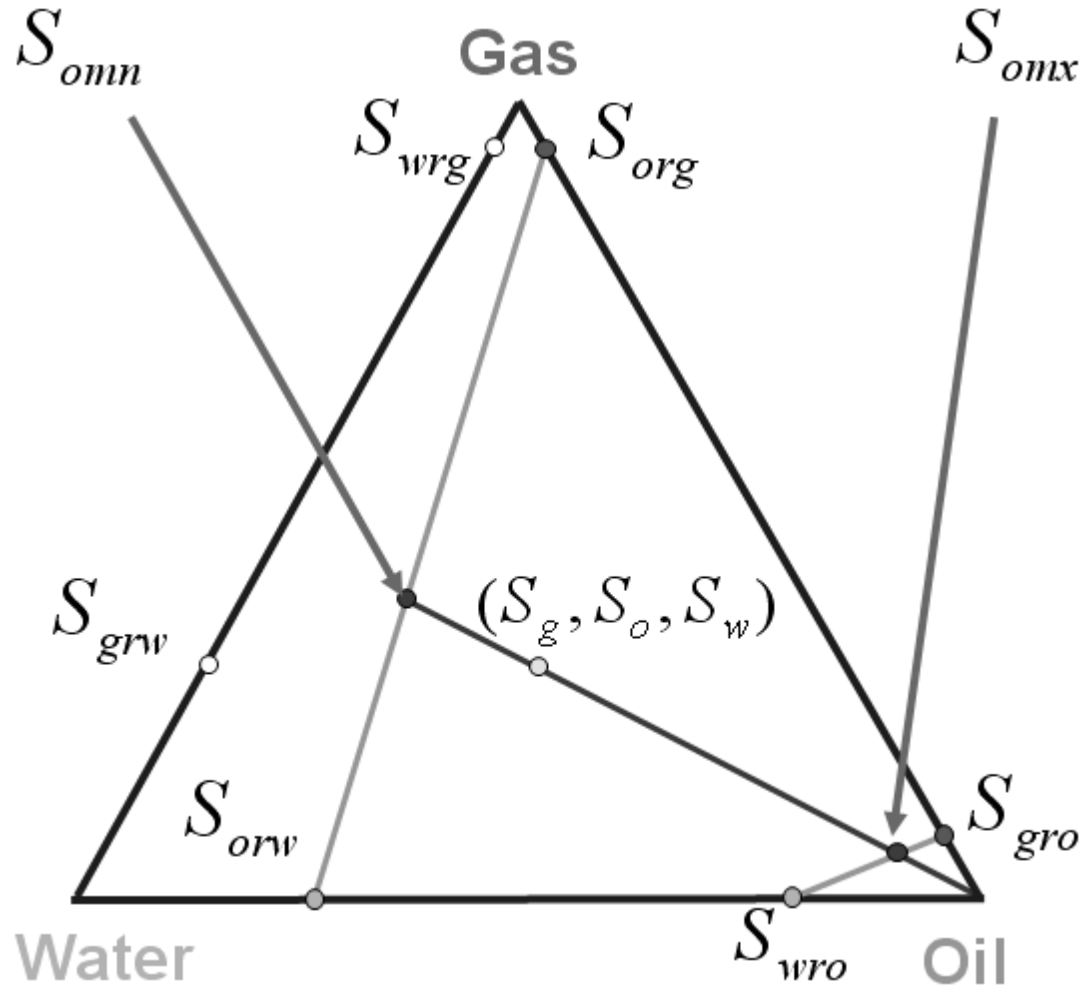


Figure 3.5. Ternary diagram showing the linear interpolation used to calculate S_{omn} and S_{omx}

It is assumed that a three-phase saturation condition exists as depicted by the point denoted by (S_g, S_o, S_w) in the center of the ternary diagram. A line is drawn through this point and the oil apex. This line meets the line constructed by joining the grid block points S_{orw} and S_{org} at the minimum saturation to oil flow S_{omn} and the line constructed by joining the grid block points S_{wro} and S_{gro} at the maximum saturation to oil flow S_{omx} . These oil saturation extrema are calculated according to:

$$S_{omn} = \frac{S_w S_{orw} + S_g S_{org} + S_{orw} S_{org} (S_o - 1)}{S_w (1 - S_{org}) + S_g (1 - S_{orw})} \quad \text{Eq. 3.25}$$

$$S_{omx} = \frac{S_w S_{gro} + S_g S_{wro} + S_{wro} S_{gro} (S_o - 1)}{S_w S_{gro} + S_g S_{wro}} \quad \text{Eq. 3.26}$$

using values pertaining to the grid block, and the normalized oil saturation S_{on} for the grid block is calculated according to:

$$S_{on} = \frac{S_o - S_{omn}}{S_{omx} - S_{omn}} \quad \text{Eq. 3.27}$$

The two-phase relative permeabilities $k_{row}(S_{on})$ and $k_{rog}(S_{on})$ are determined from table look-up where $k_{row}(S_o)$ has been normalized to unity extent between the table values for S_{orw} and $1 - S_{wro}$ and $k_{rog}(S_o)$ has been normalized to unity extent between the table values for S_{org} and $1 - S_{gro}$. Finally k_{ro} is calculated from:

$$k_{ro} = \frac{S_w}{S_w + S_g} \cdot k_{row} + \frac{S_g}{S_g + S_w} \cdot k_{rog} \quad \text{Eq. 3.28}$$

k_{rg} and k_{rw} are calculated in a similar fashion.

ODD3P method

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The ODD3P method provides a coupled hysteretic model for three-phase relative permeabilities and capillary pressures. This method is due to [Hustad \(2002\)](#) and extends the IKU method ([Hustad et al, 2002](#)) described above. A summary of the ODD3P method is presented below. A full description of this method may be found in [Hustad \(2002\)](#). It is recommended that this reference is read as a precursor to using this model.

Whereas the Stone-type models are based upon water-wet systems, the ODD3P method is more general with respect to wettability in that it uses pairs of two-phase relative permeability data for each of the three phases, with the wettability property being implied via the supplied input data.

The primary motivation for the development of the ODD3P method has been the difficulty in reproducing measured core flooding behavior using reservoir simulators with the traditional models of three-phase relative permeability ([Hustad, 2002](#)). The ODD3P method also ensures consistency (continuous functions) of the three-phase properties at the two-phase and single-phase boundaries with the respective two-phase properties. The saturation profiles observed in core flooding simulations have been found to be strongly dependent upon the capillary forces and the boundary conditions at the core inlet and outlet. In addition, the relative permeability has been found to depend not only upon fluid saturation but also upon saturation history. The requirements for the ODD3P method, therefore, include relative permeability and capillary pressure values which are saturation history dependent. Hence the properties assigned to the relative permeability and capillary pressure for a particular set of three-phase saturations should not be unique but instead depend upon the saturation history leading up to the current saturation condition.

Problems have also been found to occur when modeling three-phase systems if a first-contact miscible injection process causes an oil-water system to become a gas-water system due to the labelling of the hydrocarbon phase. This can result in discontinuities in relative permeabilities and capillary pressures. A further motivation for the ODD3P method, therefore, has been to ensure that relative permeability and capillary pressure are consistent and continuous for the flow processes which transition from any two-phase state to another two-phase state via a three-phase state.

In this model an important distinction is made between the end-point saturations and residual saturations ([Hustad, 2002](#)). The end-point saturation is defined as the lowest value of saturation for which the phase is

continuous and also corresponds to the saturation where the relative permeability is zero. In contrast, the residual saturation is defined as the saturation at which the phase becomes immobile. This condition may arise for different values of capillary pressure according to the pressure gradient of the phase or how high or low a capillary pressure is achieved by the flow conditions. The relative permeability need not be zero at the residual saturation, but for a phase not to flow, the phase pressure gradients must be zero. On the basis of these definitions, therefore, the residual saturation is greater than or equal to the end-point saturation.

End-point and residual saturation data will typically incorporate information about the processes which have been subjected to upscaling. For this model, the end-point saturation applied to a simulation model grid block may be related to the upscaled processes by including a representation of the sweep efficiency of the grid block using a Land-type formulation ([Land, 1968](#)).

Overview

The ODD3P method provides a three-phase relative permeability model which is based upon the saturation functions of the contributing three pairs of two-phase relative permeability saturation functions. This method extends the normalized saturation approach employed by the IKU method ([Hustad et al, 2002](#)), which is described above. In the ODD3P method, relative permeability and capillary pressure hysteresis effects are incorporated together with grid-block end-point scaling which can optionally include miscible effects.

Instead of the traditional specification of drainage and imbibition saturation functions and the explicit selection of the wetting phase, three sets of saturation functions are employed which are based upon the following ideas:

- Originally, in terms of geological time, formations which currently contain hydrocarbon fluids contained only water. Over a period of millions of years, hydrocarbon fluids gradually penetrated these formations until a point of equilibrium was reached between water, oil and gas. A set of primary saturation functions is used to represent this initial state of the reservoir and the initial production of hydrocarbon fluids. These also define the initial departure of the reservoir from its equilibrium conditions.
- When the formation is exploited commercially, an initial process of sometimes unassisted oil and gas production is usually followed by injection to maintain recovery. Typically the injection process will consist of alternating water and gas (WAG) injection which is one mechanism for improving oil recovery. At some point the saturation behavior will cease to follow the primary saturation functions and enter a hysteresis loop. Two sets of hysteresis saturation functions are used to model the hysteresis behavior experienced by the reservoir when fluid saturations increase and decrease as a function of the cyclical injection processes.

The hysteresis curves are specified in terms of increasing and decreasing saturation directions instead of the more usual notation of drainage and imbibition. For example, the relative permeability to water for the water-oil system is expressed both in terms of the increasing and decreasing water saturation. This corresponds to processes which result in the water saturation increasing and decreasing respectively with respect to the oil saturation. Conversely, the relative permeability to oil for the oil-water system is expressed in terms of increasing and decreasing oil saturation with respect to the water saturation.

The use of primary saturation functions which are distinct from the secondary and tertiary hysteresis saturation functions permits the modeling of the relative permeability behavior where, for example, the decreasing saturation characteristics of relative permeability are different for primary and tertiary processes. One complexity of this model is establishing the appropriate criteria for the relative permeabilities and capillary pressures to transition from the primary to the hysteresis saturation functions.

Three-phase formulation

The three-phase relative permeabilities and capillary pressures are based upon pairs of representative two-phase properties which are combined through a saturation weighting scheme. Each phase property is made dependent upon its own normalized saturation in order to ensure consistency (Hustad *et al*, 2002). Zero relative permeabilities are ensured at and below the end-point saturations. The gas-water data is represented by the oil-water data in order to avoid the difficulties associated with representing the hydrocarbon-water system when gas and oil are miscible. This is especially important for first-contact miscible floods in which phase labeling changes between gas and oil require that the functions of relative permeability and capillary pressure remain continuous. This approach also avoids the potential interpolation complexities associated with the use of separate saturation functions tabulated against different sets of saturation values.

Unlike the choice of the different Stone methods for the selection of three-phase relative permeability models and the separate selection of the different methods for modeling hysteresis effects, for example, Jargon or Killough, the ODD3P method incorporates implicitly both three-phase relative permeability and capillary pressure hysteresis. In addition, no explicit specification of the wetting phase is required. Instead, the wetting phase is implicit in the characteristics of the tables of relative permeability and capillary pressure expressed as functions of gas, oil and water saturation for the primary, secondary hysteresis and tertiary hysteresis processes.

Saturation process modes (directions)

The primary, secondary hysteresis and tertiary hysteresis processes refer to the direction of the phases' saturations. The directions of the gas, oil and water saturations are defined with respect to the change in their respective grid-block saturations. The relative permeability and capillary pressure data are based upon three pairs of two phase data such that, for each two-phase pair, $S_i = 1 - S_j$ and $S_k = 0$ where the subscripts i, j and k denote the three phases subject to the constraint that $i \neq j \neq k$. The saturation directions for each phase may be summarized as follows:

Primary

S_w decreasing

S_g increasing

$S_o = 1 - S_g$ or $S_o = 1 - S_w$

Secondary hysteresis

S_w increasing

S_g decreasing

$S_o = 1 - S_g$ increasing

$S_o = 1 - S_w$ decreasing

Tertiary hysteresis

S_w decreasing

S_g increasing

$S_o = 1 - S_g$ decreasing

$S_o = 1 - S_w$ increasing

For a given saturation process mode, the gas saturation direction will be in the opposing direction to that of the water saturation. For the increasing and decreasing hysteresis saturation directions, the oil saturation

direction will be in the opposing direction to the water or gas saturation against which it has been measured. For example, for the secondary hysteresis process, gas saturation is decreasing; therefore, if the oil saturation is measured or modeled against gas saturation, the measurements for increasing oil saturation will be used for the secondary hysteresis process.

Availability and limitations

The ODD3P model is currently only available for Eclipse 300 compositional models using an equation of state solution method.

ODD3P model definition

The ODD3P three-phase relative permeability and capillary pressure hysteresis model is selected by specifying the **ODD3P** keyword in the PROPS section. The use of the default three-phase relative permeability model and the other permeability models **STONE** (an alias for **STONE2**) and **STONE1**, **STONE2**, **IKU3P** and **ODD3P** are mutually exclusive.

ODD3P saturation functions

Saturation function keywords

The relative permeabilities and capillary pressures are expressed as a function of gas, oil and water saturation using the **SGF3**, **SOF3** and **SWF3** keywords respectively. In general, three sets of these tables will be required per saturation region corresponding to the primary, secondary hysteresis and tertiary hysteresis saturation processes.

In order to avoid discontinuities due to phase labelling changes in miscible models, the gas-water relative permeability data are replaced by the oil-water relative permeability data. Therefore, the relative permeability to water data for a water-oil system k_{rwo} (column 2 of the **SWF3** keyword) are used to represent the relative permeability to water for a gas-water system k_{rgw} (column 3 of the **SWF3** keyword). Similarly, the relative permeability to oil data for an oil-water system k_{row} (column 2 of the **SOF3** keyword) are used to represent the relative permeability to gas for a gas-water system k_{rgw} (column 3 of the **SGF3** keyword). It follows that the data in column 3 of the **SWF3** and **SGF3** keywords may be defaulted, subject to the caveat that sufficient information is specified in terms of initial values of 0.0 and final values of 1.0 in order to infer the residual and maximum saturations to be applied to the gas-water and water-gas systems. An example of the specification of these three keywords is shown below in abbreviated form.

SGF3				
---	Sg	kr _{go} (Sg)	kr _{gw} (Sg)	Pc _{og} (Sg)
	0.00000	0.000000E+00	0.000000E+00	0.000000E+00
	0.00010	0.000000E+00	0.000000E+00	0.000000E+00
	0.00034	0.141600E-13	1*	0.197385E-02
	0.00049	0.610900E-13	1*	0.276339E-02
	...			
	0.83000	0.502886E+00	1*	0.404639E-01
	0.91000	0.726646E+00	1*	0.542808E-01
	0.95000	0.865329E+00	0.100000E+01	0.800395E+00
	0.96000	0.900000E+00	0.100000E+01	0.986923E+00
/				

SOF3			
--	So	k _{row} (So)	k _{rog} (So)
	0.00000	0.000000E+00	0.000000E+00
	0.04000	0.140000E-02	0.000000E+00
	0.09000	0.315000E-02	0.383300E-06
	0.17000	0.595000E-02	0.455400E-04
	0.20000	0.700000E-02	0.333655E-03
	...		
	0.82000	0.670000E+00	0.360146E+00
	0.85000	0.700000E+00	0.433916E+00

/				
0.87000 0.700000E+00 0.483097E+00				
0.95000 0.700000E+00 0.765334E+00				
0.99767 0.700000E+00 0.987923E+00				
0.99922 0.700000E+00 0.995944E+00				
0.99951 0.700000E+00 0.997451E+00				
0.99966 0.700000E+00 0.998230E+00				
0.99990 0.700000E+00 0.100000E+01				
/				
SWF3				
--	Sw	krwo (Sw)	krwg (Sw)	Pcwo (Sw)
	0.05000	0.000000E+00	0.000000E+00	0.986923E+00
	0.15000	0.000000E+00	1*	0.986923E+00
	0.18000	0.150000E-02	1*	0.493462E-01
	...			
	0.80000	0.500000E+00	1*	0.102640E-01
	0.99990	0.999750E+00	0.100000E+01	0.888300E-02
	1.00000	0.100000E+01	0.100000E+01	0.888231E-02
/				

In practice it is envisaged that the relative permeability and capillary pressure data will be obtained either from laboratory measurements of cores or from network simulation models where grain size distributions and pore wettabilities will be representative of the reservoir being modeled. The data are obtained from three discrete sets of two phases measurements or simulations, that is, in the absence of the third phase. In particular it should be noted that, for the **SOE3** keyword, the relative permeability to oil for an oil-gas system should be defined in the absence of connate water.

Saturation function region numbers

The association between saturation tables and saturation region numbers follows the usual pattern. However, whereas in the traditional hysteresis models, the drainage and imbibition saturation tables are associated with saturation regions using the keywords **SATNUM** and **IMBNUM** respectively, for the ODD3P method, the keywords **PSTNUM**, **ISTNUM** and **DSTNUM** are used to define the tables associated with the primary, secondary hysteresis and tertiary hysteresis saturations respectively. Internally, the **SATNUM** and **IMBNUM** keywords are aliases of the **PSTNUM** and **ISTNUM** keywords respectively and may be used as alternatives. This may be useful when converting existing data sets to work with the ODD3P model. However, there is no equivalent internal alias for the **DSTNUM** keyword. It is recommended, therefore, that for the purposes of clarity the matching trio of **PSTNUM**, **ISTNUM** and **DSTNUM** keywords should be used wherever possible.

The saturation region keyword **PSTNUM** refers to the primary saturation tables, **ISTNUM** refers to the tables containing the secondary hysteresis saturation functions, and **DSTNUM** refers to the tables containing the tertiary hysteresis saturation functions. The selection of the appropriate combination of saturation functions for a given process direction is made internally according to the current saturations and the history of the saturation directions thus far in the simulation.

The **PSTNUM**, **ISTNUM** and **DSTNUM** keywords are optional but are required if hysteresis is to be modeled. These may be used to define the entire set of grid blocks or some subset via the usual set of array assignment and manipulation operators. For grid blocks for which one or more of these keywords have not been specified, the following defaulting process will be applied:

- **PSTNUM**: Primary saturation function region number defaults to 1.
- **ISTNUM**: Secondary hysteresis saturation function region number defaults to **PSTNUM**.
- **DSTNUM**: Tertiary hysteresis saturation function region number defaults to **ISTNUM**.

Although the ODD3P method is implicitly hysteretic, hysteresis may be switched off for a given grid block by defining the values of **ISTNUM** and **DSTNUM** to be equal. In addition, the modeling of separate primary, secondary and tertiary processes may be switched off by specifying the values of **ISTNUM** and **DSTNUM**

not only to be equal to each other but also equal to the value of PSTNUM. The only constraint upon the values of PSTNUM, ISTNUM and DSTNUM for a given grid block, therefore, is that if PSTNUM is equal to ISTNUM or DSTNUM, then ISTNUM and DSTNUM must also be equal. Failure to comply with this constraint will cause an error to be reported during the checking of the input data.

Saturation function data constraints

Because of the complexity of the ODD3P model, there is a number of specific input data constraints associated with the specification of the saturation function data for this model. These are summarized below.

- For a specified pair of hysteresis curves, the hysteresis value of the end-point saturation S_{irj} is taken as the lower value of the two S_{irj} values for the secondary and tertiary hysteresis curves, where S_{irj} is defined to be the maximum tabulated saturation value in S_i for which k_{rij} is zero, with S_i measured in the increasing sense. In cases where the secondary and tertiary hysteresis values of S_{irj} are found to be different, which is permissible, a warning is issued. This is to identify possible mismatches between the pairs of curves specified for secondary and tertiary hysteresis operations.
- Each pair of hysteresis curves is required to form a closed hysteresis loop in the region enclosed between their bounding residual saturations. This may be expressed as follows:
 - For the increasing and decreasing relative permeability functions $k_{rij}(S_i(inc))$ and $k_{rij}(S_i(dec))$, the values should be equal at S_{irj} and $1-S_{jri}$.
 - Similarly for the increasing and decreasing relative permeability functions $k_{rji}(S_j(inc))$ and $k_{rji}(S_j(dec))$, the values should be equal at S_{jri} and $1-S_{irj}$.
 - Corresponding constraints apply to the increasing and decreasing hysteresis oil-water and gas-oil capillary pressures.

If the above conditions are not satisfied, appropriate warning messages will be issued. Failure to comply with these conditions will lead to inconsistent or unpredictable results.

- In general, it is expected that the relative permeability functions will be monotonically increasing over the range of the closed loop hysteresis curves. Similarly, it is expected that the capillary pressures will vary monotonically over this range. Although these conditions are not mandatory warning messages will be issued if they are not met.
- The first saturation table entries for the secondary and tertiary hysteresis curves are checked for equal initial values of saturation and a warning is issued if this test is not satisfied. This is to identify possible mismatches between pairs of curves specified for secondary and tertiary hysteresis operations.
- For the primary saturation [SGF3](#) tables, the initial gas saturation value should be 0.0. Similarly for the primary [SWF3](#) tables, the final water saturation value should be 1.0. In most cases, for the primary tables, negative values of capillary pressure are not required, and a warning is issued if negative values are encountered.
- In some cases, the primary capillary pressure curve may intersect the secondary hysteresis or tertiary hysteresis capillary pressure curves; such intersections do not prevent correct operation of the model. For the 2014.1 version, warnings are no longer issued if any such conditions are encountered. However, these warnings may be reinstated by setting [item 307](#) of the `OPTIONS3` keyword to 1 in the `PROPS` or prior sections.
- If the secondary and tertiary hysteresis capillary pressure curves intersect within the hysteresis saturation extent, the hysteresis modeling will fail and this will be reported as an error condition during the input data checking.

- There are no constraints on the intersection of the primary, secondary hysteresis and tertiary hysteresis relative permeability curves.

End-point scaling

For the ODD3P method, when the **ENDSCALE** keyword has been specified in the **RUNSPEC** Section, end-point scaling is accomplished by applying scaling via grid-block end-point saturations using the **PROPS** Section keywords **PSORG**, **PSGRO**, **PSORW**, **PSWRO**, **PSGRW** and **PSWRG** for the primary saturation functions and the corresponding keywords **HSORG**, **HSGRO**, **HSORW**, **HSWRO**, **HSGRW** and **HSWRG** for the hysteresis saturation functions. Only one set of hysteresis end-point scaling keywords is required because the hysteresis saturation functions form closed loop curves. For a given grid block, the following constraints apply to the specification of these data values:

Residual Saturation	Primary Constraint	Hysteresis Constraint	Primary-Hysteresis Constraint
SORW		$HSORW > 0.0$	$PSORW \leq HSORW$
SWRO	$PSWRO > 0.0$	$HSWRO > 0.0$	$PSWRO = HSWRO$
SORG	$PSORG > 0.0$	$HSORG > 0.0$	$PSORG = HSORG$
SGRO	$PSGRO \leq 0.001$	$HSGRO > 0.0$	$PSGRO < HSGRO$
SGRW		$HSGRW > 0.0$	$PSGRW < HSGRW$
SWRG	$PSWRG > 0.0$	$HSWRG > 0.0$	$PSWRG = HSWRG$

Table 3.5: End-Point Scaling Constraints

If **ENDSCALE** has not been specified, the grid-block end-point values will be taken from those of the appropriate saturation table. Similarly if **ENDSCALE** has been specified but not all end-point values have been defined for each grid block using the above keywords, the unspecified values will default to those of the appropriate saturation table. The above constraint checking is performed after any defaults have been applied. Failure to comply with the above input data constraints will lead to warning messages.

Capillary pressure calculation control

The calculation of the capillary pressure is controlled by the **PROPS** section table keywords **PCODD3P**, **PCODD3PG** and **PCODD3PW** which should be specified for each saturation region. In practice, only those tables which are associated with the primary saturation region numbers are used; those associated with the secondary and tertiary hysteresis region numbers may be defaulted. The **PCODD3P** keyword consists of a single row of seven columns of data per saturation region. These are defined as follows:

1. Capillary pressure calculation type. This provides thirteen functional relationships for the calculation of capillary pressure which are numbered 0 to 12.
2. The constant δ_g in the gas saturation dependent capillary pressure weighting function.
3. The first exponent α_g in the gas saturation dependent capillary pressure weighting function.
4. The second exponent β_g in the gas saturation dependent capillary pressure weighting function.
5. The constant δ_w in the water saturation dependent capillary pressure weighting function.
6. The first exponent α_w in the water saturation dependent capillary pressure weighting function.
7. The second exponent β_w in the water saturation dependent capillary pressure weighting function.

The data specified in columns 2 to 7 are only required if the functional relationship type is between 1 and 8. For these types, the capillary pressure weighting functions $H(S_g)$ and $F(S_w)$ are calculated as shown below:

$$H(S_g) = \frac{\delta_g S_g^{\alpha_g}}{S_g^{\alpha_g} + (1 - S_g)^{\beta_g}} \quad \text{Eq. 3.29}$$

and

$$F(S_w) = \frac{\delta_w S_w^{\alpha_w}}{S_w^{\alpha_w} + (1 - S_w)^{\beta_w}} \quad \text{Eq. 3.30}$$

The *representative* capillary pressures, that is, the values obtained from table look-up using the normalized saturations are corrected according to the following equations:

$$P_{cow} = \hat{P}_{cow} + H \cdot R \quad \text{Eq. 3.31}$$

$$P_{cgo} = \hat{P}_{cgo} + F \cdot R \quad \text{Eq. 3.32}$$

$$R = \hat{P}_{cgw} - \hat{P}_{cgo} - \hat{P}_{cow} \quad \text{Eq. 3.33}$$

where the circumflex denotes the **representative** (IFT scaled) capillary pressures. The corrected capillary pressures may be made functions of one of two pairs of saturations as shown below:

Saturation Dependence	Oil-Water System	Gas-Oil System	Gas-Water System
1	$\hat{P}_{cow}(S_o)$	$\hat{P}_{cgo}(S_g)$	$\hat{P}_{cgw}(S_g)$
2	$\hat{P}_{cow}(S_w)$	$\hat{P}_{cgo}(S_o)$	$\hat{P}_{cgw}(S_w)$

Table 3.6: Saturation functions

The calculation control parameter defined above in column 1 of the [PCODD3P](#) keyword selects the functional relationship type to calculate the capillary pressures as shown below:

Type	Calculation of oil-water and gas-oil capillary pressures
0	$P_{cow} = 0.0, P_{cgo} = 0.0$
1&2	$P_{cow} = \hat{P}_{cow}(S_o) + H \cdot R, P_{cgo} = \hat{P}_{cgo}(S_g) + F \cdot R$
3&4	$P_{cow} = \hat{P}_{cow}(S_o) + H \cdot R, P_{cgo} = \hat{P}_{cgo}(S_o) + F \cdot R$
5&6	$P_{cow} = \hat{P}_{cow}(S_w) + H \cdot R, P_{cgo} = \hat{P}_{cgo}(S_g) + F \cdot R$
7&8	$P_{cow} = \hat{P}_{cow}(S_w) + H \cdot R, P_{cgo} = \hat{P}_{cgo}(S_o) + F \cdot R$
9	$P_{cow} = \hat{P}_{cow}(S_o), P_{cgo} = \hat{P}_{cgw}(S_w) - \hat{P}_{cow}(S_o)$
10	$P_{cow} = \frac{S_g}{S_g + S_o} \hat{P}_{cgw}(S_w) + \frac{S_o}{S_g + S_o} \hat{P}_{cow}(S_w), P_{cgo} = \frac{S_w}{S_o + S_w} \hat{P}_{cgw}(S_g) + \frac{S_o}{S_o + S_w} \hat{P}_{cgo}(S_g)$
11	$P_{cow} = \frac{S_g}{S_g + S_o} \hat{P}_{cgw}(S_w) + \frac{S_o}{S_g + S_o} \hat{P}_{cow}(S_w), P_{cgo} = \hat{P}_{cgo}(S_g)$

Type	Calculation of oil-water and gas-oil capillary pressures
12	$P_{cow} = \frac{S_g(S_o[\hat{P}_{cgw}(\tilde{S}_w) - \hat{P}_{cgo}(\tilde{S}_{wg})] + S_w[\hat{P}_{cgw}(\tilde{S}_{ww}) - \hat{P}_{cgw}(\tilde{S}_{wg})]) + S_o(S_o + S_w)\hat{P}_{cow}(\tilde{S}_{ww})}{S_o(S_g + S_o + S_w)}$ $P_{cgo} = \left[\frac{S_o}{(S_o + S_w)} + \frac{S_g S_w}{(S_o + S_w)(S_g + S_o + S_w)} \right] \cdot \hat{P}_{cgo}(\tilde{S}_g) +$ $\left[\frac{S_w}{(S_o + S_w)} + \frac{S_g^2 S_w}{S_o(S_o + S_w)(S_g + S_o + S_w)} \right] \cdot \hat{P}_{cgw}(\tilde{S}_g) -$ $\left[\frac{S_g S_w}{(S_g + S_o + S_w)} \right] \cdot \hat{P}_{cgw}(\tilde{S}_w) - \left[\frac{S_w}{(S_g + S_o + S_w)} \right] \cdot \hat{P}_{cow}(\tilde{S}_w)$

Table 3.7: Functional relationship types for the calculation of capillary pressures

For functional relationship types 1 to 8, R is calculated as shown below:

Type	Calculation of value R
1	$R = \hat{P}_{cgw}(S_g) - \hat{P}_{cgo}(S_g) - \hat{P}_{cow}(S_o)$
2	$R = \hat{P}_{cgw}(S_w) - \hat{P}_{cgo}(S_g) - \hat{P}_{cow}(S_o)$
3	$R = \hat{P}_{cgw}(S_g) - \hat{P}_{cgo}(S_o) - \hat{P}_{cow}(S_o)$
4	$R = \hat{P}_{cgw}(S_w) - \hat{P}_{cgo}(S_o) - \hat{P}_{cow}(S_o)$
5	$R = \hat{P}_{cgw}(S_g) - \hat{P}_{cgo}(S_g) - \hat{P}_{cow}(S_w)$
6	$R = \hat{P}_{cgw}(S_w) - \hat{P}_{cgo}(S_g) - \hat{P}_{cow}(S_w)$
7	$R = \hat{P}_{cgw}(S_g) - \hat{P}_{cgo}(S_o) - \hat{P}_{cow}(S_w)$
8	$R = \hat{P}_{cgw}(S_w) - \hat{P}_{cgo}(S_o) - \hat{P}_{cow}(S_w)$

Table 3.8: Functional relationship types and the calculation of R

The functional relationship types 9, 10 and 11 correspond to cases where R is equal to 0.0.

Capillary pressure functional dependence calculation control parameter options 9, 10 and 11 correspond to cases where R is equal to 0.0. For options 10, 11 and 12 the saturation weightings are obtained from the grid block saturations S_i . Option 12 incorporates saturation weighting based upon all three grid block saturations (S-Weighted Model). This model has been developed by StatoilHydro ASA ([StatoilHydro ASA, 2010](#)) and results obtained from this model are described in [Hustad and Browning \(2009\)](#) and compared with alternative models above.

For ultimate flexibility the capillary pressure weighting functions $H(S_g)$ and $F(S_w)$ may be defined explicitly using the keywords `PCODD3PG` and `PCODD3PW` which tabulate these functions against gas and water saturation respectively. Either of these keywords may optionally be specified. If the `PCODD3PG` keyword is specified, the values defined in columns 2, 3 and 4 of the `PCODD3P` keyword will be ignored. Similarly, if the `PCODD3PW` keyword is specified, the values defined in columns 5, 6 and 7 of the `PCODD3P` keyword will be ignored.

End-point scaling calculation control

The end-point scaling calculation is controlled by the PROPS section table keyword EPSODD3P. This keyword provides control over the application of a sweep efficiency component to the process-dependent end-point saturations. The following Land formulation (*Hustad, 2002*) is used for this purpose and calculates a modified grid-block end-point saturation S_{irj} according to the expression:

$$S_{irj} = \frac{S_i^{max}}{1 + \frac{S_i^{max}}{1 + S_{irj}^p} - \frac{S_i^{max}}{1 - S_{jri}^p}} \quad \text{Eq. 3.34}$$

where

S_i^{max} is the maximum grid block saturation and

S_{irj}^p and S_{iri}^p are the process-dependent grid-block end-point saturations and

$i \neq j$, and S_i^{max} are subject to the constraint $S_i^{max} \leq 1 - S_{jri}^p$

The Land formulation may be applied individually to any or all of the oil, gas and water saturations.

Saturation direction indicators

During the calculation process, six saturation process direction indicators are employed to monitor the switching of the process directions from primary to hysteresis and between increasing and decreasing saturations. These six saturation indicators are specified per grid block and may possess one of four values:

1. Using primary saturation functions with switching to hysteresis not permitted.
2. Using primary saturation functions with switching to hysteresis permitted.
3. Using decreasing saturations.
4. Using increasing saturations.

Internally, these saturation direction indicators are assigned an initial value of 1. When, during the course of the simulation, the grid-block saturations come within the hysteresis extent as defined by the hysteresis grid-block end-points, the indicators are assigned a value of 2. Subsequently when a saturation process direction change occurs these indicators are assigned a value of 3 or 4 corresponding to whether the saturation direction is increasing or decreasing respectively. Thereafter, the saturation direction indicators will switch between the values of 3 and 4 in response to changes in the phase's saturation direction.

In some special cases, it may be helpful to initialize these saturation direction indicators to values other than 1, for example, when the results of some experiment or some other simulation provide details for the initial saturation states. Under these circumstances, it is possible to initialize these saturation direction pointers to values in the range of 1 to 4 on a per grid-block basis using the REGION Section keywords [SDROW](#), [SDRWO](#), [SDROG](#), [SDRGO](#), [SDRGW](#) and [SDRWG](#). When only partial array specifications are provided for these arrays, the unspecified portions of the arrays will be initialized to 1. Considerable care should be exercised when using this facility. In general, the initial is at ion of these arrays to non-default values will not be necessary and these arrays need not be specified in the input data.

Reporting facilities

In view of the complexity of the ODD3P method, a considerable number of array reporting facilities are provided both as solution arrays and for output to the Summary file on a per grid-block basis. The arrays

which are provided for output are listed below with the characters i and j signifying gas G , oil, O and water W , noting that $i \neq j$. The following grid-block arrays are considered to be useful for the general monitoring of the calculation process.

- KRijR: Representative relative permeabilities.
- SDRij: Saturation process direction indicators.
- SiNRM: Normalized saturations.
- SiRjB: Residual saturations.

When specifying the required outputs in the input data file, the characters i and j should be replaced with the specific phase pairs which are required. For output on a per grid-block basis as Summary vectors, the array names above should be pre-appended with a B and defined in the SUMMARY section for the selected grid-block indices.

Mixed water relative permeability modeling

An alternative model, that takes into account the dissolution of gases, may be used to calculate the water relative permeability during a CO2SOL, H2SSOL or GASSOL simulation. This optional mixing model is activated using the keyword MIXWPERM.

In this case, the standard water relative permeability is corrected to take into account the hydrocarbon moles dissolved in the water phase, using the expression:

$$\widetilde{K}_w = \frac{m_w K_w + \sum_i (m_i)_{Aq} [\alpha_i(P, T) K_o + \beta_i(P, T) K_g]}{m_w + \sum_k (m_k)_{Aq}}$$

in which

$\widetilde{K}_w$ denotes the new water relative permeability obtained following the mixing calculation,

K_w denotes the standard water relative permeability prior to mixing,

K_o and K_g denote, respectively, the oil and gas phase permeabilities,

m_w represents the water component molar density,

$(m_k)_{Aq}$ and $(m_i)_{Aq}$ denote, respectively, the aqueous molar density of components k and i defined by CNAMEs, with indices k and i each spanning all of the soluble hydrocarbon components.

Finally, denoting pressure and temperature (the latter in degrees Celsius) using P and T respectively, the oil and gas activity functions for component i are represented, respectively, by:

$$\alpha_i(P, T) = a_{i1}(P) + a_{i2}(P)T + a_{i3}(P)T^2$$

$$\beta_i(P, T) = b_{i1}(P) + b_{i2}(P)T + b_{i3}(P)T^2$$

The coefficients a_{i1} , a_{i2} and a_{i3} are interpolated, respectively, from columns 2, 3 and 4 of the MIXACTO keyword input table for component i .

Similarly, coefficients b_{i1} , b_{i2} and b_{i3} are interpolated, respectively, from columns 2, 3 and 4 of the MIXACTG keyword input table for component i .

	Eclipse 100
x	Eclipse 300

Note that the sum $\alpha_i + \beta_i$ must not exceed unity for the given pressure and temperature, for any component i . This ensures that the new water relative permeability following mixing is bounded by unity. If $\alpha_i + \beta_i$ exceeds unity, the simulation will stop and the activity function coefficients must be reduced accordingly.

Table end-points

In the equilibration calculation (see keyword [EQUIL](#)) the water and gas saturation values above and below the transition zones are determined from the maximum and minimum saturation values in the water and gas saturation tables. The oil saturation values are thus defined consequentially from these values. The initial saturations in each zone are shown in the following figure.

GAS ZONE	$S_g = S_{g \max}$ $S_w = S_{wco}$ $S_o = 1 - S_{g \max} - S_{wco}$
GAS - OIL TRANSITION ZONE	
OIL ZONE	$S_g = S_{gco}$ $S_w = S_{wco}$ $S_o = 1 - S_{gco} - S_{wco}$
OIL-WATER TRANSITION ZONE	
WATER ZONE	$S_g = S_{gco}$ $S_w = S_{w \max}$ $S_o = 1 - S_{gco} - S_{w \max}$

Figure 3.6. Initial saturations for each zone

In the case where the two-dimensional alternatives of the saturation functions are used ([SWF32D](#), [SGF32D](#), [PCW32D](#), [PCG32D](#)) the equilibration process is done as follows, based on the depth of the cell:

GAS	$S_w = S_{wco}$ $S_o = 0$ or from PCG32D (TZ) $S_g = 1 - S_o - S_{wco}$
OIL	$S_g = 0$ $S_o = 1 - S_{wco}$ or from PCW32D (TZ) $S_w = 1 - S_o$
WATER	$S_g = 0$ $S_o = 0$ or from PCW32D (TZ) $S_w = 1 - S_o$

Figure 3.7. Equilibration process based on the depth of the cell

Note: This equilibration process does not handle the situation where the water zone overlaps with the gas zone.

Consistency requirements

The oil, water and gas saturation tables, for each particular saturation table number region, must obey certain consistency requirements. These are detailed below.

- $S_{g \max}$ must not exceed $1 - S_{wco}$

If this condition is violated, the gas saturation in the gas cap will be reset to $1 - S_{wco}$, to prevent a negative oil saturation.

Normally, if there is no oil in the gas cap, $S_{g \max} = 1 - S_{wco}$.

- S_{gco} must not exceed $1 - S_{w \max}$

If this condition is violated, the gas saturation in the water zone will be re-set to $1 - S_{w \max}$, to prevent a negative oil saturation.

Normally, there is no initial free gas below the gas cap and the water zone is fully saturated with water, thus $S_{gco} = 0$ and $1 - S_{w \max} = 0$.

- $S_{o \max}$ must equal $1 - S_{wco} - S_{gco}$
- $k_{row}(S_{o \max})$ must equal $k_{rog}(S_{o \max})$
- $k_{rw}(S_w = 0) = k_{rg}(S_g = 0) = k_{row}(S_o = 0) = k_{rog}(S_o = 0) = 0$.

Otherwise, phases can be mobile even at zero saturation and there is nothing to stop saturations going negative.

In **oil-wet system** modeling, in order to apply correctly the Killough wetting phase model on the oil phase (option 7, Item 2 of [EHYSTR](#) keyword), the consistency requirements are slightly modified:

- $S_{g \max}$ must equal $1 - S_{oco}$.
- $S_{o \max}$ is normally equal to 1.0.
- $S_{w \max}$ must equal $1 - S_{oco}$.

Mobile fluid requirements

In order to ensure that at least one fluid remains mobile in the three-phase ternary diagram, Eclipse 100 also checks for the following:

- $S_{owcr} + S_{wcr} < 1$
- $S_{ogcr} + S_{gcr} + S_{wco} < 1$

Eclipse issues a warning if these conditions are not satisfied. When [SCALECRS](#) is used, this would lead to an error because in an oil-water run, for example, K_{rw} would be scaled between S_{wcr} and $1 - S_{owcr}$ that is a null range if the first condition is not satisfied.

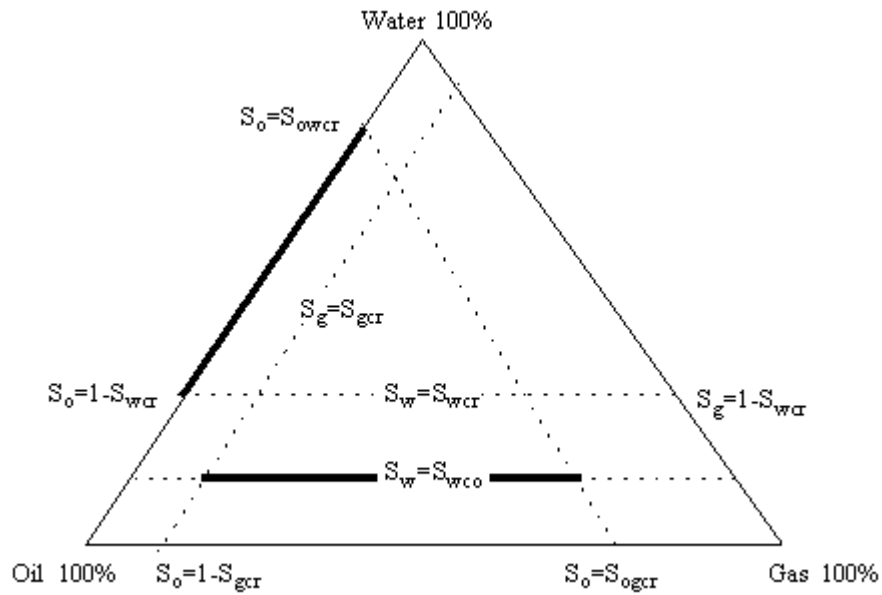


Figure 3.8. Ternary diagram showing mobile fluid end-points

Near critical oil and gas relative permeabilities

Eclipse 300

In compositional mode it is possible for a cell to make a direct transition from an oil to a gas, without passing through a series of intermediate saturations. This could result in a jump in the relative permeability. To prevent this, the oil and gas relative permeabilities are interpolated, as described below. This interpolation is not used or required when **ISGAS** is specified. It will be disabled if **NOMIX** is used.

Consider a system with mobile hydrocarbon, initially regarded as an oil, but under gas injection. The effect of gas injection is to move the phase envelope to the left, and the system critical temperature can pass through the reservoir temperature.

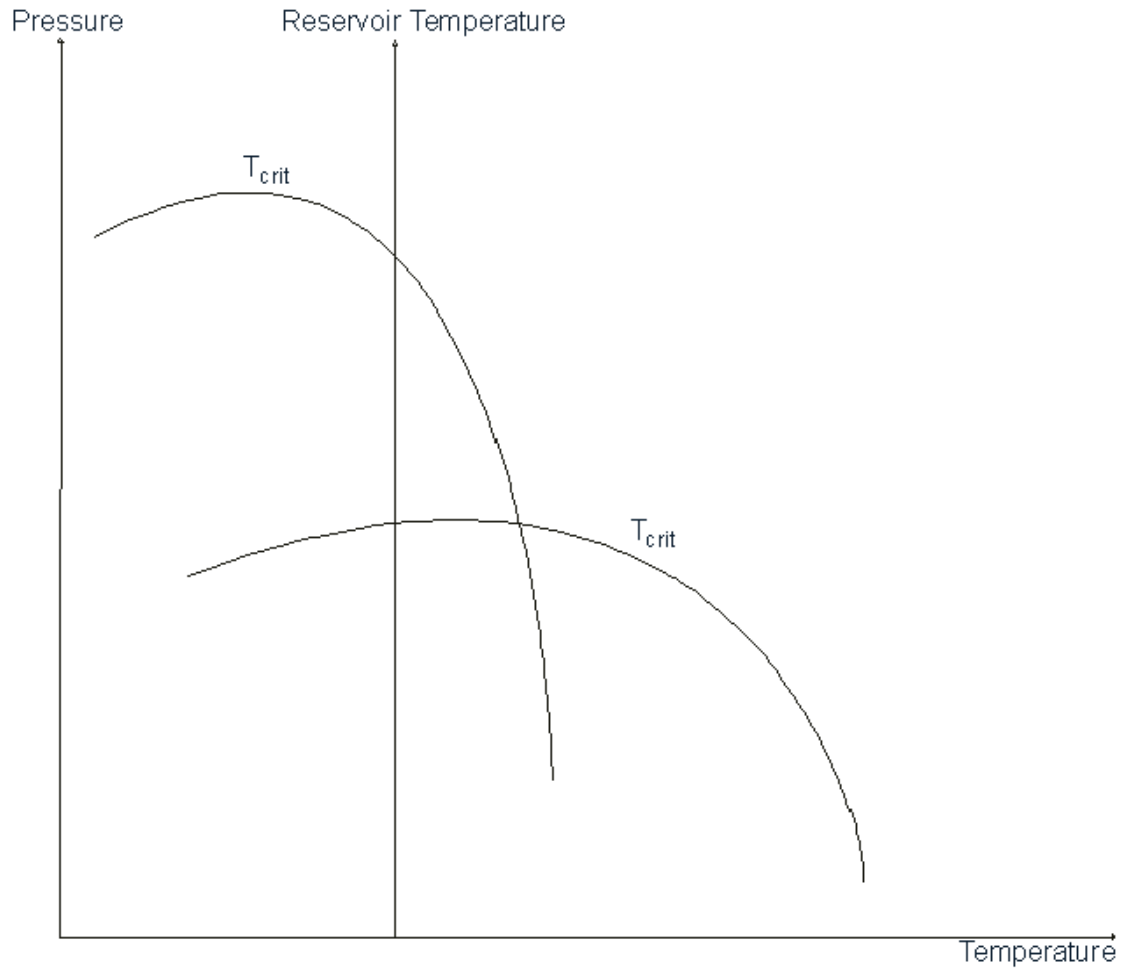


Figure 3.9. The effect of gas injection

There must be continuity between the hydrocarbon relative permeability, K_{ro} when the system is an oil and K_{rg} when the system is a gas, for arbitrary S_w .

To do this there is a water-hydrocarbon relative permeability for single phase hydrocarbon states. Oil-gas and gas-oil relative permeabilities are only relevant if both oil and gas are present. This water-hydrocarbon K_{rh} must tend to the user-entered K_{ro} when the composition is oil-like, and to K_{rg} when it is gas-like.

To set up a variable to interpolate between these two values, we define the pseudo-critical temperature T^{crit} which is given by the Li correlation as:

$$T^{\text{crit}} = \frac{\sum_j T_{cj} \cdot V_{cj} \cdot z_j}{\sum_j V_{cj} z_j} \quad \text{Eq. 3.35}$$

Then define f as:

$$f = \frac{T^{\text{crit}}}{T^{\text{res}}} \quad \text{Eq. 3.36}$$

f will clearly be 1 if the reservoir temperature T^{res} is the critical temperature, but becomes greater than 1 for an oil ($T^{\text{crit}} > T^{\text{res}}$) and less than 1 for a gas. If we choose values at which the system is to be regarded completely as an oil or a gas, such as $f_o = 1.25$ and $f_g = 0.75$ (these values may be user-defined by the `LILIM` keyword), then an interpolating function may be defined as

$$E = \frac{f - f_g}{f_o - f_g} \quad \text{Eq. 3.37}$$

in the region $f_g < f < f_o$, and 0 or 1 outside this range. This goes from 0 when the system is gas-like (f below f_g) to 1 when the system is oil-like (f above f_o), with a linear interpolation in between. The hydrocarbon-water relative permeability may then be defined as

$$K_{rhw} = EK_{row}^u + (1-E)K_{rgw}^u \quad \text{Eq. 3.38}$$

where K_{row}^u and K_{rgw}^u are the user-input oil and gas relative permeabilities in water, with their endpoints scaled, to force the endpoints of K_{rhw} to vary between those of K_{row}^u and K_{rgw}^u as E varies.

This is used for both oil and gas phases. A cell will make a transition from oil-filled to gas-filled when T^{crit} passes T^{res} , but the change from K_{row} to K_{rgw} will be continuous.

The oil-in-gas and gas-in-oil relative permeabilities are only used in the case when both oil and gas are present. To obtain the full oil and gas relative permeabilities in the case in which oil, gas and water are present either the default or the Stone method is used for oil, and the default method for gas. In the case in which the default method is used for oil, this yields:

$$K_{ro} = gK_{rhw}(S_w) + (1-g)K_{rog}(S_o) \quad \text{Eq. 3.39}$$

$$g = \frac{S_w - S_{wc}}{S_w - S_{wc} + S_g} \quad \text{Eq. 3.40}$$

$$K_{rg} = hK_{rhw}(S_w) + (1-h)K_{rgo}(S_g) \quad \text{Eq. 3.41}$$

$$h = \frac{S_w - S_{wc}}{S_w - S_{wc} + S_o} \quad \text{Eq. 3.42}$$

At connate water $S_w = S_{wc}$, the interpolation parameters defined by equations 3.40 and 3.41 are zero, $g = h = 0$, and the oil and gas relative permeabilities become

$$K_{ro} = K_{rog}(S_o) \quad \text{Eq. 3.43}$$

$$K_{rg} = K_{rgo}(S_g) \quad \text{Eq. 3.44}$$

Therefore, to ensure there is no discontinuity when oil becomes gas, or vice versa, it is necessary for the oil and gas relative permeabilities to be the same at the connate water saturation. This can be accomplished using similar interpolation to that used in equation 3.38:

$$K_{rog} = EK_{rog}^u + (1-E)K_{rgo}^u \quad \text{Eq. 3.45}$$

$$K_{rgo} = EK_{rog}^u + (1-E)K_{rgo}^u \quad \text{Eq. 3.46}$$

where K_{rog}^u and K_{rgo}^u are the user-input oil and gas relative permeabilities.

This second scaling of the relative permeabilities at the connate water is controlled by the `FORMOPTS` keyword options `HCSCAL` and `NOHCSCAL`.

These equations can be simplified for the gas relative permeability, since a single user-defined table is defined for both gas-in-oil and gas-in-water $K_{rgo}^u = K_{rgw}^u = K_{rg}^u$. Therefore equations 3.41, 3.38 and 3.46 can be written as

$$K_{rg} = E(hK_{row}^u + (1-h)K_{rog}^u) + (1-E)K_{rg}^u \quad \text{Eq. 3.47}$$

Saturation table scaling

x	Eclipse 100
x	Eclipse 300

The Eclipse saturation table end-point scaling option provides a mechanism for redefining values for the connate, critical and maximum saturations in the saturation tables describing the flow of the reservoir fluids. The scaling facility is useful for modeling reservoirs which contain an initial depth variation of either the connate or critical saturations for one or more of the phases present. It has applications in the use of pseudo functions and in general cases where the saturation function data for the reservoir fluids depend on a normalized saturation variable.

The end-point scaling option is enabled by the [ENDSCALE](#) keyword in the RUNSPEC section. Rescaling is applied separately to the relative permeability and capillary pressure curves. In general, rescaling consists of a **two-point** transformation, where two saturation values in the tables are moved to new positions. There is also an option to apply a **three-point** rescaling to the relative permeability curves only.

In a three-phase model, there are usually eight, and in some cases ten, saturation table end-points which may be identified:

Keyword	Description
SWL	The connate water saturation. This is the smallest water saturation entry in a water saturation table.
SWCR	The critical water saturation. This is the highest water saturation for which the water is immobile.
SWU	The maximum water saturation. This is the largest water saturation entry in a water saturation table.
SGL	The connate gas saturation. In Eclipse 100 this is the smallest gas saturation entry in a gas saturation table. In Eclipse 300 this is set to zero by default irrespective of the values specified in the saturation functions; if item 103 of the OPTIONS3 keyword is set to a non-zero value, non-zero connate gas saturations are permitted.
SGCR	The critical gas saturation. This is the highest gas saturation for which the gas is immobile.
SGU	The maximum gas saturation. This is the largest gas saturation entry in a gas saturation table.
SOWCR	The critical oil-in-water saturation. This is the highest oil saturation for which the oil is immobile in an oil-water system.
SOGCR	The critical oil-in-gas saturation. This is the highest oil saturation for which the oil is immobile in an oil-gas-connate water system.
SGWCR	The scaled critical gas-in-water saturation. This is the highest gas saturation for which the gas is immobile in a gas-water system.
SWGCR	The scaled critical water-in-gas saturation. This is the highest water saturation for which the water is immobile in a water-gas system.

Table 3.9: Saturation table end points

Eclipse 300

Note: The [SGWCR](#) and the [SWGCR](#) saturation table end-points are only available in Eclipse 300. They are only used by the [IKU3P](#) three-phase relative permeability model and by the [BAKER1](#) and the [BAKER2](#) three-phase relative permeability models where the Baker gas model ([SGWCR](#)) or water model ([SWGCR](#)) has been specified.

The saturation end-points taken from the saturation tables in the input data file are referred to as the unscaled saturation end points.

The end-point scaling option enables new values to be defined for any of the above saturation end points for each grid cell, subject to maintaining a consistent set of saturation tables within each grid cell. The appropriate subset of the above set of end-points is applicable in two phase runs. The new values defined using the end-point scaling option are referred to as the scaled end-points. When relative permeability and capillary pressure values need to be computed at a particular saturation, a linear transformation is used to determine the equivalent saturation to be used for look-up in the unscaled table.

For example, in a grid cell whose water saturation is SW , scaled connate water saturation is $SWCO$, scaled maximum water saturation is SWU , which belongs to a saturation region whose unscaled connate water is S_{wco} and whose unscaled maximum water saturation is S_{wmax} , the water capillary pressure P_{cow} is evaluated in the saturation table at the saturation, S_w' , where

$$S_w' = S_{wco} + \frac{(SW - SWCO)(S_{wmax} - S_{wco})}{SWU - SWCO}$$

so that P_{cow} is evaluated by lookup in the input capillary pressure table using

$$P_{cow}(SW) = P_{cow}(S_w')(\text{table})$$

In addition it is possible to scale the relative permeability and capillary pressure values using the keywords [KRW](#) and [PCW](#) for example.

If the hysteresis option is active, the above set of keywords is used to scale the drainage relative permeability curves. The imbibition curves can be scaled by using an analogous family of keywords:

- [ISWL](#), [ISWCR](#), [ISWU](#)
- [ISGL](#), [ISGCR](#), [ISGU](#)
- [ISOWCR](#), [ISOGCR](#)

Note:

If one or more end-point scaling arrays are not specified, the values are inferred from their associated table values. For [SGU](#) this can lead to differences between Eclipse 100 and Eclipse 300 in cases where the maximum gas saturation table entry is less than $(1 - S_{wco} - S_{ogco})$. This is because Eclipse 100 uses the maximum gas saturation table entry value whereas Eclipse 300 calculates the maximum gas saturation according to $(1 - S_{wco} - S_{ogco})$. The Eclipse 100 behaviour, however, can be replicated in Eclipse 300 by setting [item 357](#) of the `OPTIONS3` keyword to 1.

Scaling of capillary pressure functions

The end points of the oil-water capillary pressure functions are taken to be the connate water and maximum water saturations. The values of P_{cow} at the saturation nodes [SWL](#) and [SWU](#) are preserved during the scaling process. Similarly, the gas-oil capillary pressure end points are [SGL](#) and [SGU](#).

It is possible to modify the water (or gas) connate saturation for the capillary pressure scaling without modifying the connate saturation used for the oil relative permeability scaling.

The keywords [SWLPC](#) and [SGLPC](#) are used to input the connate saturation in this case. If both of the [SWL](#) and [SWLPC](#) keywords are present, the capillary pressure scaling will use [SWLPC](#) and the relative permeability scaling will use [SWL](#).

Eclipse 100

Scaling of the capillary pressure values (vertical scaling)

It is possible to scale the maximum capillary pressure value on a grid block by grid block basis.

The maximum values are specified using the [PCW](#) and [PCG](#) keywords for the oil-water and gas-oil capillary pressures respectively.

In the water-oil case:

$$P_c = P_{ct} \frac{PCW}{P_{cm}} \quad \text{Eq. 3.48}$$

where

P_{ct} capillary pressure from the table

P_{cm} maximum P_c in the table - at $S_w = S_{wco}$, connate water (this is the entry capillary pressure in the table)

PCW maximum P_c from the [PCW](#) data.

Scaling of relative permeability functions

There are two options available for the scaling of relative permeabilities. In the default case, the scaling process preserves relative permeabilities at two saturation nodes. The following end points are assumed for each phase relative permeability:

K_{rw} SWCR and SWU

K_{rg} SGCR and SGU

K_{row} SOWCR and (1.0-SWL-SGL)

K_{rog} SOGCR and (1.0-SWL-SGL)

An alternative scaling of relative permeabilities can be invoked by using the [SCALECRS](#) keyword in the PROPS section. The second form of scaling preserves the relative permeabilities at three saturation nodes. In three-phase, oil-water or oil-gas runs, or runs which use the miscible flood option, the following end points are used:

K_{rw} SWCR, (1.0-SOWCR-SGL) and SWU

K_{rg} SGCR, (1.0-SOGCR-SWL) and SGU

K_{row} SOWCR, (1.0-SOWCR -SGL) and (1.0-SWL-SGL)

K_{rog} SOGCR, (1.0-SGCR -SWL) and (1.0-SWL-SGL)

In gas-water runs, the following end points are used if the second form of relative permeability scaling is selected:

K_{rw} SWCR, (1.0-SGCR) and SWU

K_{rg} SGCR, (1.0-SWCR) and SGU

The second method for relative permeability scaling can be interpreted in two phase runs as preserving the values of relative permeabilities at both ends of the two-phase mobile region. Convergence problems can arise in certain cases when using this form of scaling if the intermediate end point begins to approach the maximum saturation for one of the phases. For example, in an oil-water problem in which the residual oil saturation to water tends to zero at a certain depth, the saturation (1.0-SOWCR) begins to approach SWU and this can in turn lead to a discontinuity in the water relative permeability as $S_w \rightarrow S_{w \max}$.

Eclipse 100

You may limit this sharpness by using the [SCALELIM](#) keyword which sets an upper limit on the value of (1.0 - SOWCR) to be used in scaling the water relative permeability function. The SCALELIM keyword is optional and at present can only be applied to water relative permeabilities.

For example, we consider the scaling of the water relative permeability function $K_{rw}(SW)$ at a water saturation SW where the scaled and unscaled critical water saturations are $SWCR$ and S_{wcr} , scaled and unscaled oil-in-water critical saturations are $SOWCR$, and S_{owcr} , scaled and unscaled connate gas saturations are SGL and S_{gco} , and scaled and unscaled maximum water saturations are SWU and S_{wmax} .

With the two point scaling method, the water relative permeability $K_{rw}(SW)$ is then evaluated in the input saturation table at the new saturation S_w' , where

$$S_w' = S_{wcr} + \frac{(SW - SWCR)(S_{wmax} - S_{wcr})}{SWU - SWCR} \quad \text{Eq. 3.49}$$

so that K_{rw} is evaluated by lookup in the input table using

$K_{rw}(SW) = K_{rw}(S_w')(table)$ for $SWCR \leq SW \leq SWU$. For $SW \leq SWCR$ then $K_{rw}(SW) = 0$ and for $SW \geq SWU$ then $K_{rw}(SW) = K_{rwmax}(table)$.

With the three point scaling method, if SR is the displacing critical saturation of the associated phase saturation (SR),

$SR = 1 - SOWCR - SGL$ in water/oil or gas/oil/water runs

$SR = 1 - SGCR$ in gas/water runs,

and S_r is the value of the displacing critical saturation in the table

$S_r = 1 - S_{owcr} - S_{gco}$ in water/oil or gas/oil/water runs

$S_r = 1 - S_{gcr}$ in gas/water runs.

The two cases are:

$SWCR \leq SW \leq SR$

$$S_w' = S_{wcr} + \frac{(SW - SWCR)(S_r - S_{wcr})}{SR - SWCR} \quad \text{Eq. 3.50}$$

$SR \leq SW \leq SWU$

$$S_w' = S_r + \frac{(SW - SR)(S_{wmax} - S_r)}{SWU - SR} \quad \text{Eq. 3.51}$$

and also for

$$SW \leq SWCR \quad \text{Eq. 3.52}$$

then

$$K_{rw}(SW) = 0 \quad \text{Eq. 3.53}$$

, and for

$$SW \geq SWU \quad \text{Eq. 3.54}$$

then

$$K_{rw}(SW) = K_{rw\max}(\text{table}) \quad \text{Eq. 3.55}$$

The two-point and three-point scaling equations are similar for K_{rg} , K_{row} and K_{rog} with alternate definitions of the critical and displacing critical saturations.

Scaling of the relative permeability values (vertical scaling)

It is possible to scale the relative permeability at the maximum phase saturation and at the critical/residual saturation of the associated phase.

The scaled relative permeabilities can be specified on a block by block basis, or alternatively the depth variation can be specified using the `ENKRVD` and `ENDNUM` keywords. The `KRW`, `KRG`, `KRO` keywords and their derivatives are used to set the relative permeabilities at the maximum phase saturation, while the `KRWR`, `KRGR`, `KROW`, `KROG` keywords and their derivatives are used to set the relative permeabilities at the critical/residual saturation of the associated phase.

The resultant scaling of the **water** relative permeability is shown below. **Gas** and **oil** relative permeabilities are scaled in an **equivalent** manner.

If the `KRWR` keyword has not been used, the `KRW` keyword has the effect of scaling the relative permeability value calculated from the appropriate saturation table after the scaled saturation end points have been accounted for. Hence:

$$K_{rw} = K_{rw}(\text{table}) \left(\frac{KRW(\text{grid block})}{K_{rw\max}(\text{table})} \right) \quad \text{Eq. 3.56}$$

The $K_{rw\max}(\text{table})$ is taken to be the value at either the maximum saturation of the saturation table or at `SWU` if this has been specified.

If the `KRWR` keyword has been used, then the scaling will honor the K_r at the critical saturation (SR) of the associated phase.

$SR = 1 - SOWCR - SGL$ in water/oil or gas/oil/water runs

$SR = 1 - SGCR$ in gas/water runs

Hence the two cases are:

$$SWCR \leq SW \leq SR$$

$$K_{rw}(S) = K_{rw}(S') \frac{KRWR(\text{grid block})}{K_{rw}(S_r)(\text{table})} \quad \text{Eq. 3.57}$$

$$SR \leq SW \leq SWU$$

$$K_{rw}(S) = KRWR + \frac{(K_{rw}(S')(\text{table}) - K_{rw}(S_r)(\text{table}))}{(K_{rw\max}(\text{table}) - K_{rw}(S_r)(\text{table}))} (KRW - KRWR) \quad \text{Eq. 3.58}$$

If the value of $K_{rw\max} = K_{rw}(SR)$, a linear function is assumed between `KRWR` and `KRW`.

The two-point scaling method is illustrated in the following figure:

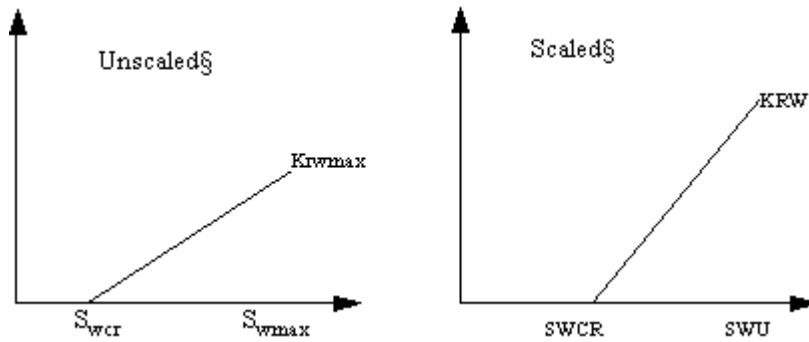


Figure 3.10. Two-point scaling

and the three-point scaling method is illustrated in the following figure:

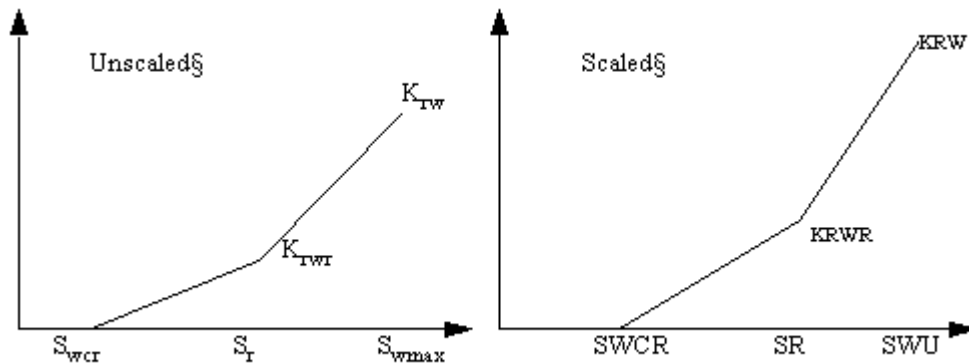


Figure 3.11. Three-point scaling

The [KRWR](#), [KRGR](#), [KRORW](#), [KRORG](#) keywords and their directional and hysteretic specializations can be used to set the relative permeabilities at the critical/residual saturation for both two-point scaling ([SCALECRS](#) keyword item 1 set to NO, which is the default if this keyword is not specified) and for the alternative three-point scaling ([SCALECRS](#) keyword item 1 set to YES). However, the impact of these keywords upon the resultant scaling will typically be different for the two-point and three-point scaling methods.

For the two-point scaling of water relative permeability, the value specified by [KRWR](#) will be preserved at the table saturation $S_w = 1 - S_{sowcr} - S_{gco}$ but noting that this value will be modified by the two-point horizontal scaling applied to S_w if $S_{wcr} \neq SWCR$ and/or if $S_{wmax} \neq SWU$ because of the resultant two-point transformation in saturation space.

In contrast for the three-point scaling, the value specified by [KRWR](#) will be preserved at the transformed saturation $SW = 1 - SOWCR - SGL$. Corresponding scaling differences between two and three-point scaling apply also to [KRGR](#), [KRORW](#) and [KRORG](#).

Considerable care should be exercised when using the critical/residual vertical scaling end-points. For example, for two-point scaling, if $S_{wmax} = 1 - S_{sowcr} - S_{gco}$, it will not be possible to preserve both the values specified by [KRWR](#) and [KRW](#) at the same point in saturation space. Similarly, for three-point scaling,

if $SWU = 1 - SOWCR - SGU$ it will not be possible to preserve both values specified by KRWR and KRW at the same point in saturation space.

In such cases, in general, the value specified by KRW will be selected; however, due to small differences in notionally equal floating point values, the value specified by KRWR may be selected instead. It is recommended that the EPSDEBUG or EPSDBGS keyword should be used to view the tables following the application of end-point scaling.

Eclipse 100

For the two-point scaling method (SCALECRS keyword item 1 set to NO, which is the default if this keyword is not specified), if the unscaled critical/residual and maximum relative permeabilities are equal, the scaled critical/residual relative permeability will be set equal to the scaled maximum relative permeability irrespective of whether this value has been specified. For example, if in the input unscaled water saturation functions, $k_{rwr} = k_{rwmax}$, where k_{rwr} is the value of $k_{rw}(S_w)$ at $S_w = 1 - S_{sowcr} - S_{gco}$ and k_{rwmax} is the value of $k_{rw}(S_w)$ at $S_w = S_{wmax}$, the value of KRWR will be assumed to be equal to that of KRW.

Miscellaneous points

The end-point scaling option is activated using the keyword ENDSCALE in the RUNSPEC section. The directional scaling options are activated by specifying the appropriate arguments of ENDSCALE.

The end point saturations may be supplied in one of two ways:

- by cell, using the keywords SWL, SWCR, SWU, SGL, SGCR, SGU, SOWCR, SOGCR; or,
- by depth, using the ENPTVD keyword.

The cell keywords need not specify data for the whole field - any values that are not set do not cause an error, (as for PORO or DX), but imply that they are to be obtained from the default values for the cell, or from ENPTVD, if present.

If SWL, SOGCR and ENPTVD are used, values taken from ENPTVD are used to fill in any data that is not set, as long as the endpoint table number is not 0. End point table numbers are set in the REGIONS section, using ENDNUM, and default to 1, but a value of 0 may be entered.

The overall process is thus:

1. If keywords SWL and SOGCR are present, use the value
2. If not, and ENPTVD is present with non-zero ENDNUM, use this
3. Otherwise, use defaults obtained from tables.

The values of the end-point arrays SWL and SOGCR may be printed out for each cell using RPTSOL with the appropriate mnemonic, and output for plotting with Petrel using RPTRST.

Eclipse 100

If the Hysteresis model is active, the imbibition end points can be specified using the IMPTVD keyword.

When the end-point scaling option is activated, the scaling keywords are entirely optional. If a particular end point keyword is not present in the input data file, the unscaled end points are substituted and used in the scaling calculations.

By default only scaled quantities are reported in any INIT file generated. To display the actual end points used in the simulation for each grid cell the keyword FILLEPS should be added.

The directional capability of the scaling option may be used to control the saturation functions which apply to the different faces of each grid cell. The facility is similar to the directional relative permeability option

and allows saturation end points to be defined on the X,Y and Z faces of each cell or, in the highly directional case, on the +X,-X,+Y,-Y,+Z and -Z faces of each cell. Note that the saturation table look-ups required by the well model and equilibration algorithm are performed using the [SATNUM](#) table for each cell. If directional scaling is active, the isotropic end points are used to rescale the saturation tables within the well model and the equilibration algorithm. Note that if the directional form of an end point is not explicitly defined for a given cell, then it is taken from the relevant table, and is **not** defaulted to the isotropic form (except for pseudo-critical end points arising from the initial fluid mobility correction enabled by entering [MOBILE](#) in keyword [EQLOPTS](#)).

Note: When directional scaling is used, the isotropic form must also be supplied if scaling is required in equilibration and well block flows.

Data checking is performed on the rescaled saturation tables in each cell. The checks ensure that the connate, critical and maximum saturations are in increasing order and that the rescaled tables are consistent for each phase. In the case of directional scaling, the data checking is performed on the rescaled tables for each grid face.

The scaling calculation is performed on the saturation tables used for flow between grid cells and well connections. If, however, the saturation table for the well connection differs from that of the connecting cell then the scaling option is disabled. In this case, any scaling required in the connection saturation table should be done using the [COMPRP](#) keyword. If the table numbers are equal then the end points used are those of the connecting cell. These end points will then be overridden by any subsequent end points defined using the [COMPRP](#) keyword.

Eclipse 100 In the present version of Eclipse 100 the [SWLPC](#) and [SGLPC](#) values cannot be supplied using the [ENPTVD](#) depth type keywords.

Eclipse 100 The end-point scaling option should not be used at present with the vertical equilibrium option.

Eclipse 100 The unscaled and scaled maximum oil saturations used for the two-point end-point scaling of the oil relative permeability to water and gas are calculated according to $S_{omax} = 1 - S_{wco}$ and $SOU = 1 - SWL$ respectively. These can be modified to include connate gas; that is, $S_{omax} = 1 - S_{wco} - S_{gco}$ and $SOU = 1 - SWL - SGL$ respectively by setting [item 205](#) of the [OPTIONS](#) keyword to 1.

Eclipse 300 The saturation table end-points can depend on temperature in a thermal simulation using the keywords [ENKRVT](#), [ENPCVT](#) and [ENPTVT](#).

Eclipse 300 The saturation table end-points can depend on composition using the keywords [ENKRVC](#), [ENPCVC](#) and [ENPTVC](#).

Combined grid-block and temperature-dependent table end-points

Eclipse 300 For thermal models, it is possible to combine the grid-block end-points, which are temperature independent, with the temperature-dependent table end-points. In general, for the end-point scaling option, the grid-block end-points are obtained from the end-point arrays, for saturation, relative permeability and capillary pressure end-points respectively. The end-point arrays are specified using keywords such as [SWL](#), [SWCR](#), [SWU](#), [KRW](#) and [PCW](#), possibly in combination with the depth-dependent end-point tables [ENPTVD](#), [ENKRVD](#) and [ENPCVD](#). The defaulted values are furnished from the end-points obtained from the saturation tables.

If the [ENDSCALE](#) keyword is specified with item 5 set to a value of 1 or 2, it is possible either to supersede or combine respectively these values with those specified in the temperature-dependent end-point tables [ENPTVT](#), [ENKRVT](#) and [ENPCVT](#) for saturation, relative permeability and capillary pressure end-points respectively.

If item 5 is set to 0 or defaulted, the use of grid block end-points and temperature-dependent table end-points is mutually exclusive. This is the only option available for versions of Eclipse 300 prior to the 2011.1 version and is the default option for the 2011.1 version.

If item 5 is set to 1, a temperature dependent end-point, where specified, will supersede the grid block value.

If item 5 is set to 2, a temperature dependent table end-point, where specified, will be combined with the grid block value such that the resultant temperature dependent grid block end-point $Ep_{gb}(T)$ is given by:

$$Ep_{gb}(T) = Ep_t(T) \times \frac{Ep_{gb}}{Ep_t(T_1)} \quad \text{Eq. 3.59}$$

where

Ep is used to signify any of the saturation, relative permeability or capillary pressure end-points

$Ep_t(T)$ is the end-point value obtained from the temperature-dependent table for the grid-block temperature T

Ep_{gb} is the temperature-independent grid-block end-point value

$Ep_t(T_1)$ is the end-point value from the temperature-dependent table at temperature T_1 which corresponds to the first temperature entry in the table.

This form of scaling ensures that the original grid-block end-point is honored at the lowest temperature in the table.

Any end-point in any of the temperature-dependent end-point tables may be defined to be temperature independent by specifying the values in all rows of the appropriate column to be default values (1*). The consequence of this defaulting will depend upon the value of item 5 of the `ENDSCALE` keyword. If this value is 0, the combination of grid-block and temperature-dependent table end-points is not permitted; if only the temperature-dependent tables are specified, the defaults will be supplied from the saturation table end-points. However, if item 5 of the `ENDSCALE` keyword is 1 or 2, the defaults will be supplied from the grid-block end-points, except for the relative permeabilities at critical and residual saturations. If these are defaulted, these will be scaled according to the current values of the maximum relative permeabilities. The value of item 5 is ignored for non-thermal models.

Output of scaled end-points

Eclipse 300

When the saturation end-point scaling depends on temperature or composition (`ENPTVT` and `ENPTVC` keywords), the scaled values can be output using the following keywords and mnemonics.

Refer to [Processed scaled end-points](#) in the *Eclipse Reference Manual*.

To the PRINT file:

```
RPTSOL
PSWL PSWCR PSWU PSGL PSGCR PSGU PSOWCR PSOGCR /
RPTSCHED
PSWL PSWCR PSWU PSGL PSGCR PSGU PSOWCR PSOGCR /
```

To the SUMMARY file

Block keyword	Description
BPSWL	Block scaled minimum water saturation.
BPSWCR	Block scaled critical water saturation.
BPSWU	Block scaled maximum water saturation.
BPSGL	Block scaled minimum gas saturation.
BPSGCR	Block scaled critical gas saturation.
BPSGU	Block scaled maximum gas saturation.
BPSOWCR	Block scaled critical oil-in-water saturation.
BPSOGCR	Block scaled critical oil-in-gas saturation.

To the RESTART file:

```
RPTRST
PSWL PSWCR PSWU PSGL PSGCR PSGU PSOWCR PSOGCR /
```

When the relative permeability end-point scaling depends on temperature or composition ([ENKRVT](#) and [ENKRVC](#) keywords), the scaled values can be output using the following keywords and mnemonics.

To the PRINT file:

```
RPTSOL
PKRO PKRG PKRW PKRGR PKRWR PKRORG PKRORW /
RPTSCHED
PKRO PKRG PKRW PKRGR PKRWR PKRORG PKRORW /
```

To the SUMMARY file

Block keyword	Description
BPKRG	Block scaled gas relative permeability at maximum gas saturation.
BPKRGR	Block scaled gas relative permeability at residual oil saturation.
BPKRO	Block scaled oil relative permeability at maximum oil saturation.
BPKRORG	Block scaled oil relative permeability at critical gas saturation.
BPKRORW	Block scaled oil relative permeability at critical water saturation.
BPKRW	Block scaled water relative permeability at maximum water saturation.
BPKRWR	Block scaled water relative permeability at residual oil saturation.

To the RESTART file:

```
RPTRST
PKRO PKRG PKRW PKRGR PKRWR PKRORG PKRORW /
```

When the capillary pressure end-point scaling depends on temperature or composition ([ENPCVT](#) and [ENPCVC](#) keywords), the scaled values can be output using the following keywords and mnemonics.

To the PRINT file:

```
RPTSOL
PPCG PPCW /
```

```
RPTSCHED
PPCG PPCW /
```

To the SUMMARY file

Block keyword	Description
BPPCG	Block scaled maximum gas-oil capillary pressure.
BPPCW	Block scaled maximum water-oil capillary pressure.

To the RESTART file:

```
RPTRST
PPCG PPCW /
```

The initial fluid mobility correction

In the production of a reservoir model it is important to ensure that the model is initialized with the correct mobile quantities of each fluid as well as the correct quantities of total fluid-in-place. If this is not done the recovery factors of each fluid are predicted erroneously.

The fine scale equilibration algorithm predicts too small an initial mobile oil-in-place even though the fluid volumes-in-place are predicted correctly. To see why the mobile fluids-in-place are underestimated consider the fine scale equilibration of an oil-water system. The variation of water saturation with depth for a typical grid cell that intersects the oil-water transition zone is shown in the following figure.

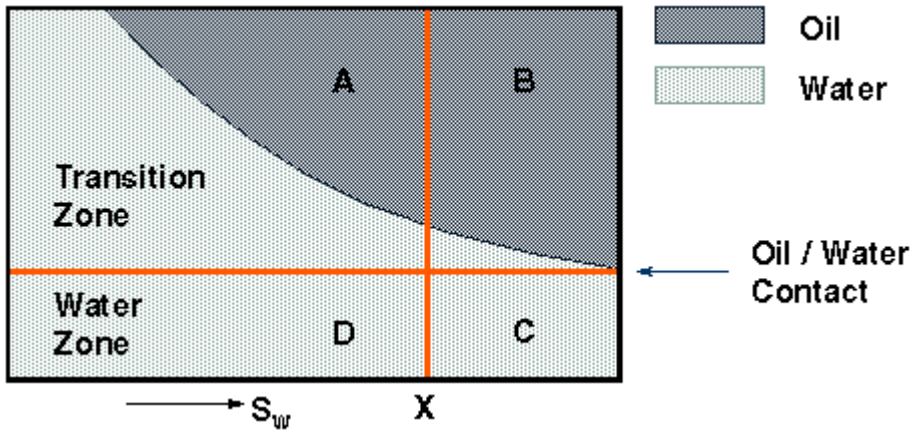


Figure 3.12. Water saturation versus depth for a typical grid cell that intersects the oil-water transition zone

The water saturation at the value X is $1.0 - SOWCR$. V is the pore volume.

The initial volume of oil-in-place is $V(A + B)$

The correct volume of mobile oil-in-place is VA

The volume of mobile oil-in-place calculated by the fine scale equilibration procedure is $V(A - C)$, which leads to an underestimate of the total volume of mobile oil of VC .

The initial volume of mobile oil-in-place is underestimated in cases where there is a negligible transition zone. Note that the problem may also arise in the calculation of the initial mobile volumes of other phases.

A facility based on the end-point scaling option is available in Eclipse to provide correct estimates of both the total and mobile fluids-in-place. The method uses the fine scale equilibration option and computes pseudo values for the critical saturation of each phase in each grid cell. The procedure adopted is as follows:

- For each of the fine scale integration layers ($i=1,...,N$) the saturation, S_{pi} , of each phase, p , and its critical saturation $S_{p,cr,i}$ is determined
- The volume of mobile phase within each layer, M_{pi} , is then found

$$M_{pi} = 0 \text{ if } S_{pi} < S_{p,cr,i}$$

$$M_{pi} = V_i(S_{pi} - S_{p,cr,i}) \text{ if } S_{pi} > S_{p,cr,i}$$

where V_i is the pore volume of layer i

- The total mobile phase volume, M_p , in each cell is then determined using $M_p = \sum_i M_{pi}$. The total phase volume, $T_p = \sum_i V_i S_{pi}$.
- The average phase saturation in the cell is $S_p' = T_p / (\sum_i V_i)$ and the average mobile phase saturation is $M_p' = M_p / (\sum_i V_i)$.
- For cells in a which a mobility-immobility transition occurs due to the saturation variation with depth, the pseudo-critical phase saturation is set to

$$S'_{p,cr,i} = S'_p - M'_p \quad \text{Eq. 3.60}$$

- The remaining cells in the grid contain either mobile or immobile fluid throughout their depth. In this case

$$S'_{p,cr,i} = (\sum_i V_i S_{p,cr,i}) / (\sum_i V_i) \quad \text{Eq. 3.61}$$

This calculation of each pseudo critical phase saturation ensures that the critical saturations are only modified in the vicinity of the fluid contacts in the initial state. During the simulation phase, the relative permeability curves are rescaled using the pseudo critical saturations computed above. The pseudo critical phase saturations are assumed to be isotropic; if directional scaling is enabled, then the directional counterparts of the pseudo critical phase saturations are set equal to their isotropic form.

The mobile fluid-in-place calculation is activated using the `MOBILE` flag in the keyword `EQLOPTS` of the `RUNSPEC` section. In a three-phase run, the initial mobile fluid-in-place correction is applied to all the critical end points `SOWCR`, `SOGCR`, `SWCR` and `SGCR`. In an oil-water system it is applied only to `SOWCR` and `SWCR`. In a gas-oil system it is applied only to `SOGCR` and `SGCR`. In a gas-water system it is applied only to `SWCR` and `SGCR`. Output of the processed end points used in the simulation can be requested with the keywords `SOWCR`, `SOGCR`, `SWCR` and `SGCR` (items 24-27) in the `RPTSOL` keyword (the output is headed by keywords `PSOWCR`, `PSOGCR`, `PSWCR` and `PSGCR`).

End-point variations through a transition zone

In some reservoirs it is necessary to model a depth variation of the critical saturation of one phase such that the critical saturation equals the phase saturation providing the phase saturation is less than a threshold (See figure 3.13). Although the `ENPTVD` keyword could be used to model the profile, a more convenient method, which also gives the correct initial mobile fluids-in-place, is to initially specify a constant critical saturation with depth and set $S'_{p,cr,i} = S'_p - M'_p$ regardless of the phase mobility in each cell. This calculation can be invoked by using the `TZONE` keyword in the `PROPS` section.

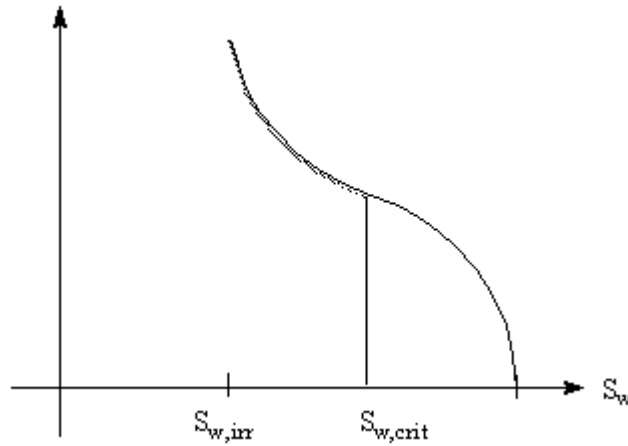


Figure 3.13. Modeling a depth variation of critical saturation

To use the facility and achieve the result shown, specify the keyword **ENDSCALE** in the **RUNSPEC** section, turn on the second data item in the **TZONE** keyword, and set the critical water saturation to $S_{w,crit}^*$. Output of the processed end points used in the simulation can be requested with the mnemonics **SOWCR**, **SOGCR**, **SWCR** and **SGCR** (switches 24-27) in the **RPTSOL** keyword at all depths.

The **TZONE** keyword allows control of the calculation for each phase separately; if the water switch is set true then the critical water saturation will be corrected to the initial water saturation where this is below the input critical saturation.

Leverett J-function

The Leverett J-function is a dimensionless group that allows the capillary pressure function to be correlated with the rock properties.

$$J(S_w) = \frac{P_c}{\sigma} \left(\frac{K}{\phi} \right)^{\frac{1}{2}} \quad \text{Eq. 3.62}$$

where

P_c capillary pressure

σ surface tension

K rock permeability

ϕ rock porosity.

If either the **JFUNC** or the **JFUNCR** keyword is used in the **GRID** section then the capillary pressure is calculated as follows:

$$P_c = F \cdot J(S) \quad \text{Eq. 3.63}$$

where

$J(S)$ J-function. This is input as a function of saturation in the column for capillary pressure in the **SWFN** or **SGFN** table.

F Scaling Factor given by:

$$F = \sigma \cdot \left(\frac{\varphi}{K} \right)^{\frac{1}{2}} \cdot U_{\text{const}} \quad \text{Eq. 3.64}$$

where

σ Surface tension - specified in the `JFUNC` or the `JFUNCR` keyword

φ cell porosity

K cell permeability

U_{const} constant depending on the unit system employed:

= 0.318316 (METRIC units, pressure in bars, perm in mD)

= 4.61678 (FIELD units, pressure in psi, perm in mD)

= 0.314153 (LAB units, pressure in atm, perm in mD)

= 0.314153 (PVT-M units, pressure in atm, perm in mD)

The permeability direction to be used in the calculation may be specified using item 6 of the `JFUNC` or the `JFUNCR` keywords. If this is defaulted, the permeability will be calculated from $K = (K_x + K_y)/2$, unless the model is a cross section, in which case K will be selected from the appropriate horizontal permeability. For a one-dimensional model, the permeability value K is obtained from the permeability corresponding to the direction of the row or column of cells. In addition, irrespective of the permeability direction specified:

- if the numbers of cells in the x and y directions are both equal to 1, the z-direction permeability values will be used
- if the number of cells in the x direction only is equal to 1, the y-direction permeability values will be used
- if the number of cells in the y direction only is equal to 1 the x-direction permeabilities will be used.

Eclipse 100

`JFUNC` and `JFUNCR` have an additional option U which allows user-defined permeabilities specified in the `PERMJFUN` keyword to be used instead of directional permeabilities.

The `JFUNC` or the `JFUNCR` keyword in the `GRID` section causes the scaling factor F to be computed from the grid data and stored in the `PCW` or `PCG` array. The scaling factor can be output by using the mnemonic `ENDPT` in the `RPTPROPS` keyword.

Eclipse 100

Note that, if the surface tension is required to be a function of pressure, the pressure dependent capillary pressure option (`STOW` and `STOG` keywords) can be used to modify the surface tension. In this case:

$$P_c = J(S) \cdot \sigma \cdot \frac{\sigma(P)}{\sigma_{\text{ref}}} \cdot \left(\frac{\varphi}{K} \right)^{\frac{1}{2}} \cdot U_{\text{const}} \quad \text{Eq. 3.65}$$

where $\sigma(P)$ and σ_{ref} are input using the `STOG` or `STOW` keywords.

Matching initial water distribution

The water-oil capillary scaling facility can be used to honor an input water distribution map.

The required water distribution is entered using the `SWATINIT` keyword in the PROPS section. Eclipse automatically computes the PCW array such that:

$$P_{cw}(S_{w \text{ init}}) = P_{oil} - P_{wat} \quad \text{Eq. 3.66}$$

for the appropriate equilibration region, where:

P_{oil} is the oil phase pressure

P_{wat} is the water phase pressure

$S_{w \text{ init}}$ is the required water saturation from `SWATINIT`

If $P_{oil} - P_{wat} < 0$, the `PCW` value is left unchanged and the equilibrated S_w value will be the maximum water saturation, `SWU`. The phase pressures P_{wat} and P_{gas} are determined by equilibration using `EQUIL` and the supplied PVT data. See `OPTIONS` keyword [item 74](#) and `OPTIONS3` keyword [item 143](#) for a different scaling cases.

If the water distribution and the input capillary pressure curves are very inconsistent, the scaling required may be extreme. The scaled maximum capillary pressure can be output to the PRINT file using the mnemonic `SWATINIT` (item 30) in the `RPTSOL` keyword, and is automatically output to the INIT file. These arrays should be checked to ensure that the scaling is physically reasonable. The `PCWMAX` keyword could be used to limit the maximum capillary pressure scaling.

If the `SWATINIT` keyword is used with the J-function option (`JFUNC` or the `JFUNCRCR` keyword) then in regions of the field where has been set the J-function data will be ignored and the `SWATINIT` saturations honored instead. If the `PCW` keyword is used, then the `SWATINIT` distribution will override the input `PCW/IPCW` where $P_{oil} - P_{wat} > 0$.

In a gas condensate study with no initial liquid phase, where the gas-oil contact coincides with the oil-water contact in the `EQUIL` keyword, then `PCW` is taken as the hydrocarbon capillary pressure and the `SWATINIT` distribution will override the input `PCW` where $P_{gas} - P_{wat} > 0$.

Consistency requirements

The oil, water and gas saturation tables, for each particular saturation table number region, must obey certain consistency requirements. Essentially these are described for unscaled tables in "[Consistency requirements](#)". The same requirements apply to the scaled endpoints.

- `SGU` must not exceed $1.0 - \text{SWL}$.

If this condition is violated, the gas saturation in the gas cap is reset to $1.0 - \text{SWL}$, to prevent a negative oil saturation. Normally, if there is no oil in the gas cap, $\text{SGU} = 1.0 - \text{SWL}$.

- `SGL` must not exceed $1.0 - \text{SWU}$

If this condition is violated, the gas saturation in the water zone is reset to $1.0 - \text{SWU}$, to prevent a negative oil saturation. Normally, there is no initial free gas below the gas cap and the water zone is fully saturated with water, thus $\text{SGL} = 0.0$ and $1.0 - \text{SGL} = 1.0$

Otherwise, phases can be mobile even at zero saturation and there is nothing to stop saturations going negative.

The $k_{rw\text{max}}(\text{table})$ is taken to be the value at either the maximum saturation of the saturation table or at `SWU` if this has been specified.

If the **KRWR** keyword has been used, then the scaling honors the K_r at the critical saturation (SR) of the associated (that is, displacing) phase. When the alternative three-point end point scaling method is invoked using keyword **SCALECRS**, the scaling uses the three nodes **SWCR**, **SR** and **SWU** where

$SR = 1 - SOWCR(\text{grid block}) - SGL(\text{grid block})$ in water/oil or gas/oil/water runs

$SR = 1 - SGCR(\text{grid block})$ in gas/water runs

Eclipse 100

In order to ensure that at least one fluid remains mobile in the three-phase ternary diagram, Eclipse also checks for the following:

- $SOWCR + SWCR < 1.0$
- $SOGCR + SGCR + SWL < 1.0$

Eclipse issues a warning if these conditions are not satisfied. When **SCALECRS** is used, this would lead to an error because in an oil-water run, for example, K_{rw} would be scaled between **SWCR** and $1.0 - SOWCR$, which is a null range if the first condition is not satisfied.

Note: In some cases when consistency warning messages involving **SOWCR** and **SOGCR** are reported by Eclipse, these are referred to using the alternative names **SOCRW** and **SOCRG** respectively. (It should be noted that these are not accepted as alternative keyword names.)

Example of end-point scaling

The following will produce the same results in a run:

With end- point scaling

```

SGL
50*0.00 50*0.00 /
SGCR
50*0.10 50*0.20 /
SGU
50*0.99 50*0.98 /
SOWCR
50*0.05 50*0.22 /
SOGCR
50*0.08 50*0.21 /
SWL
50*0.01 50*0.02 /
SWCR
50*0.20 50*0.22 /
SWU
50*1.0 50*1.0 /
SGFN
0.0 0.0 0.0
1.0 1.0 0.5 /
SOF3
0.00 0.0 0.0
1.00 0.8 0.8 /
SWFN
0.0 0.0 1.0
1.0 1.0 0.0 /

```

With multiple tables

```

SGFN
0.0 0.0 0.0
0.1 0.0 1*
0.99 1.0 0.5 /

```

```
0.0 0.0 0.0
0.2 0.0 1*
0.98 1.0 0.5 /
SOF3
0.00 0.0 0.0
0.05 0.0 0.0
0.08 1* 0.0
0.99 0.8 0.8 /
0.00 0.0 0.0
0.21 0.0 0.0
0.22 0.0 1*
0.98 0.8 0.8 /
SWFN
0.01 0.0 1.0
0.2 0.0 1*
1.0 1.0 0.0 /
0.02 0.0 1.0
0.22 0.0 1*
1.0 1.0 0.0 /
REGIONS
SATNUM
50*1 50*2 /
```

Miscibility and surface tension effects

	Eclipse 100
x	Eclipse 300

The surface tension between the oil and gas phases can be used as a measure of how miscible the two fluids are. The Eclipse 300 MISCIBLE model follows that of [Coats \(1979\)](#), where parameters that are functions of surface tension are used to interpolate properties such as relative permeability and capillary pressure between immiscible and miscible values:

$$K_{ro} = F_k \cdot K_{ro}^{imm} + (1 - F_k) \cdot K_{ro}^{mis} \quad \text{Eq. 3.67}$$

$$K_{rg} = F_k \cdot K_{rg}^{imm} + (1 - F_k) \cdot K_{rg}^{mis} \quad \text{Eq. 3.68}$$

$$P_{cog} = F_p \cdot P_{cog}^{imm} \quad \text{Eq. 3.69}$$

K_{ro}^{imm} , K_{rg}^{imm} and P_{cog}^{imm} are the immiscible oil and gas relative permeabilities and capillary pressure obtained from the user-defined rock curves (for example [SGFN](#)).

The miscible capillary pressure is taken to be zero.

K_{ro}^{mis} and K_{rg}^{mis} are the miscible oil and gas relative permeabilities. These can be specified as user-defined rock curves using the [MISCNUM](#) keyword. If these are not specified, then by default they are taken to be straight lines. These straight line relative permeability curves connect the zero point to the maximum point on the immiscible curve. That is the miscible relative permeabilities are defined as

$$K_{ro}^{mis} = K_{ro, \max}^{imm} \left(\frac{S_o}{S_{o, \max}^{imm}} \right) \quad \text{Eq. 3.70}$$

$$K_{rg}^{mis} = K_{rg, \max}^{imm} \left(\frac{S_g}{S_{g, \max}^{imm}} \right) \quad \text{Eq. 3.71}$$

where $S_{o, \max}^{imm}$ and $S_{g, \max}^{imm}$ are the maximum saturations on the immiscible oil and gas relative permeability curves respectively and $K_{ro, \max}^{imm}$ and $K_{rg, \max}^{imm}$ are the corresponding maximum relative permeability values. This implies that $K_{ro, \max}^{mis} = K_{ro, \max}^{imm}$ and $K_{rg, \max}^{mis} = K_{rg, \max}^{imm}$.

The interpolation parameters F_k and F_p are functions of the surface tension σ between the oil and gas phases. For high surface tension the parameters tend to 1 and the flow becomes immiscible. For low surface tension the parameters tend to 0 and the flow becomes miscible.

In order to add the relative permeability curves sensibly, the critical saturations must also be scaled so that both the immiscible and miscible relative permeabilities have the same endpoints:

$$S_{ocr} = F_k \cdot S_{ocr}^{imm} + (1 - F_k) \cdot S_{ocr}^{mis} \quad \text{Eq. 3.72}$$

$$S_{gcr} = F_k \cdot S_{gcr}^{imm} + (1 - F_k) \cdot S_{gcr}^{mis} \quad \text{Eq. 3.73}$$

S_{ocr}^{imm} and S_{gcr}^{imm} are the critical immiscible saturations, generally obtained from the user-defined rock curves, though in the case of hysteresis they may be the critical point of a scanning curve (see ["Hysteresis"](#) for further information).

The critical saturations for miscible flow S_{ocr}^{mis} and S_{gcr}^{mis} are by default zero (unless the [MISCNUM](#) keyword is used), so these equations simplify to:

$$S_{ocr} = F_k \cdot S_{ocr}^{imm} \quad \text{Eq. 3.74}$$

$$S_{gcr} = F_k \cdot S_{gcr}^{imm} \quad \text{Eq. 3.75}$$

The relative permeabilities defined by equations 3.67 and 3.68 are two phase (oil-gas) relative permeabilities. When there are three phases present, these values are used to calculate three phase relative permeabilities, as described in "Saturation Functions".

Note the similarity between this model and that provided by the Velocity Dependent Relative Permeability option: see "Velocity-dependent relative permeabilities". Note also that the length of timestepping is of great importance in simulating surface tension effects.

Relative permeability interpolation parameter

The relative permeability interpolation parameter $F_k(\sigma)$ can either be entered as a table using the `FVST` keyword, or by using Coats's formula:

$$F_k = \min\left[1, \left(\frac{\sigma}{\sigma_0}\right)^N\right] \quad \text{Eq. 3.76}$$

where

σ_0 is the reference surface tension, described in "Reference surface tension",

N is the exponent set by keyword `MISCEXP`.

Therefore the relative permeabilities defined in equations 3.67 and 3.68 approach the miscible values $K_r \rightarrow K_r^{mis}$ as the surface tension tends to zero, $\sigma \rightarrow 0$, and equal the immiscible values $K_r = K_r^{imm}$ when the surface tension is greater than or equal to the reference surface tension $\sigma \geq \sigma_0$.

For single-phase cells, a value of 1 is used for the interpolation parameter.

Values of F_k can be output to the print file or restart file using the argument `FMISC` with the `RPTSOL`, `RPTSCHEd`, or `RPTRST` keywords. Block values can be output to the summary file using the `BFMISC` keyword.

Capillary pressure interpolation parameter

The capillary pressure interpolation parameter $F_p(\sigma)$ is given as a linear function of the surface tension:

$$F_p = \min\left[\frac{\sigma_1}{\sigma_0}, \frac{\sigma}{\sigma_0}\right] \quad \text{Eq. 3.77}$$

where σ_1 is the maximum surface tension for capillary pressure calculation, which defaults to $\sigma_1 = \sigma_0$ but can be set larger than this value using the third data item in the `MISCSTR` or `MISCSTRR` keyword.

Therefore, the capillary pressure defined in equation 3.69 equals the miscible value of zero $P_{cog} = 0$ when the surface tension is zero, $\sigma = 0$, and equals the immiscible value $P_{cog} = P_{cog}^{imm}$ when the surface tension is equal to the reference surface tension $\sigma = \sigma_0$.

For surface tensions greater than the reference surface tension $\sigma \geq \sigma_0$, the capillary pressure can be extrapolated beyond the immiscible values, with the interpolation parameter being allowed to exceed 1:

$$F_p \leq \frac{\sigma_1}{\sigma_0}$$

For single-phase cells a value of 1 will be used for the interpolation parameter.

Values of F_p can be output to the print file or restart file using argument `FPC` with the `RPTSOL`, `RPTSCHED`, or `RPTRST` keywords.

Evaluation of surface tensions

Surface tensions can either be entered as a function of pressure using the `STVP` keyword, or can be obtained as functions of the oil and gas compositions using the Macleod-Sugden correlation:

$$\sigma = \left[\sum_{i=1}^{N_{\text{comps}}} [P]_i (b_L^m x_i - b_V^m y_i) \right]^4 \quad \text{Eq. 3.78}$$

where

σ is the surface tension (always in dynes/cm) and

(x_i, y_i) are the liquid and vapor mole fractions.

the liquid and vapor phase molar densities (b_L^m, b_V^m) are converted into gm-M/cc for this evaluation.

the parachors $[P]_i$ should be provided by the `PARACHOR` keyword.

For one-phase states the surface tension is not used (and the reference surface tension will be reported). The value of σ will go sharply to zero as the phase compositions and phase molar densities become equal, such as occurs near the critical point.

Warnings of un-physically large values of surface tension can be requested using the second data item in the `MISCSTR` or `MISCSTRR` keyword.

Values of surface tension can be output to the print file or restart file using the argument `STEN` with the `RPTSOL`, `RPTSCHED`, or `RPTRST` keywords. Block values can be output to the summary file using the `BSTEN` keyword.

Reference surface tension

The reference surface tension σ_0 can be defined or calculated in various ways:

- By evaluating the Macleod-Sugden correlation, equation 3.78 at a reference pressure, specified either by the `MISCSTRR` or `MISCSTRP` keyword.
- By specifying a value directly, using either the `MISCSTRR` or `MISCSTR` keyword.
- By taking the maximum value from the `STVP` table.
- By evaluating the Macleod-Sugden correlation equation 3.78 at the saturation pressure in each grid block.

Use of the MISCNUM option

In the default treatment of the variation of relative permeabilities with surface tension a straight line relative permeability curve is used as the limit as $\sigma \rightarrow 0$. This is a natural limit, as the two phases become identical and flow at the same velocity. It also gives us continuity of flow as the system makes a transition to the one hydrocarbon phase regime.

However, there are cases where another limit may be used. When straight line relative permeabilities are used, there may be excessive numerical dispersion on coarse full-field grids, so that the smearing of the fronts offsets the gain in recovery expected from the approach to miscibility. There is also some experimental evidence (*Bardon and Longeron, 1980*) that relative permeabilities may be significantly different from straight lines even at very low surface tensions.

To model such cases the `MISCNUM` keyword may be used. This allows a set of relative permeability tables to be nominated to be used rather than the straight line default as $\sigma \rightarrow 0$. As in the default case the tables are scaled to the same critical saturation. In the limit of zero surface tension only the `MISCNUM` curve will be used.

Care is need when using this model - in particular when a transition is made from a near miscible two phase hydrocarbon state to a one phase hydrocarbon state. If the `MISCNUM` curves are far from a straight line this may involve a flow discontinuity which may make it difficult for the simulator to converge the flow equations.

Hysteresis

x	Eclipse 100
x	Eclipse 300

This chapter describes the hysteresis option. This option enables different saturation functions to be specified for drainage (decreasing wetting phase saturation) and imbibition (increasing wetting phase saturation) processes.

You supply two saturation function table numbers for each cell, using the [SATNUM](#) and [IMBNUM](#) keywords in the REGIONS section. These provide respectively the primary drainage and pendular imbibition curves.

The primary drainage curve is for a process which starts at the maximum possible wetting phase saturation, S_{wmaxd} . This value will depend upon the end-points of the saturation tables specified using the [SATNUM](#) keyword. If the wetting phase saturation decreases to S_{wmin} , this primary drainage curve is used.

In a similar way, if the initial saturation is S_{wmin} , and the wetting phase saturation increases to S_{wmaxi} , the imbibition table data will be used. The maximum wetting phase saturation which can be reached, S_{wmaxi} , is determined from the end-points of the tables specified using the [IMBNUM](#) keyword, and will generally be less than S_{wmaxd} .

If the drainage or imbibition process is reversed at some point, the data used does not simply run back over its previous values but instead follows a scanning curve generated by the hysteresis model employed. The way in which this hysteresis process is treated in Eclipse follows the techniques described by [Aziz and Settari \(1979\)](#), [Killough \(1976\)](#), [Carlson \(1981\)](#), [Land \(1968\)](#), and Jargon (this hysteresis method is unpublished).

Using the hysteresis option

The Hysteresis option in Eclipse is enabled by specifying the item 'HYSTER' in the keyword [SATOPTS](#) in the RUNSPEC section. The normal saturation table numbers in the REGIONS section are taken as defining the drainage curves. A second set of keywords, [IMBNUM](#), or the directional variants, are used to specify the tables to be used for each cell to supply imbibition curves. If the imbibition and drainage table numbers for a cell are the same, no hysteresis will occur for that cell.

The mnemonics [IMBNUM](#), or the directional variants, in the [RPTREGS](#) keyword will control the output of the entered imbibition table region number data.

In Eclipse it is possible to apply hysteresis to either the relative permeabilities, capillary pressures, or both.

The keyword [EHYSTR](#) in the PROPS section is specific to hysteresis. This keyword sets the values of two parameters that determine the form of the scanning curves for capillary pressure hysteresis and wetting phase relative permeability hysteresis, and selects one of a choice of models for relative permeability hysteresis. It also allows you to control the application of hysteresis to either the relative permeability, capillary pressure, or both. The default is for hysteresis in both relative permeability and capillary pressure.

There is also a choice of options in the [EHYSTR](#) keyword to apply the Killough hysteresis model to the wetting phase, and also to model an oil-wet system.

Features of the Eclipse hysteresis model include:

- The reservoir is always equilibrated using the drainage curves.
- If the Hysteresis option is used in a non-VE run with a [COMPRP](#) or [COMPVE](#) keyword present, the well connections must use the same saturation tables for both imbibition and drainage.
- For wells, the drainage and imbibition table numbers may be specified using the connection keywords [COMPDAT](#) and [COMPIMB](#), and also with the [WELLCOMP](#) keyword.

- Capillary pressure hysteresis is always applied using the Killough method ([Killough, 1976](#)).

Eclipse 100

The hysteresis option cannot be used in combination with the vertical equilibrium option.

Eclipse 100

The hysteresis option can be used in combination with the surfactant model but hysteresis will only be applied to the gas phase.

Eclipse 100

Relative permeability hysteresis may be modeled using the [Carlson \(1981\)](#) or [Killough \(1976\)](#) methods but not the method of Jargon.

Eclipse 300

The hysteresis option cannot be used in combination with the thermal option.

Relative permeability hysteresis in the non-wetting phase

A typical pair of relative permeability curves for a non-wetting phase is shown in figure 3.14. The curve 1 to 2 represents the user-supplied drainage relative permeability table, and the curve 2 to 3 represents the user-supplied imbibition relative permeability table. (Note that the non-wetting phase saturation increases from right to left in this diagram). The critical saturation of the imbibition curve, S_{ncri} , is greater than that of the drainage curve, S_{ncrd} . The two curves **must** meet at the maximum saturation value, S_{nmax} .

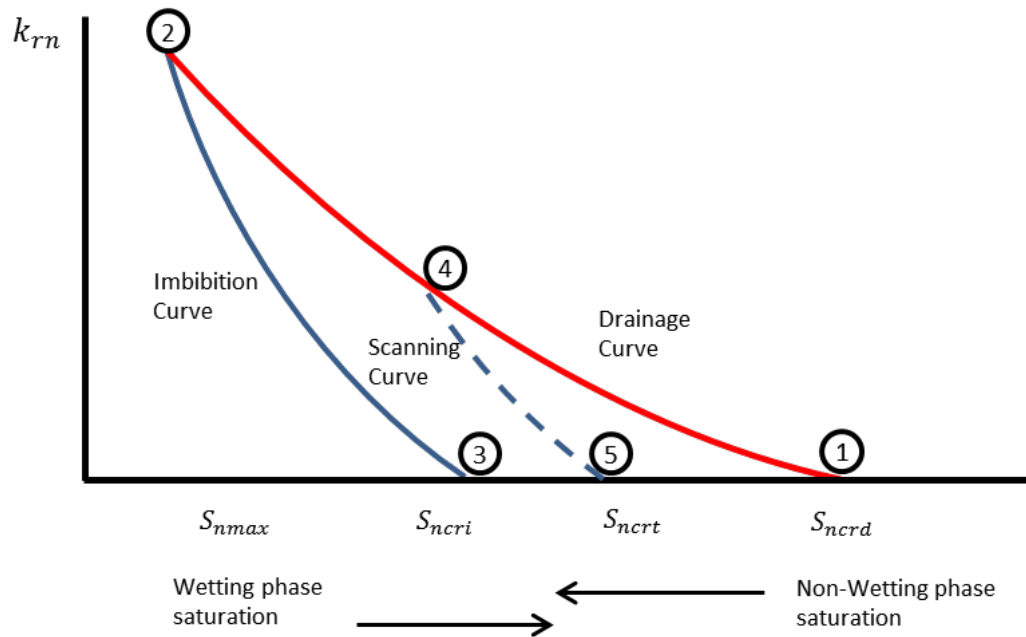


Figure 3.14. A typical pair of relative permeability curves for a non-wetting phase

The primary drainage curve is for a process which starts at the maximum possible wetting phase saturation, S_{wmaxd} . (This value will depend upon the end points of the saturation tables specified using the [SATNUM](#) keyword.) If the wetting phase saturation decreases to S_{wmin} , this primary drainage curve is used.

In a similar way, if the initial saturation is S_{wmin} , and the wetting phase saturation increases to S_{wmaxi} , the imbibition table data will be used. (The maximum wetting phase saturation which can be reached, S_{wmaxi} , is determined from the endpoints of the tables specified using the [IMBNUM](#) keyword, and will generally be less than S_{wmaxd} .)

If the drainage or imbibition process is reversed at some point, the data used does not simply run back over its previous values but runs along a **scanning curve**.

Consider a drainage process starting at point 1. If a full drainage process is carried out, the bounding drainage curve is followed to point 2. If an imbibition process then occurs, the water saturation increasing, the bounding imbibition curve is followed to point 3, the imbibition critical saturation.

But suppose that the drainage process is reversed at some intermediate point 4. A scanning curve results (curve 4 to 5 in the diagram). The critical saturation remaining at point 5 is the trapped critical saturation, S_{ncrt} , which is a function of the maximum non-wetting phase saturation reached in the run, S_{hy} .

If a further drainage process begins from any point on the scanning curve 5 to 4, the same scanning curve is retraced until S_{hy} is reached, at which point the drainage curve is rejoined. S_{hy} is updated during the run, so that further imbibition processes would occur along the appropriate scanning curves.

Eclipse 100

There is a choice of two methods for the generation of scanning curves from a given value of S_{hy} using Carlson's method (Carlson, 1981) or Killough's method (Killough, 1976). The choice of method is governed by item 2 in keyword EHYSTR.

Eclipse 300

There is a choice of three methods for the generation of scanning curves from a given value of S_{hy} using Carlson's method or Killough's method or J. Jargon's method (unpublished). The choice of method is governed by item 2 in keyword EHYSTR.

Carlson's method for relative permeability hysteresis

Carlson's method produces a scanning curve that is parallel to the imbibition curve. It can be visualized by shifting the imbibition curve horizontally until it cuts the drainage curve at the saturation S_{hy} . When this method is chosen, it is important to ensure that the imbibition curve is always steeper than the drainage curve at the same K_r value. If this is not the case, the scanning curve could cross to the right of the drainage curve, which may produce a negative value of S_{ncrt} .

Killough's method for relative permeability hysteresis

Killough's method does not have such a simple geometric interpretation. For a given value of S_{hy} the trapped critical saturation is calculated as

$$S_{ncrt} = S_{ncrd} + \frac{S_{hy} - S_{ncrd}}{1 + C(S_{hy} - S_{ncrd})} \quad \text{Eq. 3.79}$$

where

$$C = \frac{1}{S_{ncri} - S_{ncrd}} - \frac{1}{S_{n \max} - S_{ncrd}} \quad \text{Eq. 3.80}$$

(Killough's formulae have been adapted to allow for non-zero values of S_{ncrd} .)

The relative permeability for a particular saturation S_n on the scanning curve is

$$K_{rn}(S_n) = \frac{K_{rni}(S_{norm})K_{rnd}(S_{hy})}{K_{rnd}(S_{n \max})} \quad \text{Eq. 3.81}$$

where K_{rni} and K_{rnd} represent the relative permeability values on the bounding imbibition and drainage curves respectively, and

$$S_{norm} = S_{ncri} + \frac{(S_n - S_{ncrt})(S_{n \max} - S_{ncri})}{S_{hy} - S_{ncrt}} \quad \text{Eq. 3.82}$$

With Killough's method S_{ncrt} will always lie between S_{ncrd} and S_{ncri} . But if the drainage and imbibition curves are made to coincide, the scanning curve will not necessarily follow this combined curve, except at its end points.

Eclipse 300

If the **DRAINAGE** keyword is specified, the scanning curve will be forced always to lie below the drainage curve.

The Killough method described above has been modified slightly to overcome an un-physical behavior when S_{hy} is small. As S_{hy} tends to S_{ncrd} in equation 3.79, tends to $S_{ncS_{ncrt}rd}$, which leads to an infinite imbibition relative permeability gradient. This in turn can cause convergence problems, while having little material effect on the simulation results. Therefore equation 3.79 has been modified as follows:

$$S_{ncrt} = S_{ncrd} + \frac{S_{hy} - S_{ncrd}}{A + C(S_{hy} - S_{ncrd})} \quad \text{Eq. 3.83}$$

where

$$A = 1 + a \cdot (S_{n\max} - S_{hy}) \quad \text{Eq. 3.84}$$

The parameter a is input through the fourth data item in the **EHYSTR** keyword and defaults to 0.1.

Jargon's method for relative permeability hysteresis

Eclipse 300

In this model the trapped saturation S_{ncrt} is constructed by moving the drainage critical saturation towards the imbibition critical saturation by the same fraction that the hysteresis saturation has moved towards the maximum non-wetting saturation

$$S_{ncrt} = S_{ncrd} + \frac{(S_{ncri} - S_{ncrd})(S_{hy} - S_{ncrd})}{(S_{n\max} - S_{ncrd})} \quad \text{Eq. 3.85}$$

A function which represents the ratio of imbibition and drainage curves as a function of the saturation value scaled between the drainage curve end point and the maximum saturation is determined

$$R(X) = \frac{K_{ri}(S_n)}{K_{rd}(S_n)} \quad \text{Eq. 3.86}$$

where X is defined from the non-wetting saturation S_n by

$$X = \frac{S_n - S_{ncrd}}{S_{n\max} - S_{ncrd}} \quad \text{Eq. 3.87}$$

Clearly X lies between 0 and 1 and R goes from 0 at $X = 0$ to 1 at $X = 1$, where the two curves meet. If the drainage curve is always above the imbibition curve, R also must lie between 0 and 1.

The scanning relative permeability value is then constructed by multiplying the drainage curve value by the function R , where R is evaluated at an X -value which reflects the fractional distance of the current saturation between the drainage curve end point and the hysteresis saturation

$$K_{rn}(S_n) = R(X_s)K_{rd}(S_n) \quad \text{Eq. 3.88}$$

$$X_s = \frac{S_n - S_{ncrd}}{S_{hy} - S_{ncrd}} \quad \text{Eq. 3.89}$$

This can also be written as

$$K_{rn}(S_n) = \frac{K_{ri}(\hat{S})}{K_{rd}(\hat{S})} K_{rnd}(S_n)$$

where

$$\hat{S} = S_{ncrd} + (S_n - S_{ncrd}) \frac{(S_{nmax} - S_{ncrd})}{(S_{hy} - S_{ncrd})}$$

Differences in hysteresis treatment in three-phase cases

Eclipse 300

The treatment of relative permeability hysteresis differs between the Stone's method and the default option.

In the [STONE2](#) case, K_{row} is obtained at $S_n = 1 - S_w$, equal to the hydrocarbon saturation in the block. It is thus natural to take the maximum historical hydrocarbon saturation as S_{nhy} and to apply hysteresis to the K_{row} part of the Stone II expression only. Gas relative permeability occurs in K_{rg} only.

In the default case, both K_{row} and K_{rog} are obtained at the grid block oil saturation, and hysteresis is applied using this variable. The K_{rog} end points should normally be the same in the drainage and imbibition curves.

Consistency requirements

Certain consistency relationships between the end-points on the imbibition and drainage curves should be met in order to use the above models. As discussed above, the imbibition and drainage curves should meet at the maximum saturation S_{nmax} . Clearly also $S_{ncri} > S_{ncrd}$ is required.

Eclipse 100

Eclipse 100 checks these consistency relationships if the [HYSTCHCK](#) keyword is included in the PROPS section.

Relative permeability hysteresis in the wetting phase

There is an option to use the Killough model for wetting phase hysteresis. Otherwise the same curve is used to obtain the wetting phase relative permeability in both drainage and imbibition processes and for which either the drainage or the imbibition curve can be selected. You make these modeling choices using item 2 of the [EHYSTR](#) keyword.

A typical pair of wetting phase relative permeability curves suitable for the Killough model are shown in figure 3.15. The curve 1 to 2 represents the user-supplied drainage relative permeability table, and the curve 2 to 3 represents the user-supplied imbibition relative permeability table. The two curves **must** meet at the connate saturation ($S_{wco} = 1 - S_{nmax}$). The maximum saturation on the imbibition curve is $1 - S_{ncri}$.

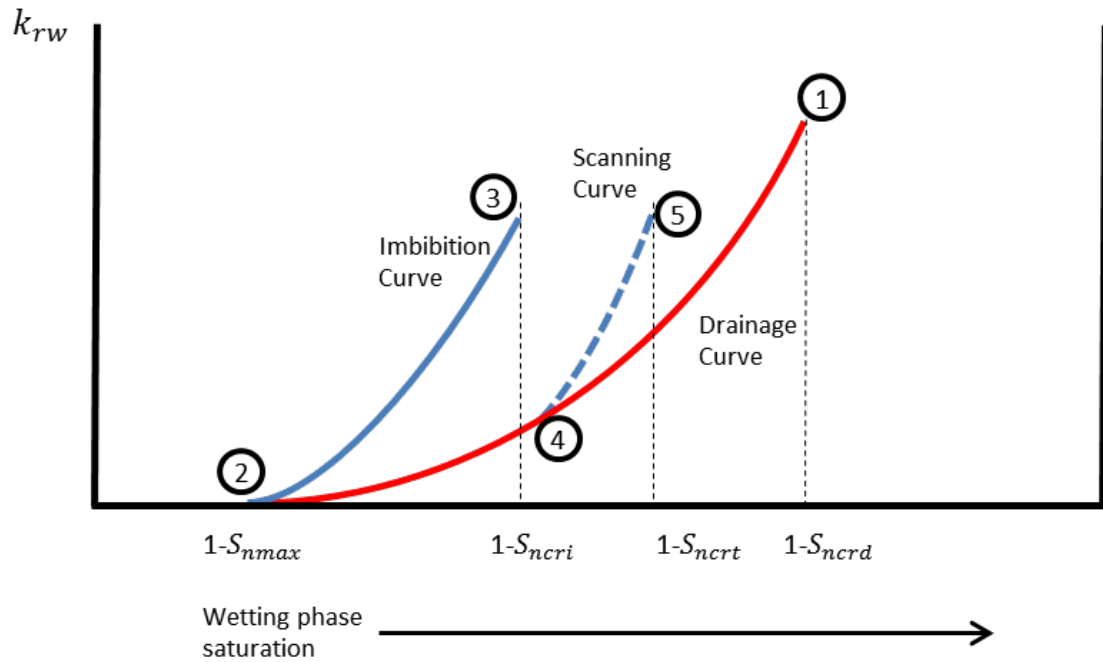


Figure 3.15. A typical pair of relative permeability curves for a wetting phase

An initial drainage process would follow the drainage curve (point 1 to point 2). An imbibition process starting at point 2, $S_w = S_{wco} = 1 - S_{nmax}$, follows the bounding imbibition curve (point 2 to point 3). Point 3, $S_w = 1 - S_{ncri}$, is the maximum wetting phase saturation that can be reached starting from S_{wco} , since the trapped non-wetting phase saturation is S_{ncri} .

An imbibition process that starts from an intermediate saturation (point 4) will follow a scanning curve (point 4 to point 5). The saturation at point 4 is $S_w = 1 - S_{hy}$, where S_{hy} is the maximum non-wetting phase saturation reached. The maximum saturation that can be reached on the scanning curve (point 5) is $S_w = 1 - S_{ncrt}$, where S_{ncrt} is the trapped critical saturation of the non-wetting phase, as defined in the previous section.

If a further drainage process begins from any point on the scanning curve, the same scanning curve is retraced until point 4 is reached, where the drainage curve is rejoined.

Killough's method for calculating the scanning curves uses some of the quantities derived in the previous section for the non-wetting phase. The trapped critical non-wetting phase saturation S_{ncrt} is determined for the particular value of S_{hy} . The wetting phase relative permeability at the complementary saturation is calculated, thus fixing the position of point 5 via

$$K_{rw}(1 - S_{ncrt}) = K_{rwd}(1 - S_{ncrt}) + (K_{rwi}(1 - S_{ncri}) - K_{rwd}(1 - S_{ncri})) \left(\frac{S_{ncrt} - S_{ncrd}}{S_{ncri} - S_{ncrd}} \right)^A \quad \text{Eq. 3.90}$$

where the exponent A is a curvature parameter entered in item 3 of the keyword EHYSTR

K_{rwd} and K_{rwi} represent the wetting phase relative permeability values on the bounding drainage and imbibition curves respectively.

The relative permeability for a particular saturation S_w on the scanning curve is given by

$$K_{rw}(S_w) = K_{rwd}(1 - S_{hy}) + \frac{(K_{rw}(1 - S_{ncrt}) - K_{rwd}(1 - S_{hy}))K_{rwi}(1 - S_{norm})}{K_{rwi}(1 - S_{ncrt})} \quad \text{Eq. 3.91}$$

where S_{norm} is the function of $S_n = 1 - S_w$ defined in the non-wetting phase hysteresis section.

In some cases it has been observed that the path of the wetting phase relative permeability scanning curve is not contained within the region formed by the drainage and imbibition bounding curves. An example of this behavior can be observed in figure 3.16. Typically, the scanning curve, which is constructed from a transformed replica of the imbibition curve, will depart initially from this region if its initial gradient is less than that of the drainage curve at the point of departure. This tends to occur when the initial gradient of the imbibition curve is itself small.

If Item 13 of the **EHYSTR** keyword is set to 1, the scanning curve will be constructed from a reduced portion of the imbibition curve in order to mitigate this effect. For cases where the scanning curve would have otherwise remained within the region of the bounding curves this correction has very little effect. This modification is only applied to the wetting phase relative permeability hysteresis scanning curves, that is, when item 2 of the **EHYSTR** is set to 4 or 7. In extreme cases where very low initial gradients are being modeled, this modification may still result in scanning curves which depart marginally from within the region of the bounding curves.

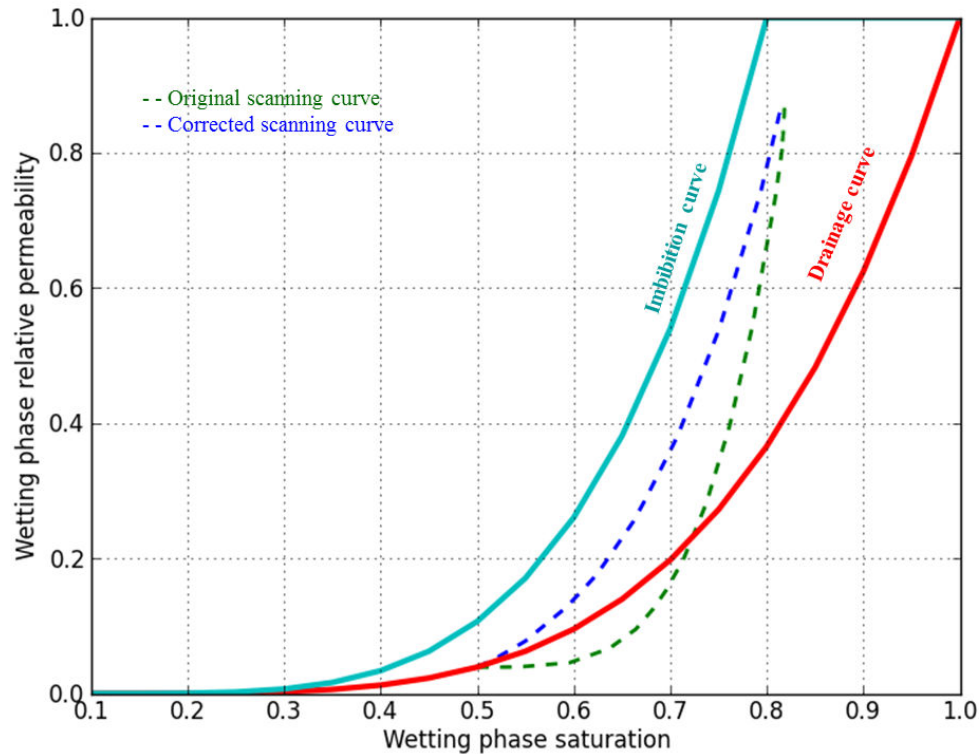


Figure 3.16. Correction of the Killough scanning curve for the wetting phase relative permeability

As with Killough's non-wetting phase hysteresis model, if the drainage and imbibition curves are made to coincide the scanning curve will in general only meet this combined curve at its end points (points 4 and 5 in figure 3.15).

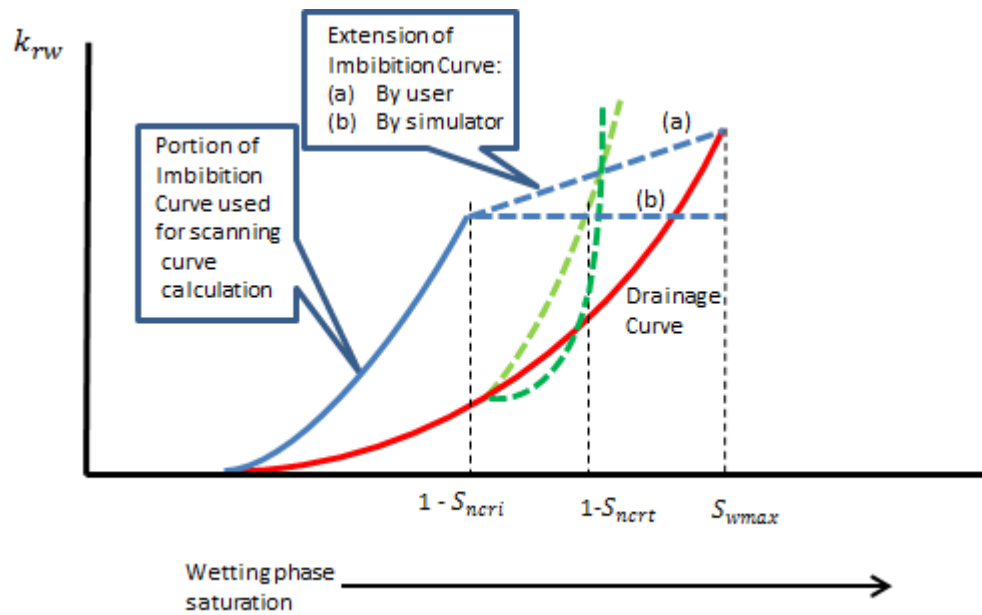


Figure 3.17. Overshoot of the Killough scanning curve for the wetting phase relative permeability

When modeling relative permeability hysteresis in the wetting phase using the Killough hysteresis model, it is important to note that the construction of the hysteresis wetting phase scanning curve is based upon the properties of the non-wetting phase critical saturations in addition to the wetting phase saturation functions. The target point for the scanning curve is determined by the non-wetting phase trapped critical saturation (point 5 in figure 3.15) and occurs at a wetting phase saturation of $1 - S_{ncrt}$ for oil relative permeability to water and water relative permeability and at $1 - S_{ncrt} - S_{wco}$ for oil relative permeability to gas. For example, for a water-wet system and for water relative permeability hysteresis, only the portions of the water relative permeability imbibition and drainage curves up to water saturations of $1 - S_{ocri}$ and $1 - S_{ocrd}$ respectively are used for constructing the scanning curve. If the imbibition and drainage curves extend beyond these saturations, for example, to form a closed loop at the maximum attainable saturation, the scanning curve may deviate beyond the region bounded by the drainage and imbibition curves as the wetting phase saturation approaches its maximum attainable value. This is illustrated in figure 3.17 where only the portion of the imbibition curve up to $1 - S_{ncri}$ has been used for constructing the wetting phase scanning curve, in accordance with Killough model. However, in curve (a), the imbibition curve has been extended in the input data table to the maximum wetting phase saturation to form a closed loop with the drainage curve. On the other hand, if the last saturation entry in the table were $1 - S_{ncri}$, internally the saturation function would be extended horizontally for saturations up to S_{wmax} , as shown in curve (b). In the case of figure 3.17, both scanning curves, (the original Killough model (dark green), and the scanning curve correction for Killough model (light green)), appear to deviate outside the region bounded by the imbibition curve whereas, for Killough model, this extension of the imbibition curve beyond $1 - S_{ncri}$ can arguably be considered an artefact because this portion of the imbibition curve is not used by the Killough model for constructing the wetting phase scanning curve.

Note: The Carlson hysteresis model (and also the Killough hysteresis model if the wetting phase scanning curve correction has been specified) use inverse lookup of the imbibition curve relative permeability to construct the hysteresis scanning curve, that is, saturation is looked up as a function of relative permeability. This requires that the imbibition curve (both before and after scaling if end-point scaling has been specified) should be monotonically increasing over the saturation extent from critical to maximum

saturation. Otherwise, the inverse lookup will not be uniquely defined and undesired results may occur during the calculation of the scanning curves. Moreover, if three-point scaling is specified ([SCALECRS](#) keyword), the relative permeability at the residual saturation of the displacing phase should occur at a saturation which is less than the maximum saturation in both the unscaled and scaled tables. For example, for water permeability to oil, the following criteria should be satisfied:

$$(1 - S_{owcr} - S_{gco}) < S_{wmax}(1 - SOWCR - SGL) < SWU \quad \text{Eq. 3.92}$$

where:

S_{owcr} is the saturation table derived critical oil saturation in water.

S_{gco} is the saturation table derived connate gas saturation.

S_{wmax} is the saturation table derived maximum water saturation.

$SOWCR$ is the scaled oil saturation in water end-point.

SGL is the scaled minimum gas saturation end-point.

SWU is the scaled maximum water saturation end-point.

Modeling oil wet systems

It is possible to apply the non-wetting phase hysteresis models (Carlson, Killough or Jargon) to the water phase. This enables the modeling of trapped water by oil in an oil wet rock.

In the standard case, the `SATNUM` curves are used when S_w is decreasing, the `IMBNUM` curves when S_w is increasing. This still applies in an oil-wet case.

The gas phase will use the appropriate non-wetting phase model. The oil relative permeabilities will be taken from the drainage curves unless the Killough wetting phase model has been specified.

In order to be coherent with the physical meaning of an oil-wet rock, the saturation function data must satisfy certain relationships if the Killough wetting phase model is applied to oil (option 7, item 2 of [EHYSTR](#) keyword). In these cases it is required that $S_{g\max} = 1.0 - S_{oco}$ and also $S_{w\max} = 1.0 - S_{oco}$, while $S_{o\max}$ can be equal to 1.0, where S_{oco} is the **connate** oil saturation.

Note: One has to be aware that if $S_{oco} > 0$ then $S_{w\max} < 1$ even in the aquifer.

This option is selected by setting the second data item in the `EHYSTR` keyword to 5 for the Carlson model and 6 for the Killough model. If option 7 is selected, the Killough model is applied to both the wetting and non-wetting phases.

Capillary pressure hysteresis

Given the method described in the previous relative permeability sections for calculating S_{ncrt} , the trapped non-wetting phase saturation, this section describes capillary pressure hysteresis. Again, the data available are the primary drainage and imbibition curves.

In the three-phase and gas-water cases both water and gas capillary pressure hysteresis occurs.

Water capillary pressure

Consider the water capillary pressure case shown below.

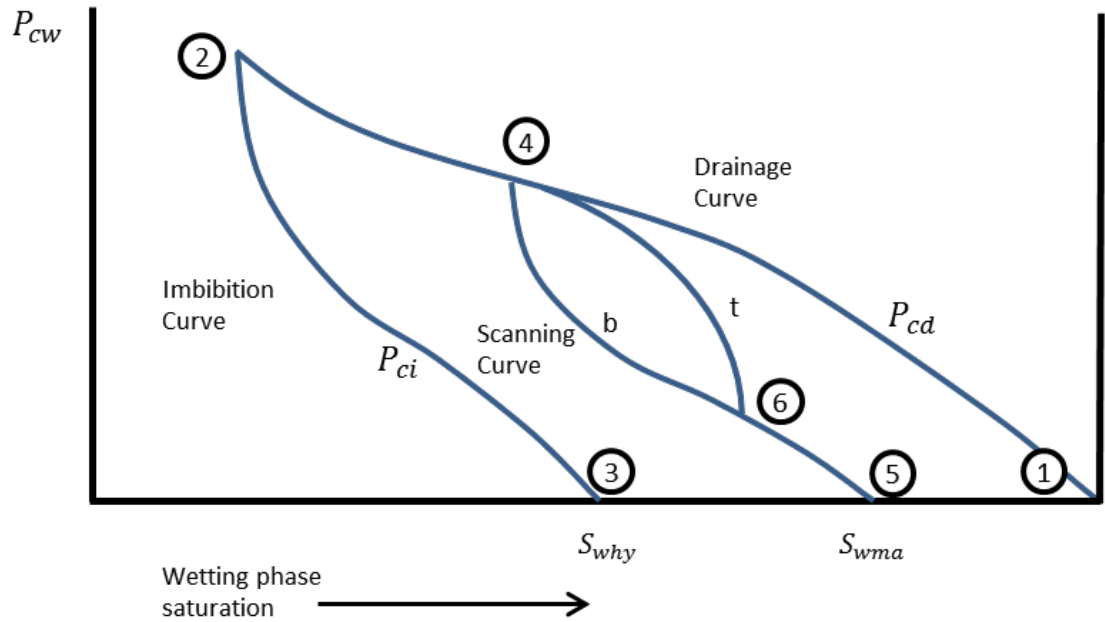


Figure 3.18. The water capillary pressure case

Again, suppose the drainage process starting at point 1 is reversed at an intermediate saturation S_{hy} (point 4). The capillary pressure again traverses a scanning curve from point 4 to the trapped critical non-wetting phase saturation at point 5. This curve is formed as a weighted average of the drainage and imbibition curves, starting with 100% of the drainage curve at point 4. The form used is that suggested by Killough, namely

$$P_c = P_{cd} + F(P_{ci} - P_{cd}) \quad \text{Eq. 3.93}$$

with

$$F = \left(\frac{1}{S_w - S_{why} + E} - \frac{1}{E} \right) / \left(\frac{1}{S_{wma} - S_{why} + E} - \frac{1}{E} \right) \quad \text{Eq. 3.94}$$

in which

E is a curvature parameter of the order of 0.1.

S_{why} is the water saturation at the hysteresis reversal point 4, that is the minimum historical water saturation in the cell

S_{wma} is the maximum water saturation attainable allowing for the trapped non-wetting phase saturation (that is at $S_n = S_{ncrt}$).

S_{wma} is $1 - (S_{ocr} - S_{gcrt})$, the critical gas saturation reflecting the trapping saturation due to relative permeability hysteresis (depending on the choice of Carlson's Killough's or Jargon's method).

This occurs when $S_w \rightarrow S_{why}$, $F \rightarrow 0$ and $P_{cw} \rightarrow P_{cwi}$ and when $S_w \rightarrow S_w^{\max}$, $F \rightarrow 1$ and $P_{cw} \rightarrow P_{cwi}$. The curvature parameter E can be specified by item 1 of the **EHYSTR** keyword.

The imbibition capillary pressure P_{ci} in equation 3.93 is obtained by looking up the capillary pressure on the imbibition curve after scaling between the end-points S_{wcrit} and S_{wma} , where S_{wcrit} is the critical water saturation on the imbibition curve.

Suppose that a second reversal, back to a drainage process, occurs at point 6, rather than continuing to S_{wma} . Unlike the case of relative permeability hysteresis, this does not necessarily re-traverse the scanning curve (marked by a 'b' in the above diagram), but may follow a new drainage to imbibition to drainage scanning curve that returns to the point 4, marked by a 't' in the above diagram. The form of this is:

$$P_c = P_{ci} + G(P_{cd} - P_{ci}) \quad \text{Eq. 3.95}$$

with

$$G = \left(\frac{1}{S_{dep} - S_w + E} - \frac{1}{E} \right) / \left(\frac{1}{S_{dep} - S_{why} + E} - \frac{1}{E} \right) \quad \text{Eq. 3.96}$$

in which

S_{why} the water saturation at the hysteresis reversal point 4

S_{dep} is the departure saturation, the point at which $G = 0$.

In Eclipse, the default is for the imbibition scanning curve 'b' described by equation 3.93 to be **retraversed** when this reversal back to drainage occurs.

Eclipse 100

To obtain the **new** scanning curve 't' described by equation 3.95 for secondary drainage, item 6 in the **EHYSTR** keyword should be set to 'NEW'. However, the occurrence of multiple reversals onto different scanning curves may inhibit convergence in some cases.

The case in which an imbibition process is reversed onto an imbibition to drainage scanning curve is entirely analogous. (When the initial water saturation is less than the critical drainage saturation, the initial process must be imbibition.) The form of the scanning curve is

$$P_c = P_{ci} + F(P_{cd} - P_{ci}) \quad \text{Eq. 3.97}$$

with

$$F = \left(\frac{1}{S_{why} - S_w + E} - \frac{1}{E} \right) / \left(\frac{1}{S_{why} - S_{wcrd} + E} - \frac{1}{E} \right) \quad \text{Eq. 3.98}$$

in which

S_{why} is the water saturation at the hysteresis reversal point. In this case it is the minimum historical water saturation in the run.

S_{wcrd} is the critical water saturation on the drainage curve, which is the same for the scanning curves

When a second reversal back to imbibition occurs at a new departure point S_{dep} on the imbibition to drainage scanning curve, a second scanning curve is then followed during further imbibition from S_{dep} to S_{why} .

Gas capillary pressure

Similar expressions are used for gas capillary pressure hysteresis. In this case the gas saturation is the non-wetting saturation, and drainage to imbibition scanning curves return to the trapped critical gas saturation.

Consider the gas capillary pressure case shown below.

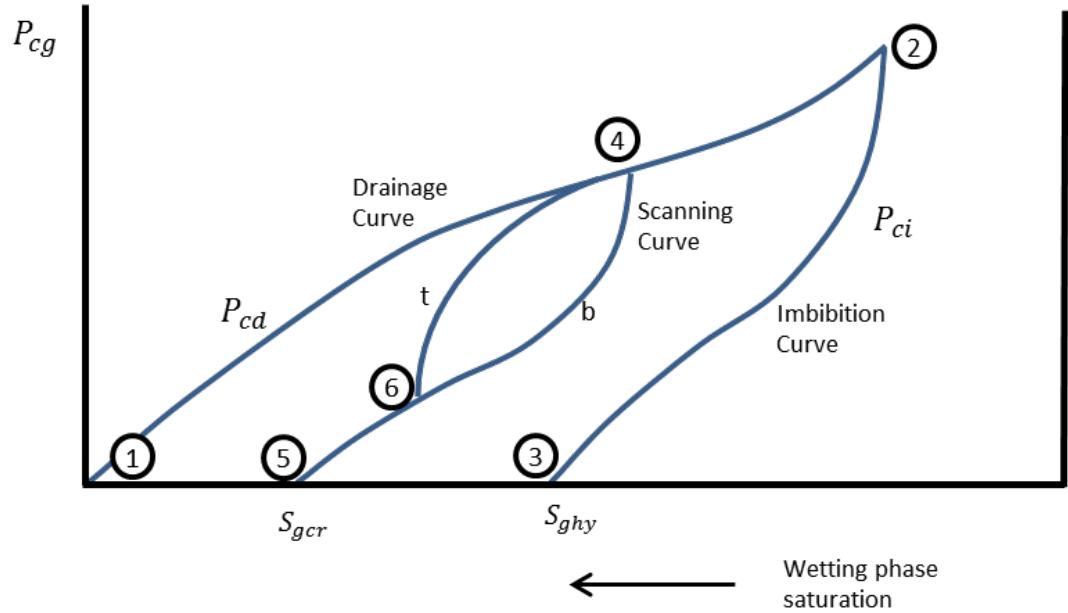


Figure 3.19. The gas capillary pressure case

Analogously to the case of P_{cw} ,

$$P_c = P_{cd} + F(P_{ci} - P_{cd}) \quad \text{Eq. 3.99}$$

with

$$F = \left(\frac{1}{S_{ghy} - S_g + E} - \frac{1}{E} \right) / \left(\frac{1}{S_{ghy} - S_{gcr} + E} - \frac{1}{E} \right) \quad \text{Eq. 3.100}$$

in which

E is a curvature parameter of the order of 0.1.

S_{ghy} is the gas saturation at the hysteresis reversal point 4

S_{wma} is the maximum water saturation attainable allowing for the trapped non-wetting phase saturation (that is, at $S_n = S_{ncrt}$)

when $S_w \rightarrow S_{why}$, $F \rightarrow 0$ and $P_{cw} \rightarrow P_{c wd}$

and when $S_w \rightarrow S_w^{\max}$, $F \rightarrow 1$ and $P_{cw} \rightarrow P_{c wi}$.

Hysteresis in WAG floods

x	Eclipse 100
x	Eclipse 300

This chapter describes the WAG (Water Alternating Gas) hysteresis option.

The hysteresis models described in "Hysteresis" can be thought of as two-phase models where a phase relative permeability is typically a function of its saturation and of its history. In WAG systems the relative permeability of a phase is also dependent on the history of the third phase. For example it has been observed that the gas relative permeability is higher when displacing pure oil than when displacing a water oil mixture following a water flood.

The WAG hysteresis model aims to provide a simple method of modeling these three-phase effects. The model essentially consists of three components: a non-wetting phase model for the gas phase, a wetting phase model for water and a modification to the residual oil saturation in the STONE 1 three-phase oil relative permeability model.

The non-wetting gas phase hysteresis model is based on the theory developed by Land and Carlson. It differs from the standard Eclipse implementation of the Carlson model in that the imbibition curves are calculated analytically from the drainage curve and an input Land's parameter. In the gas phase the imbibition tables ([IMBNUM](#)) are ignored.

The wetting phase model (for the water phase) is based on input two-phase and three-phase relative permeability curves. In this case the [IMBNUM](#) curves are interpreted as the three-phase water relative permeability, that is the relative permeability following a gas flood.

It is possible to utilize the standard hysteresis models in conjunction with the WAG hysteresis model. For example, oil phase hysteresis may be modeled in the usual way while the WAG hysteresis model is active.

The Land and Carlson theory is described in [Carlson \(1981\)](#) and [Land \(1968\)](#).

Using the WAG hysteresis option

The WAG hysteresis model is activated by using the [WAGHYSTR](#) keyword. The hysteresis option must be enabled in the RUNSPEC section by specifying the item HYSTER in the [SATOPTS](#) keyword. The model is only available in 3-phase runs.

The [WAGHYSTR](#) keyword requires a record of data for each saturation region ([SATNUM](#)). For each region the first data item, Land's parameter, must be specified to describe the imbibition curves. The second data item is the reduction factor used to describe the secondary drainage curve for the non-wetting phase (gas). The keyword in addition selects which models are active, enables the Carlson model to be modified to prevent infinite gradients in the imbibition relative permeabilities, and controls the threshold for moving from the two-phase model to the three-phase model.

The [EHYSTR](#) keyword can be used alongside the [WAGHYSTR](#) keyword to control the standard two-phase hysteresis models.

Both the eventual trapped gas saturation (the saturation that would be trapped if a water/oil flood continued) and the dynamic trapped gas saturation (the saturation deemed to be trapped at a given saturation state) can be output to the PRINT, RESTART and SUMMARY files. The [SGTRAP](#) mnemonic in the [RPTSCHEM](#) keyword requests output of both arrays to the Print file. The equivalent arrays are output to the RESTART file provided a full (not graphics only) RESTART file is requested. The SUMMARY keywords BGTRP and BGTPD output grid block values to the SUMMARY files.

In Eclipse 100, [WAGHYSTR](#) cannot be specified in a restart run unless it has been specified in the base run and any intermediate restart runs.

In Eclipse 300, WAGHYSTR can be specified in a restart run without having been specified in a previous base or restart run. However, this facility should only be used after carefully considering the efficacy of such an approach, for example, in situations where the restart containing the first WAGHYSTR keyword corresponds to the commencement of a WAG injection process, this may be an appropriate course of action. However, the results for simulations where WAG hysteresis has been modeled from the outset will be different from those where WAG hysteresis is commenced at some subsequent time thereafter because the hysteretic treatments prior to this time will be different. Moreover, the saturation history used by the WAG hysteresis model will only have been initiated upon the first occurrence of the WAGHYSTR keyword. This saturation history will also be lost if the WAGHYSTR keyword is not specified in all successive restarts following its first occurrence.

In Eclipse 300, from 2014.1 onwards, an alternative WAG hysteresis model is available. This is selected by including the keyword **WAGHALT** in the RUNSPEC section. This alternative model seeks to replicate more closely the behavior of the Eclipse 100 WAG hysteresis model, both in terms of the change in saturation direction decisions and in the modeling of conventional hysteresis for those phases not subject to WAG hysteresis. The restart records written and read by this alternative WAG hysteresis model are not compatible with the default model; if the WAG hysteresis model is changed in a restart run, the previous WAG saturation history will not be preserved nor utilized.

It is important to note that the modeling of a WAG injection process does not of itself mandate that WAG hysteresis should also be modeled. If reservoir core measurements or simulations indicate significant hysteretic characteristics, it may be advisable firstly to establish which of the conventional or WAG hysteresis effects in the gas and or water phases are more significant before deciding which hysteresis model to select. Conventional hysteresis can be modeled in conjunction with WAG hysteresis and will be applied to the oil phase and also the gas and/or water phases (if these are not subject to WAG hysteresis) according to the options selected using the EHYSTR keyword.

Conventional and WAG hysteresis

There are some important differences between the specification of the relative permeability saturation functions for conventional and WAG hysteresis. For conventional hysteresis, the drainage and imbibition data are used to describe the two-phase hysteretic behavior of oil-water and gas-oil capillary pressure, water relative permeability, gas relative permeability and oil relative permeability to water and gas. For WAG plus conventional hysteresis, the bulk of this two-phase drainage and imbibition data is used in a similar way but with the following differences:

- The gas relative permeability data referred to by the drainage saturation table indices (SATNUM) are assumed to be representative of the two-phase drainage (increasing gas saturation) gas relative permeability to oil in the presence of connate water only and are used as a basis for constructing all the associated scanning curves. With the exception of the critical gas saturation, the gas relative permeability data referred to by the imbibition saturation table indices (IMBNUM) are not used for modeling WAG hysteresis. However, they may still be required for conventional hysteresis according to the hysteresis model selected.
- The water relative permeability data referred to by the drainage saturation table indices (SATNUM) represent the two-phase water relative permeability to oil in the absence of gas. For conventional hysteresis, this column of data defines the two-phase drainage water relative permeability. If WAG hysteresis is applied to the water phase, this curve will be used for both drainage and imbibition processes in the absence of gas.
- The water relative permeability data referred to by the imbibition table indices (IMBNUM) represent the three-phase water relative permeability, that is, the water relative permeability to oil in the presence of gas, measured or modeled at the maximum gas saturation encountered during the gas injection cycle of

the WAG process. For conventional hysteresis, this column of data defines the two-phase water imbibition relative permeability.

In general, the more significant difference between conventional and WAG hysteresis is the impact of successive water and gas injection cycles upon the trapped gas and hence upon the gas phase rather than the water phase relative permeability. When modeling WAG hysteresis in the water phase, careful consideration should be given to the suitability of the conventional two-phase drainage and imbibition relative permeability data for reuse with WAG hysteresis. This is complicated by the different characteristics of these saturation functions for these different types of hysteresis. For conventional hysteresis and a water-wet model, the imbibition water relative permeability curve typically exhibits a higher relative permeability value than the drainage curve at a given water saturation. However, for WAG hysteresis, the three-phase water relative permeability curve typically exhibits a lower relative permeability than the two-phase curve at a given water saturation.

Given that the selection of the WAG hysteresis option for the water phase implies presumably that the three-phase effects associated with the WAG injection process are more significant than the effects associated with conventional hysteresis, the two-phase drainage and imbibition water relative permeability data need to be adapted for use with WAG hysteresis. For conversion from conventional drainage to WAG hysteresis two-phase water relative permeability data, the possible approaches include:

- Using the existing drainage data unchanged. If the drainage and imbibition water relative permeability data do not exhibit significant hysteresis, this may be a convenient approach.
- Using some form of averaged drainage and imbibition data. If the drainage and imbibition water relative permeability data exhibit significant hysteresis, this be a suitable approach given that this data will be used for both the drainage and imbibition processes in the absence of the gas phase.

For conversion from conventional imbibition to WAG hysteresis three-phase water relative permeability data, there is no simple approach based upon substitution or adaptation of existing data. The underlying premise of WAG hysteresis in the water phase is that the water relative permeability is lower following a gas flood than when initially only water and oil is present. Hence the three-phase WAG hysteresis data should ideally be obtained from measurements or modeling of water relative permeability in the presence of the gas phase.

One seemingly possible solution to the difference between conventional and WAG hysteresis for water relative permeability would appear to be provided by interchanging the water relative permeability columns of data in the pairs of tables associated with the drainage and imbibition saturation table indices. However, this is a difficult process to accomplish because only single columns of data in pairs of otherwise unchanged tables are involved and the table saturation nodes may be different. Moreover, the suitability of such an approach is questionable given that if the drainage and imbibition two-phase data columns are interchanged neither will be necessarily representative of the WAG hysteresis two-phase or three-phase water relative permeability preceding or following a gas injection cycle respectively.

It should be noted that if WAG hysteresis is applied only to the gas phase, the existing conventional hysteresis relative permeability saturation functions can be used unmodified.

When selecting WAG hysteresis, consideration should also be given to the form of conventional hysteresis being applied to the phases not subject to WAG hysteresis. For example, in a water-wet system, if the Killough hysteresis model is applied to both the wetting and non-wetting phases and if oil is wetting to gas, the hysteresis scanning curves for the relative permeability of oil to gas (wetting phase) will be driven by the properties of the two-phase drainage and imbibition gas relative permeability (non-wetting phase). Hence it will still be necessary to specify both suitable drainage and imbibition gas relative permeability data.

For the modeling of capillary pressure hysteresis in WAG injection processes, there is limited information in the published literature pertaining to how the characteristics differ if at all from those of conventional

hysteresis. In Eclipse capillary pressure hysteresis, if selected, is modeled using the Killough hysteresis model irrespective of whether conventional or WAG hysteresis is selected.

WAG hysteresis saturation regions

For various reservoir modeling strategies, simulations may involve comparisons between conventional and WAG hysteresis. When changing from conventional to WAG hysteresis or vice versa, the use of the drainage (SATNUM) and imbibition (IMBNUM) saturation tables for both conventional and WAG hysteresis requires that the water relative permeability data associated with the imbibition saturation tables and possibly also the drainage saturation tables are changed. Alternatively separate sets of tables can be maintained such that only the drainage and saturation table indices are changed. In order to facilitate the process of changing between conventional and WAG hysteresis, two additional saturation region numbers have been introduced for representing separately the WAG hysteresis saturation functions. The keywords WH2NUM and WH3NUM can be used to specify the table numbers for the WAG hysteresis two-phase and three-phase water relative permeability functions respectively. This provides a mechanism for the coexistence of both conventional two-phase and WAG hysteresis saturation functions.

The WH2NUM and WH3NUM keywords are optional; if these are not specified, the region numbers associated with these keywords will default to those defined for the SATNUM and IMBNUM keywords respectively. For the saturation tables associated with the WH2NUM and WH3NUM keywords, only the water relative permeability functions are utilized and the remaining data may be defaulted. In cases where the conventional two-phase water relative permeability data are considered suitable for use as the WAG-hysteresis two-phase relative permeability functions, the WH2NUM keyword can be omitted.

Where the WH2NUM or WH3NUM keywords have been specified, the minimum, critical and maximum water saturations should be equal for the corresponding SATNUM and WH2NUM table indices and for the corresponding IMBNUM and WH3NUM table indices. If one or more of these conditions are not satisfied, warning messages will be issued. Irrespective of whether or not the WH2NUM or WH3NUM keywords have been specified, warning messages will also be issued for grid blocks where the WAG hysteresis designated three-phase water relative permeability function is greater than the WAG hysteresis designated two-phase water relative permeability function at one or more table saturation nodes.

These additional optional water relative permeability functions can be considered essentially as additional columns of the associated SATNUM and IMBNUM saturation tables. If the end-point scaling option has been specified, the conventional and WAG hysteresis relative permeability will both be subject to the same horizontal (saturation) scaling using the keywords SWL, SWCR etc. for the drainage and two-phase curves and using the keywords ISWL, ISWCR etc. for the imbibition and three-phase curves. However, the vertical (relative permeability) scaling for the two-phase and three-phase water relative permeability data is applied separately via the additional keywords KRW, KRWR and IKRW, IKRWR respectively.

If the WH2NUM or WH3NUM keywords have been specified and the saturation table numbers differ from those specified via the corresponding SATNUM and IMBNUM keywords respectively, the two-phase and three-phase water relative permeability data will be used instead of the drainage and imbibition relative permeability data for all water relative permeability calculations in those saturation regions for which WAG hysteresis has been applied to the water phase. This is specified via item 5 of the WAGHYSTR keyword on a per saturation region basis.

For grid blocks for which the saturation table numbers specified by the SATNUM and WH2NUM keywords are the same, if end-point scaling has been specified, the values of KRW and KRWR will be ignored if specified. If these saturation table numbers are not the same, the values of KRW and KRWR will apply to the water relative permeability table values specified via WH2NUM. Similarly, for grid blocks for which the saturation table numbers specified by the IMBNUM and WH3NUM keywords are the same, if end-point scaling has been specified, the values of IKRW and IKRWR will be ignored if specified. If these

saturation table numbers are not the same, the values of IKRWH and IKRWRH will apply to the water relative permeability table values specified using WH3NUM.

A simple example of the use of the WH2NUM and WH3NUM keywords is shown below for the family II set of saturation functions. The gas and oil saturation functions associated with the WH2NUM and WH3NUM keywords are not used and can be defaulted in their entirety. For the water saturation functions only the saturation and water relative permeability columns are used and the remaining data may be defaulted subject to the specification of the minimum number of entries required for valid table entry. A similar approach may also be employed for the Family I set of saturation tables.

```

TITLE
Example using WH2NUM and WH3NUM with Family II Saturation Functions

DIMENS
10 1 1 /
...
SATOPTS
HYSTER
/
TABDIMS
4 ...
/
PROPS
...
EHYSTR
1* 2
/
...
WAGHYSTR
4* YES /
/
/
/
SGFN
... / -- Table 1 drainage (conventional hysteresis)
... / -- Table 2 imbibition (conventional hysteresis)
/ -- Table 3 unused
/ -- Table 4 unused
SOF3
... / -- Table 1 drainage (conventional hysteresis)
... / -- Table 2 imbibition (conventional hysteresis)
/ -- Table 3 unused
/ -- Table 4 unused
SWFN
... / -- Table 1 drainage (conventional hysteresis)
... / -- Table 2 imbibition (conventional hysteresis)
... / -- Table 3 two-phase krw only used for WAG hysteresis
... / -- Table 4 three-phase krw only used for WAG hysteresis

REGIONS
...
SATNUM
10*1 /
IMBNUM
10*2 /
WH2NUM
10*3 /
WH3NUM
10*4 /
...

```

A more complex example of the use of WH2NUM and WH3NUM is shown below for a model consisting of five layers each containing three grid blocks. End-point scaling has been specified and WAG hysteresis has been applied to the water phase in SATNUM regions (and hence layers for this model) 1, 2, 3 and 4 but not 5. The WAGHYSTR keyword single row table numbers 6 to 20 are ignored because there are no SATNUM regions associated with these table numbers. The usage of WH2NUM and WH3NUM for each layer and their associated vertical scaling end-points is summarized below:

1. WH2NUM is equal to SATNUM and WH3NUM is equal to IMBNUM for this layer and hence the two-phase and three-phase water relative permeability data used for WAG hysteresis will be taken from SATNUM and IMBNUM respectively. If KRWH, KRWRH, IKRWH or IKRWRH have been specified, they will be ignored when applying water relative permeability vertical scaling because the arrays KRW, KRWR, IKRW or IKRWR will be used if specified.
2. WH2NUM is not equal to SATNUM and WH3NUM is not equal to IMBNUM for this layer and hence the two-phase and three-phase water relative permeability data used for WAG hysteresis will be taken from WH2NUM and WH3NUM respectively. If KRWH, KRWRH, IKRWH or IKRWRH have been specified, their values will be used when applying water relative permeability vertical scaling and the values of KRW, KRWR, IKRW and IKRWR will be ignored if specified.
3. WH2NUM is equal to SATNUM but WH3NUM is not equal to IMBNUM for this layer and hence the two-phase and three-phase water relative permeability data used for WAG hysteresis will be taken from SATNUM and WH3NUM respectively. If specified, KRW and KRWR will be used for the two-phase water relative permeability vertical scaling but IKRWH and IKRWRH will be used for the three-phase water relative permeability vertical scaling.
4. WH2NUM is not equal to SATNUM but WH3NUM is equal to IMBNUM for this layer and hence the two-phase and three-phase water relative permeability data used for WAG hysteresis will be taken from WH2NUM and IMBNUM respectively. If specified, KRWH and KRWRH will be used for the two-phase water relative permeability vertical scaling but IKRW and IKRWR will be used for the three-phase water relative permeability vertical scaling.
5. WH2NUM is not equal to SATNUM and WH3NUM is not equal to IMBNUM for this layer. However, WAG hysteresis has not been applied to the water phase for this layer and hence this layer will be subject to conventional hysteresis according to the items specified via the [EHYSTR](#) keyword and WH2NUM, WH3NUM, KRWH, IKRWRH and IKRW will be ignored.

```

TITLE
Example using WH2NUM and WH3NUM with EPSDEBUG

DIMENS
3 1 5 /
ENDSCALE
/
SATOPTS
HYSTER
/
TABDIMS
20...
/
PROPS
...
EHYSTR
1* 2
/
EPSDEBUG
1 3 1 1 1 4
/
WAGHYSTR
4* YES /
4* YES /
4* YES /
4* YES /
4* NO /
/ Table 6
...
/ Table 20
...
REGIONS
...
SATNUM
3*1 3*2 3*3 3*4 3*5 /

```

```

IMBNUM
3*6 3*7 3*8 3*9 3*10 /
WH2NUM
3*1 3*12 3*3 3*14 3*15 /
WH3NUM
3*6 3*17 3*18 3*9 3*20 /
...

```

If end-point scaling has been specified, the reporting of the scaled tables via the [EPSDEBUG](#) or [EPSDBGS](#) keyword will reflect the use of WH2NUM and WH3NUM for the grid blocks selected for reporting in the manner described above. Specifically, if WAG hysteresis has been applied to the water phase:

6. If WH2NUM is not equal to SATNUM, the drainage report will indicate not only the drainage table number but also the two-phase table number from which the water relative permeability data will be taken in place of the drainage table water relative permeability data. In addition, KRWH and KRWRH will be used instead of KRW and KRWR.
7. If WH3NUM is not equal to IMBNUM, the imbibition report will indicate not only the imbibition table number but also the three-phase table number from which the water relative permeability data will be taken in place of the imbibition table water relative permeability. In addition, IKRWH and IKRWRH will be used instead of IKRW and IKRWR.

Non-wetting phase model (Gas)

Two-phase model

The gas phase model is based on the theory developed by Land and Carlson. The two-phase model differs from the standard Eclipse implementation as it uses an analytic version of the Carlson model based on the input Land's parameter, and ignores the imbibition curve input using the [IMBNUM](#) keyword.

Consider a typical drainage process followed by an imbibition process (as shown in the following figure).

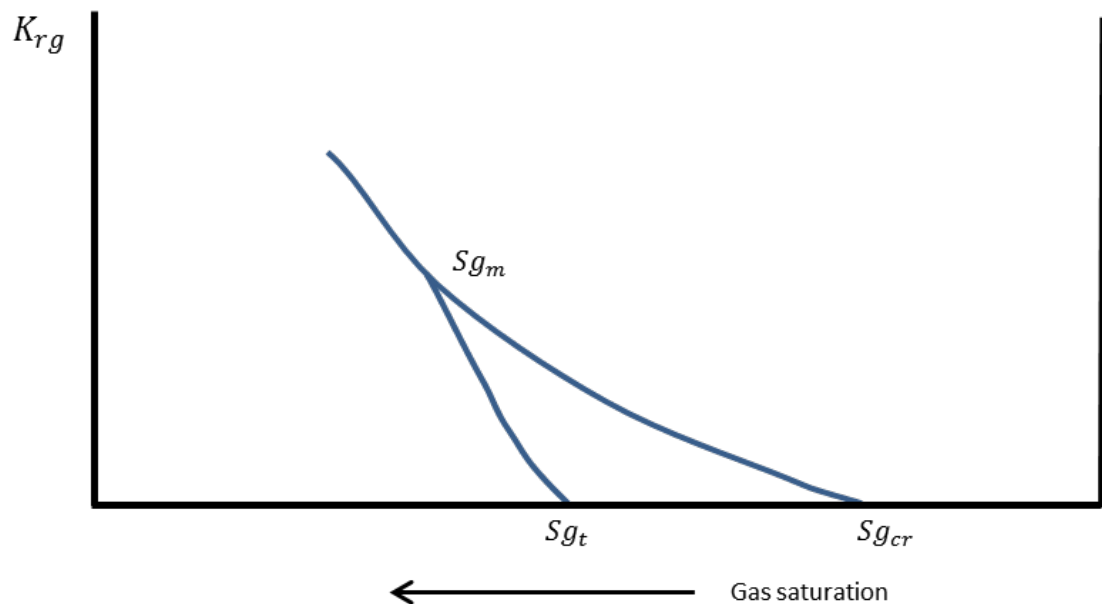


Figure 3.20. A typical drainage process followed by an imbibition process

Consider a drainage process reaching a maximum gas saturation Sg_m followed by a imbibition process leading to a trapped gas saturation Sg_{trap} . The trapped gas saturation Sg_{trap} is given by

$$Sg_{trap} = Sg_{cr} + \frac{(Sg_m - Sg_{cr})}{(1 + C \cdot (Sg_m - Sg_{cr}))} \quad \text{Eq. 3.101}$$

where

C is Land's parameter, input with the `WAGHYSTR` keyword, or specified by the transition parameter as discussed later.

Note: As for the generic hysteresis calculation, the Killough method has been modified slightly to overcome non-physical behavior when Sg_m is small. As Sg_m tends to Sg_{cr} in Equation 3.101, Sg_{trap} tends to Sg_{cr} , which leads to an infinite imbibition relative permeability gradient. This in turn can cause convergence problems, while having little material effect on the simulation results. Equation 3.101 has been modified as follows:

$$Sg_{trap} = Sg_{cr} + \frac{Sg_m - Sg_{cr}}{A + C(Sg_m - Sg_{cr})}$$

where $A = 1 + a \cdot (Sg_{max} - Sg_m)$, where Sg_{max} is the maximum attainable gas saturation. The parameter a is input through the fourth data item in the `EHYSTR` keyword and defaults to 0.1.

The gas relative permeability on the scanning curve is given by:

$$K_{rg}(Sg) = K_{rg}^{drain}(Sg_f) \quad \text{Eq. 3.102}$$

where:

$$Sg_f = Sg_{cr} + \frac{1}{2} \left\{ (Sg - Sg_{trap}) + \sqrt{(Sg - Sg_{trap})^2 + \frac{4}{C}(Sg - Sg_{trap})} \right\} \quad \text{Eq. 3.103}$$

This transformation typically has an infinite gradient dSg_f/dSg when Sg is close to Sg_{trap} . As the input drainage relative permeability functions are represented as piece-wise linear functions, the final relative permeability function also has an infinite gradient. As this causes convergence problems the transformation Sg to Sg_f is modified to a linear function near to the trapped gas saturation. The proportion of the transformation that departs for the pure Carlson equation can be controlled using the sixth data item in the `WAGHYSTR` keyword.

Note: In order to make a section of the imbibition curve linear to prevent an infinite gradient in the final relative permeability curve, the Carlson mapping algorithm is applied and the Land's parameter C in

equations 3.101 and 3.103 can be replaced by the transition parameter C^{trans} defined as

$$C^{trans} = \frac{1}{S_{ncr} - S_{ncrd}} - \frac{1}{S_{n \max} - S_{ncrd}}$$

If the gas saturation increases following an imbibition process, the gas relative permeability follows the imbibition curve provided that the model remains in two-phase mode, that is if the displacing phase is oil. The criterion for the model to remain in two-phase mode is that the water saturation at the start of the secondary drainage process must be less than the connate water saturation plus a threshold value. The threshold value is input via the seventh data item in the `WAGHYSTR` keyword, which defaults to 0.001.

Note that if the imbibition curve is retraced and the saturation reaches Sg_m then the primary drainage curve is followed for increasing gas saturation.

If the water saturation exceeds the connate saturation plus the threshold value, the three-phase model described below is used.

Dynamic trapped saturation

Sg_f in equation 3.103 can be interpreted as the free gas saturation for a given saturation state. Hence the trapped gas saturation is given by

$$Sg_{td} = Sg - Sg_f \quad \text{Eq. 3.104}$$

Three-phase model

The gas relative permeability follows a secondary drainage curve when the gas saturation increases and where the water saturation exceeds the connate value at the turnaround point. A typical secondary drainage curve is illustrated in the following figure.

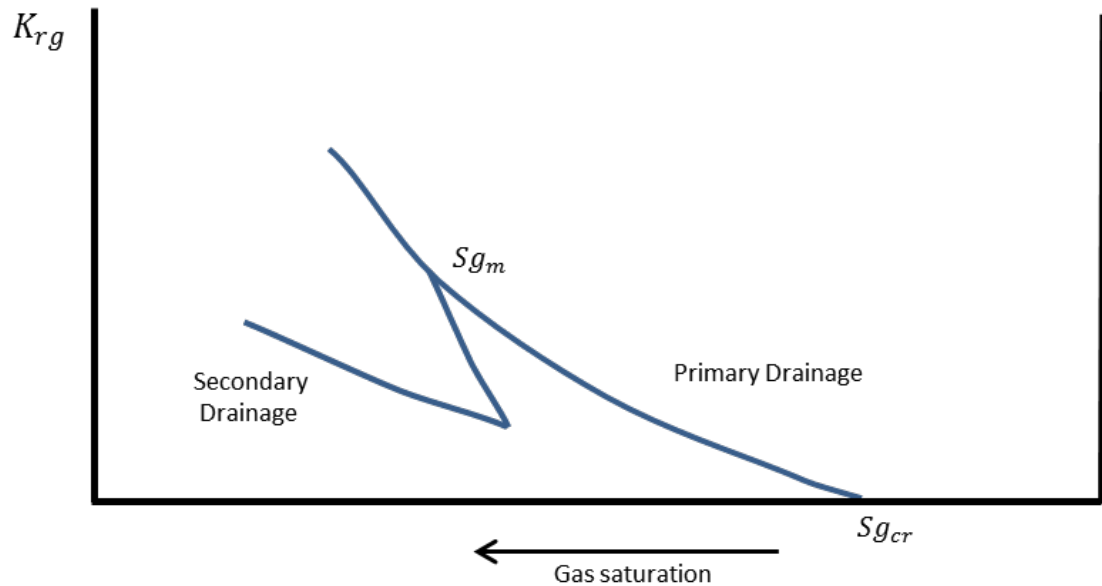


Figure 3.21. Schematic diagram showing a typical secondary drainage curve

These secondary drainage curves are calculated using the following equation

$$K_{rg}^{drain} = [K_{rg}^{input} - K_{rg}^{input}(Sg^{start})] \cdot \left[\frac{Sw_{co}}{Sw^{start}} \right]^\alpha + [K_{rg}^{imb}(Sg^{start})] \quad \text{Eq. 3.105}$$

where:

K_{rg}^{drain} is the calculated secondary drainage relative permeability as a function of Sg

K_{rg}^{input} is the input relative permeability at Sg

$K_{rg}^{input}(S_g^{start})$ is the input relative permeability at the gas saturation at the start of the secondary drainage curve

Sw_{co} is the connate water saturation

Sw^{start} is the water saturation at the start of the secondary drainage curve.

$K_{rg}^{imb}(S_g^{start})$ is the relative permeability at the start of the secondary drainage process (that is the K_r at the end of the imbibition curve)

α is the reduction exponent input in the [WAGHYSTR](#) keyword.

Subsequent imbibition curves are treated in an analogous way to an imbibition process from the primary drainage curve. The saturation is transformed using the Carlson equation, [3.103](#) and looked up on the secondary drainage curve. The trapped gas saturation is taken as the maximum historical trapped gas saturation, not the trapped gas saturation for this cycle.

If the transformed saturation Sg_f falls below the saturation at the start of the secondary drainage process, the secondary drainage curve cannot be used. In this case the transformation is modified and the primary drainage curve is used:

$$K_{rg}(Sg) = K_{rg}^{drain}(Sg_f^m) \quad \text{Eq. 3.106}$$

where Sg_f^m is given by:

$$Sg_f^m = Sg_{cr} \quad \text{where } Sg_f \leq Sg_{trap} \quad \text{Eq. 3.107}$$

$$Sg_f^m = \begin{cases} (Sg_f - Sg_{cr}) \cdot \frac{(Sg^{up} - Sg_{gcr})}{(Sg^{start} - Sg_{trf})} \\ Sg_{cr} - Sg_{trf} \cdot \frac{(Sg^{up} - Sg_{gcr})}{(Sg^{start} - Sg_{trf})} \end{cases} \quad \text{Eq. 3.108}$$

$$Sg_f > Sg_{trap}$$

where:

$$Sg_{trf} = (Sg_{trap} - Sg_{gcr}) \quad \text{Eq. 3.109}$$

Sg^{start} is the gas saturation at the start of the secondary drainage curve

Sg^{up} is the gas saturation on the drainage curve corresponding to the imbibition gas relative permeability at the start of the cycle.

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In some cases it has been observed that the gas relative permeability falls to near zero when the primary drainage curve is used to construct the secondary or subsequent drainage cycle in the manner described above. This can be prevented by using a modified version of equation [3.108](#) in which $(Sg^{start} - Sg_{trf})$ is replaced by $(Sg^{start} - Sg_{trap})$ in the denominators:

$$Sg_f^m = \begin{cases} (Sg_f - Sg_{cr}) \cdot \frac{(Sg^{up} - Sg_{gcr})}{(Sg^{start} - Sg_{trap})} \\ Sg_{cr} - Sg_{trf} \cdot \frac{(Sg^{up} - Sg_{gcr})}{(Sg^{start} - Sg_{trap})} \end{cases} \quad \text{Eq. 3.110}$$

This modified model may be selected by setting [item 197](#) of the `OPTIONS` keyword to 1.

In cases where, for one or more successive cycles of gas injection, the maximum gas saturation attained for the cycle does not exceed the maximum historical gas saturation attained on a previous cycle, the trapped gas saturation Sg_{trap} will not increase but remain constant at its value reached on the previous cycle experiencing the maximum historical gas saturation. Similarly the dynamic trapped gas saturation Sg_{td} will increase to the value of the trapped gas saturation and remain at this value until the gas saturation next exceeds the maximum historical gas saturation.

Note: The WAG (Water Alternating Gas) hysteresis option can be applied to calculate the gas trapped saturation during the CO₂ storage process. Although the WAG hysteresis option was initially developed for a three-phase system, it is also applicable in the [GASWAT](#) and [CO2STORE](#) options, which describe a two-phase system. In the GASWAT and CO2STORE options, the three-phase constitutive equations for WAG hysteresis are applied to a two-phase system.

Modification to the residual oil

If the STONE 1 three-phase oil relative permeability model is employed, (see "[Stone's first model \(Modified\)](#)") it is optionally possible to modify the SOM parameter to account for the trapped gas saturation. If this modification is employed, the assumption is made that original SOM represents the trapped hydrocarbon rather than just the trapped oil.

$$SOM_{mod} = SOM - (a \cdot Sg_{td}) \quad \text{Eq. 3.111}$$

where

SOM is the minimum residual oil saturation

Sg_{td} is the dynamic trapped gas saturation given by equation [3.104](#)

a is input as item 8 of the [WAGHYSTR](#) keyword.

Wetting phase model (Water)

The wetting phase model is based on the observation that the water relative permeability curve measured following a gas flood shows significantly less mobility. An example of the two relative permeability curves is shown in the following figure.

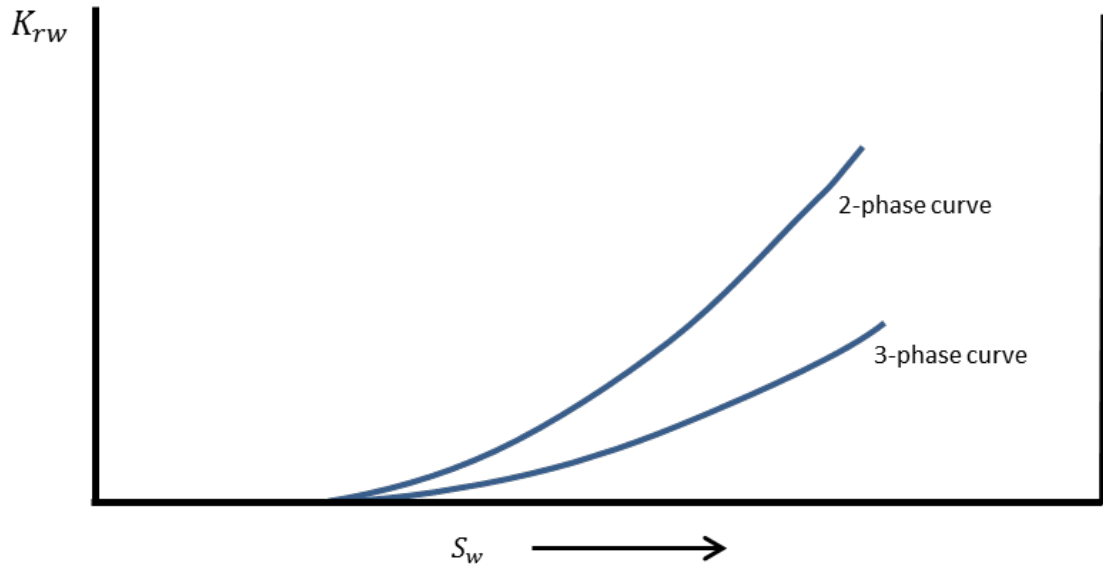


Figure 3.22. Two-phase and three-phase relative permeability curves

The two-phase curves are taken to be the drainage curves ([SATNUM](#)) and the three-phase curves to be the imbibition curves ([IMBNUM](#)). Note that these curves are not strictly drainage and imbibition but apply to the two-phase and three-phase cases respectively.

For an imbibition process, with S_w increasing, the relative permeability function used is interpolated between the two-phase and three-phase curves using the following equation

$$K_{rw}^{imb} = K_{rw2} \cdot \left(1 - \frac{S_g^{start}}{S_{gmax}}\right) + K_{rw3} \cdot \left(\frac{S_g^{start}}{S_{gmax}}\right) \quad \text{Eq. 3.112}$$

where:

K_{rw}^{imb} is the calculated imbibition relative permeability

K_{rw2} is the two-phase relative permeability at S_w

K_{rw3} is the three-phase relative permeability at S_w

S_{gmax} is the maximum attainable gas saturation, $1 - S_{wco} - S_{ogr}$

S_g^{start} is the gas saturation at the start of the imbibition process

S_{wco} is the connate water saturation

S_{ogr} is the critical oil-to-gas saturation.

A subsequent drainage relative permeability is calculated by interpolation between the imbibition curve and either the three-phase curve or the two-phase curve, depending on the gas saturation. A typical case is shown in the following figure.

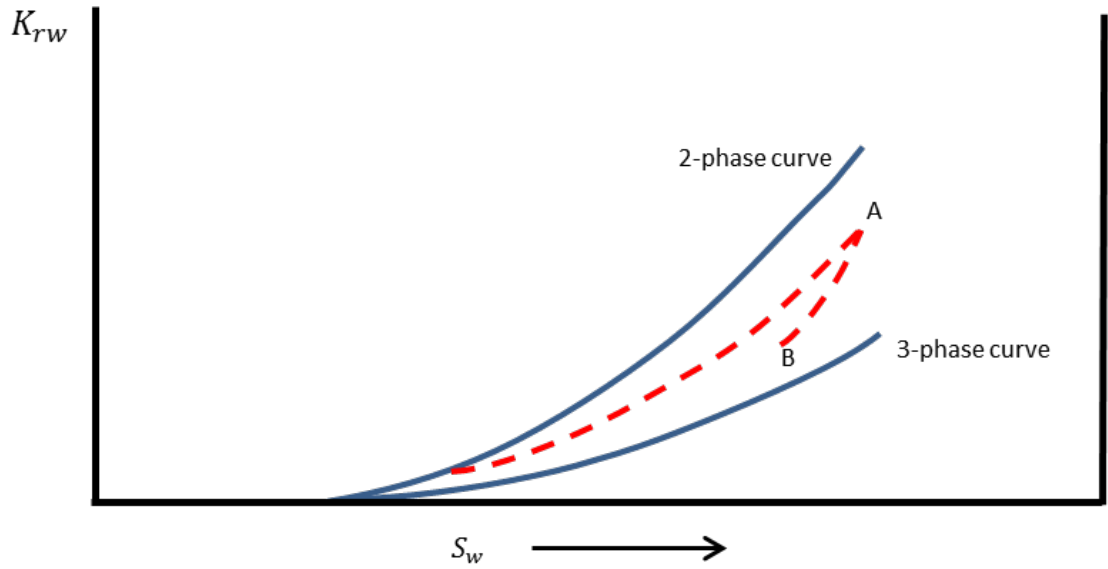


Figure 3.23. Subsequent drainage relative permeability

An initial imbibition process may reach point A. The subsequent drainage process in this case moves to point B, on a curve interpolated between the original imbibition curve and the input three-phase curve.

The drainage curve moves either side of the imbibition curve depending on the prevailing gas saturation relative to the gas saturation at the start of the drainage process S_g^{sdrain} .

If $S_g > S_g^{sdrain}$ then

$$K_{rw}^{drain} = K_{rw}^{imb} \cdot \left(1 - \frac{S_g - S_g^{sdrain}}{S_{g_{max}} - S_g^{sdrain}} \right) + K_{rw3} \cdot \left(\frac{S_g - S_g^{sdrain}}{S_{g_{max}} - S_g^{sdrain}} \right) \quad \text{Eq. 3.113}$$

If $S_g < S_g^{sdrain}$ then

$$K_{rw}^{drain} = K_{rw}^{imb} \cdot \left(1 - \frac{S_g^{sdrain} - S_g}{S_g^{sdrain}} \right) + K_{rw2} \cdot \left(\frac{S_g^{sdrain} - S_g}{S_g^{sdrain}} \right) \quad \text{Eq. 3.114}$$

where:

S_g^{sdrain} is the gas saturation at the start of the drainage process, point A

K_{rw}^{drain} is the calculated drainage curve

K_{rw}^{imb} is the previous imbibition relative permeability at S_w

K_{rw2} is the input two-phase curve at S_w

K_{rw3} is the input three-phase curve at S_w

Subsequent secondary drainage processes follow an interpolated curve in much the same manner as the primary imbibition, but using a modified S_g^{start} arranged such that the curve runs through point B.

Rock compaction

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x	Eclipse 300

Eclipse 300

Eclipse 100

Eclipse 300

By default Eclipse models rock compaction as a simple compressibility, which is entered using the [ROCK](#) keyword. Additional options are available to model more complex behavior:

- A tabular description of the compaction as a function of pressure, which can be either reversible or irreversible ("[Tabulated compaction curves](#)")
- The ability to modify the transmissibility as a function of pressure
- A hysteretic model to allow partial reflation
- A cyclic hysteretic model to allow compaction and dilation
- A water-induced compaction model ("[Water induced compaction \(Eclipse 100\)](#)")
- The Palmer-Mansoori rock model (Coal Bed Methane option only) ("[Palmer-Mansoori rock model](#)")
- The ability to modify the transmissibility as a function of porosity.

Tabulated compaction curves

The compaction of the rock can be modeled as a general function of pressure using the [ROCKTAB](#) keyword. In the case where a grid block pressure increases, the compaction can either be assumed to be reversible or irreversible.

In some cases these two reflation options (fully reversible or irreversible) may be too crude a description. A hysteretic option is available by using the [ROCKTABH](#) keyword. In this case a set of reflation curves can be input to describe the reversible reflation behavior from each starting pressure. For pressures below the starting pressure, the behavior is only partially reversible, given by the deflation curve.

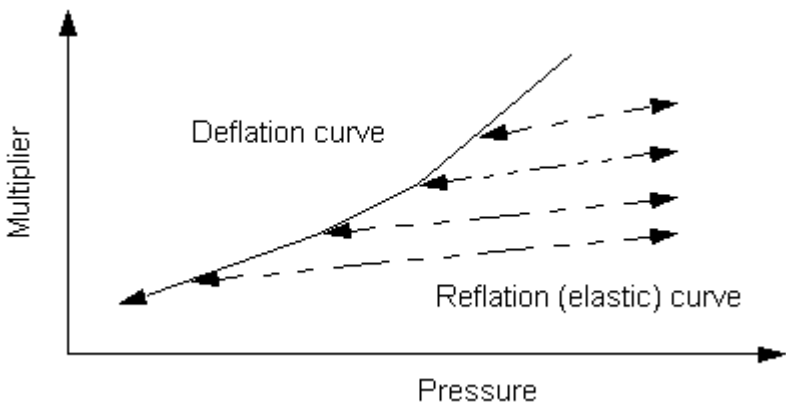


Figure 3.24. Rock Compaction: Hysteresis option

A second hysteretic option can be used to model cyclic dilation and recompaction. This model is based on [Beattie et al \(1991\)](#). The [ROCKTABH](#) keyword is again used to input curves that describe the reversible (or elastic) behavior. The reversible region is bounded at low pressures by the deflation (or compaction) curve and at high pressures by a dilation curve which describes partially reversible expansion. This option can be selected using the [ROCKCOMP](#) keyword.

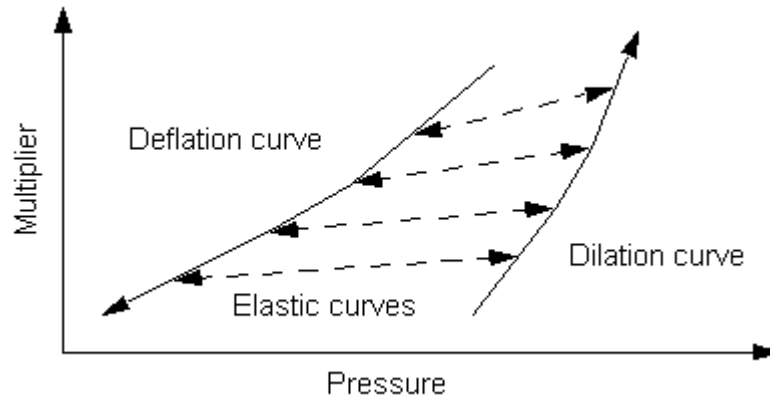


Figure 3.25. Rock Compaction: Boberg Hysteresis option

Initially it is assumed the reservoir rock is fully compacted. The initial pore volume and transmissibility multipliers will be set to the deflation curve value at the cell pressure. In general these multipliers will not be equal to one, so the initial pore volume and transmissibility will differ from the values entered in the data file. Alternatively, the rock compaction multipliers can be tabulated as functions of pressure change (ΔP), and the `ROCKOPTS` keyword can be used to store the initial pressure (see below). The initial multipliers will then be set to the deflation curve values at $\Delta P = 0$, which can be set to 1.

If the initial reservoir rock is not fully compacted, pore volume and transmissibility values can be set by finding an elastic curve with pore volume multiplier equal to one at the initial cell pressure. In this case the pore volume will be the same as the value entered in the data file, although the transmissibility may still differ. This option can be selected using the `ROCKOPTS` keyword.

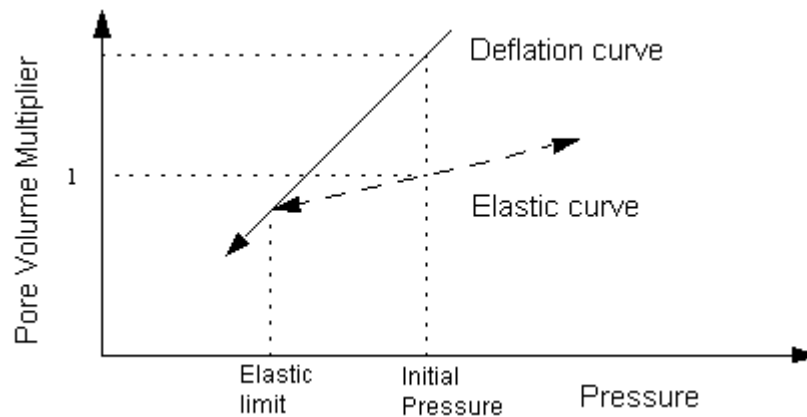


Figure 3.26. Initial multipliers

Transmissibility modification

In addition to the compaction (the reduction in the pore volume), it is possible to modify the transmissibility as a function of pressure.

At present the transmissibility modification is treated explicitly, that is the modification due to a change in pressure only affects the subsequent timestep.

If a transmissibility multiplier is specified with the [ROCKTAB](#) or [ROCKTABH](#) keywords, the well connection transmissibility factor, grid block transmissibilities and NNCs are modified accordingly.

The generalized mobility terms are updated for grid cells and well connections. Similar functionality, but based on the corresponding changes in porosity, is also available by specifying the keywords [ROCKTRMX](#), [ROCKTRMY](#) and [ROCKTRMZ](#).

In dual porosity runs, it is also possible to specify pressure tables of the sigma multiplier in order to modify the matrix-fracture coupling as a result of rock compaction. Keywords [ROCKTSIG](#) or [ROCKTHSG](#) provide the input of irreversible/reversible and hysteretic tables, respectively.

Overburden pressure

A reservoir simulator typically works in terms of pressure, as pressure is the natural variable when considering the flow. However, it is often better to consider the compaction as a function of the effective stress on the rock, which is the overburden pressure minus the fluid pressure. The [ROCKTAB](#), [ROCKTABH](#) or [ROCK2D](#) and [ROCK2DTR](#) keywords tabulate the compaction multipliers against the effective fluid pressure instead $P_{eff\ fl} = P_{fluid} - \text{Overburden}$.

The overburden pressure can be input into Eclipse as a function of depth. Note that the overburden is assumed to remain constant through time.

Typically the overburden pressure will be greater than the fluid pressure. In this case it is convenient to tabulate the compaction against the effective stress rather than the pressure. This can be achieved by using the [ROCKOPTS](#) keyword. In this case the pore volume and transmissibility multipliers are tabulated against the effective overburden pressure, $P_{eff\ ov} = \text{Overburden} - P_{fluid}$, which is $-P_{eff\ fl}$.

Reservoir pore volumes

There is scope for confusion when supplying porosities or pore volumes to Eclipse. These could either be at reference conditions (derived from core data); or at reservoir conditions (derived from log data).

Eclipse treats the input porosities (PORO) and pore volumes (PORV) as at reference conditions. With the rock compressibility model activated by the [ROCK](#) keyword, the pore volumes are then referenced at the reference pressure in the [ROCK](#) keyword. It should be borne in mind that other simulators will assume the geometric PORV represents reservoir pore volumes. The input PORV can be made equivalence to the initial reservoir pore volume in Eclipse by setting the second item of the [ROCKOPTS](#) keyword to 'STORE'.

With the rock compaction model (keywords [ROCKTAB](#), [ROCKTABH](#)), setting the second item in [ROCKOPTS](#) to 'STORE' and tabulating the [ROCKTAB](#) and [ROCKTABH](#) tables against effective pressure (which is zero at equilibrated conditions) treats the initial PORO and PORV as reservoir conditions.

Transmissibility as a function of porosity

In order to define a model which can be used to specify the dependence of transmissibility on porosity, refer to items 4 and 5 of the [ROCKCOMP](#) keyword. There are two different options which can be used:

- EXP Transmissibility is as exponential function of porosity:

$$\frac{\kappa}{\kappa_0} = e^{\left[\kappa_{\text{mult}} \left(\frac{\varphi - \varphi_0}{1 - \varphi_0} \right) \right]}$$

where

κ_0 is the initial transmissibility,

φ_0 is the initial porosity and

κ_{mult} is the user-defined multiplier factor defined by keywords, [ROCKTRMX](#), [ROCKTRMY](#) and [ROCKTRMZ](#) for a particular direction. This model is based on the general approach as described by [Pierre Lebel \(2002\)](#).

- CZ Transmissibility is a function of porosity via the Carmen-Kozeny type formula:

$$\frac{\kappa}{\kappa_0} = \frac{\varphi}{\varphi_0} \tau^{\left[\left(\frac{1-\varphi_0}{1-\varphi}\right)\right]}$$

where τ is an integer defined in item 5 of the [ROCKCOMP](#) keyword.

Items 4 and 5 of the [ROCKCOMP](#) keyword are available for use with the [ROCKTAB](#) and [ROCKTABH](#) keywords. They are currently not available for use with the [ROCK](#) keyword.

Output

Both the modified pore volumes and the transmissibilities can be output the [Print](#) and [RESTART](#) files.

Eclipse 100

In Eclipse 100, the output is controlled by the mnemonic [ROCKC](#) in keywords [RPTSCHED](#) and [RPTRST](#).

If we make the assumption that the compaction occurs in the vertical direction only, the compaction can be interpreted as a change in the height of the reservoir. When the hysteretic compaction option is used, the modified reservoir thickness can also be output.

The arguments used in the [print](#) and [RESTART](#) files are:

Argument	Description
PRES-OVB	Overburden pressure
PRES-EFF	Effective fluid pressure
PEFF-MIN	Effective minimum pressure for hysteretic compaction for ROCKTABH
PORV-MOD	Multiplier on pore volumes for ROCKTAB
TRAN-MOD	Multiplier on transmissibilities for ROCKTAB and ROCKTABH
DZ	Modified DZ, assuming compaction is vertical for ROCKTAB and ROCKTABH
DZ-TOTAL	Total modified DZ for each column of cells in the areal plane for ROCKTAB or ROCKTABH
DZ-COMP	Total compacted DZ for each column of cells in the areal plane for ROCKTABH only
DZ-TOTM	Total modified DZ for each column of cells in the areal plane for ROCKTABH only for the matrix cells in a dual porosity run only
DZ-COMPM	Total compacted DZ for each column of cells in the areal plane for ROCKTABH only for the matrix cells in a dual porosity run only
DZ-TOTF	Total modified DZ for each column of cells in the areal plane for ROCKTABH only for the matrix cells in a dual porosity run only
DZ-COMPF	Total compacted DZ for each column of cells in the areal plane for ROCKTABH only for the fracture cells in a dual porosity run only

Argument	Description
SWAT-EFF	Effective water saturation for the water induced compaction model.

Eclipse 300

In Eclipse 300, the output is controlled by the [RPTSOL](#), [RPTSCHED](#) and [RPTRST](#) keywords. The arrays are output with the name of the mnemonic used to request the output.

Argument	Description
PRES_OVB	Overburden pressure
PRES_EFF	Effective fluid pressure for ROCKTAB or ROCKTABH
PORV_MOD	Multiplier on pore volumes for ROCKTAB or ROCKTABH
PRESMIN	Minimum pressure for rock compaction hysteresis with ROCKTABH.

Water induced compaction (Eclipse 100)

Most rocks will compact to some extent as the stress on the rock is increased, when the fluid pressure falls. Some rocks, typically chinks, will exhibit additional compaction when water contacts oil bearing rock, even at constant stress. This behavior could have a significant effect on the performance of a water flood as it adds significant energy to the reservoir.

The water induced compaction option allows you to model this effect by supplying tables making the pore volume a function of both pressure and water saturation. The option allows two methods of data input:

- Two tables of pore volume multipliers, one as a function of pressure and the other as a function of water saturation. The pore volume modification is then the product of the two multipliers:

$$V = V_G \cdot M_p(P) \cdot M_s(Sw_{max} - Sw_{ini})$$

where

V is the pore volume

V_G is the geometric pore volume

$M_p(P)$ is the pore volume multiplier as a function of pressure

$M_s(S)$ is the pore volume multiplier as a function of water saturation

Sw_{max} is the maximum water saturation attained by the grid block

Sw_{ini} is the initial water saturation.

The pore volume multipliers are taken from the [ROCKTAB](#) / [ROCKTABH](#) and [ROCKTABW](#) keywords.

- A full two-dimensional table of pore volume multipliers as a function of both pressure and water saturation:

$$V = V_G \cdot M(P, Sw_{max} - Sw_{ini})$$

where $M(P, S)$ is a two-dimensional table entered using the [ROCK2D](#) and [ROCKWNOD](#) keywords.

The transmissibility modification is handled in a similar manner. If method 1 is used to input the compaction data, the transmissibility modification is the product of the pressure and saturation multipliers.

If method 2 is used, transmissibility modifiers can be entered using an analogous two dimensional table with the [ROCK2DTR](#) keyword.

Palmer-Mansoori rock model

Note: This is only available in the coal bed methane option (see the [COAL](#) keyword).

The Palmer-Mansoori rock model ([Palmer and Mansoori, 1998](#)) adjusts the pore volumes, $V(P)$, at the current pressure, P , using

$$V(P) = V(P_0) \left\{ 1 + \frac{c_m}{\varphi_0}(P - P_0) + \frac{\varepsilon_l}{\varphi_0} \left(\frac{K}{M} - 1 \right) \left(\frac{\beta P}{1 + \beta P} - \frac{\beta P_0}{1 + \beta P_0} \right) \right\}$$

where:

φ_0 is the initial porosity

P_0 is the initial pressure

$V(P_0)$ is the pore volume at the initial pressure

K is the bulk modulus

M is the constrained axial modulus

γ is the grain compressibility

β, ε_l are parameters of the match between the Langmuir curve and the change in volumetric strain due to matrix shrinkage

f is a fraction between 0 and 1

$c_m = \frac{g}{M} - \left[\frac{K}{M} + f - 1 \right] \gamma$ where g is a geometric factor (typical in the range 0–1), related to the orientation of the fracture system, see [Clarkson et al \(2010\)](#).

Following Palmer-Mansoori, assuming that permeability varies with porosity as

$$\frac{k}{k_0} = \left(\frac{\varphi}{\varphi_0} \right)^n$$

where the exponent n is typically 3, then the transmissibility multiplier is taken as

$$\left\{ 1 + \frac{c_m}{\varphi_0}(P - P_0) + \frac{\varepsilon_l}{\varphi_0} \left(\frac{K}{M} - 1 \right) \left(\frac{\beta P}{1 + \beta P} - \frac{\beta P_0}{1 + \beta P_0} \right) \right\}^n$$

This model is activated using the `PALM-MAN` option on item 1 of the `RUNSPEC` section keyword [ROCKCOMP](#). The parameters for the model are set in the `PROPS` section keyword [ROCKPAMA](#).

Eclipse 300

Cells not connected to coal cells only apply the compression term to the pore volume and permeability multipliers.

Compositional coal swelling/shrinkage model (Eclipse 300)

A modified form of the Palmer-Mansoori rock model is available by using the [ROCKPAME](#) keyword. A similar approach is described in [Mavor and Gunter \(2006\)](#). An important difference is that the model described here will not predict volume changes for undersaturated coal. Each component of the system can be assigned Langmuir curve parameters β_k, ε_k , making the swelling or shrinkage of the coal dependent on composition. The strain of one component is computed as if the coal gas is instantly in equilibrium following the extended Langmuir formula for adsorption capacity

$$e_k = \frac{\varepsilon_k \beta_k a_k P}{1 + \sum \beta_j a_j P}$$

where $a_k = L_k / (\sum L_j)$ represent the adsorbed mole fraction of component k , P the pressure and L_k are the adsorption capacities of each component (see keyword [LANGMEXT](#)).

The total strain is then computed as

$$E = \sum e_k$$

Using this in the Palmer-Mansoori equation the pore volume, $V(P, Y)$, at the current pressure, P , and gas phase composition $Y = (y_1, y_2, \dots, y_{nc})$ are adjusted using

$$V(P, Y) = V_0 \left\{ 1 + \frac{c_m}{\varphi_0} (P - P_0) + \frac{1}{\varphi_0} \left(\frac{K}{M} - 1 \right) (E - E_0) \right\}$$

where:

φ_0 is the initial porosity

P_0 is the initial pressure

V_0 is the pore volume at the initial pressure and composition

K is the bulk modulus

M is the constrained axial modulus

γ is the grain compressibility

β_k, ε_k are parameters of the match between the Langmuir curve and the change in volumetric strain due to matrix shrinkage for component k .

f is a fraction between 0 and 1

the term c_m is given as $c_m = \frac{1}{M} - \left[\frac{K}{M} + f - 1 \right] \gamma$

For undersaturated coal where the adsorbed gas is not changing the formula is modified so the strain is constant. The strain used then corresponds to the state where the gas would start to desorb.

Note also that the [ROCKPAME](#) keyword only apply to regions with cells connected to coal cells as assigned by [COALNUM](#).

Note: The rock compaction models takes the input porosities as the porosities at reservoir conditions. In areas without coal different rock compaction models can be used but some care must be taken so the input porosities are treated consistently for the entire model.
